Altera SoC Linux Intro Workshop



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Altera SW SoC Workshop Series

- SW Workshop #1 – Altera SoC SW Development Overview
- SW Workshop #2 – Introduction to Linux on Altera SoC
- SW Workshop #3 – Developing Drivers for Altera SoC Linux

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Agenda

- Essential Information Resources
- SoC Device Overview
- SoC Physical Address Map
- SoCFPGA Development Flow & Tools
- Altera SoC Linux Overview
- Components of the SoC FPGA Linux BSP
- SoC Linux Upstreaming & Driver Support
- Altera SoC Linux Boot Flow
- Das U-Boot Bootloader
- Linux Device Tree for SoC FPGA
- Take Home Lab

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Welcome. Here's What You Can Expect Today

Experienced Linux Developers

- Find a familiar embedded Linux development flow
- Overview of upstreaming and driver support for mach_SoCFPGA architecture
- Guide to SoCFPGA resources

New Linux Developers

- An exposure to the components of embedded Linux
- Essential Linux learning and documentation resources

Hardware Developers

- HW handoff to Linux build flow
- Boot and FPGA configuration for Linux
- SW implications of HW architecture

Everyone

- SoC FPGA architecture-specific information
- SoC FPGA recommendations and best practices

Focused on SoC/Nios Linux Specific Topics

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Essential Information Resources

Where to learn more…

…a non-exhaustive list



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Linux Foundation Training

Linux Developer classes are designed to help participants:

- Learn how to develop an embedded Linux product
- Become familiar with and learn to write device drivers
- Get practical experience with the Linux kernel
- Learn how to work with the Linux developer community

Developer Courses

- *LFD331* – Developing Linux Device Drivers
- *LFD405* – Building Embedded Linux with the Yocto Project
- *LFD411* – Embedded Linux Development
- *LFD414* – Introduction to Embedded Android Development
- *LFD205* – How to Participate with the Linux Community
- *LFD211* – Introduction to Linux for Developers
- *LFD262* – Developing with Git
- *LFD312* – Developing Applications for Linux
- *LFD320* – Linux Kernel Internals & Debugging
- *LFD415* – Inside Android: An Intro to Android Internals
- *LFD432* – Optimizing Linux Device Drivers for Power Efficiency

http://training.linuxfoundation.org/linux-courses/development-training

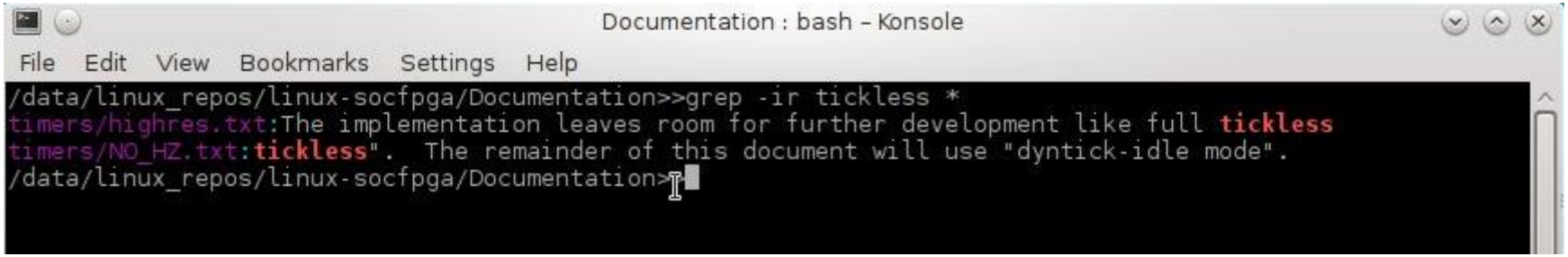
Linux Documentation Resources

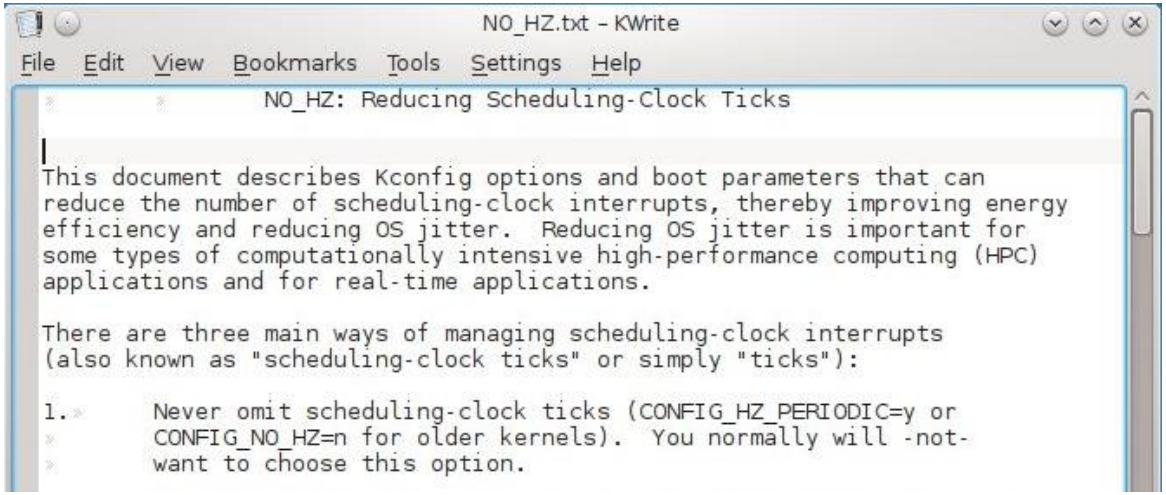
- GIT
  - Distributed revision control system to enable distributed collaboration
  - On-line documentation & training:
    - http://git-scm.com/doc
    - https://training.github.com
- Denx U-Boot Manual
  - Complete documentation from the folks who wrote Das U-Boot
    - http://www.denx.de/wiki/U-Boot/Documentation
- Free-Electrons:
  - Complete training materials posted free
    - http://free-electrons.com/docs/
- Device Tree for Dummies
  - http://events.linuxfoundation.org/sites/events/files/slides/petazzoni-device-tree-dummies.pdf

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The Two Best Sources for Linux Development Information

- Linux Kernel Documentation
  - The most complete and most essential Linux kernel documentation
  - Included with the Linux kernel source code
    - <local GIT repo>/Documentation

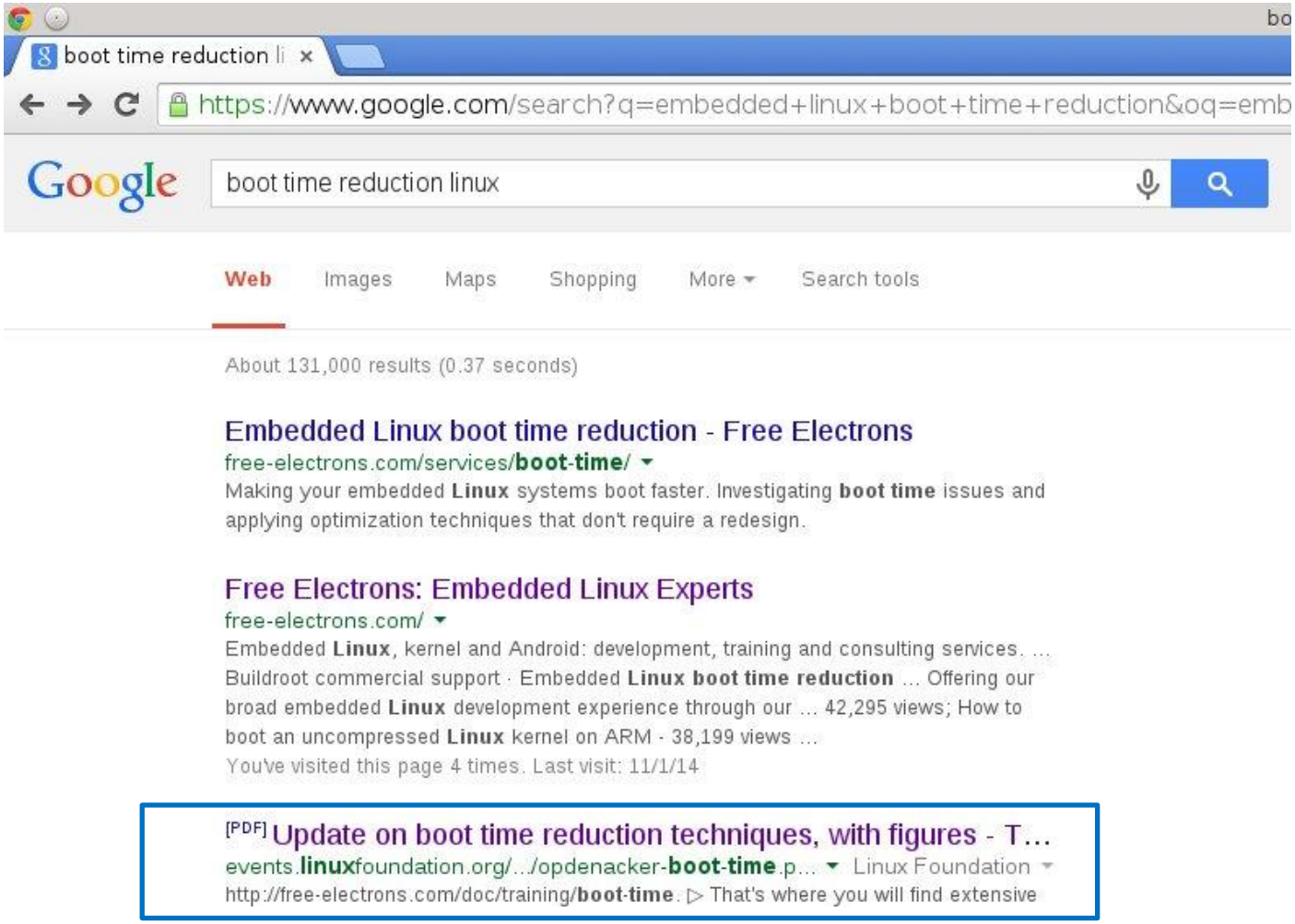




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The Two Best Sources for Linux Development Information

- An open source OS breeds open source information



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RocketBoards.org – Altera SoC Linux Community Portal

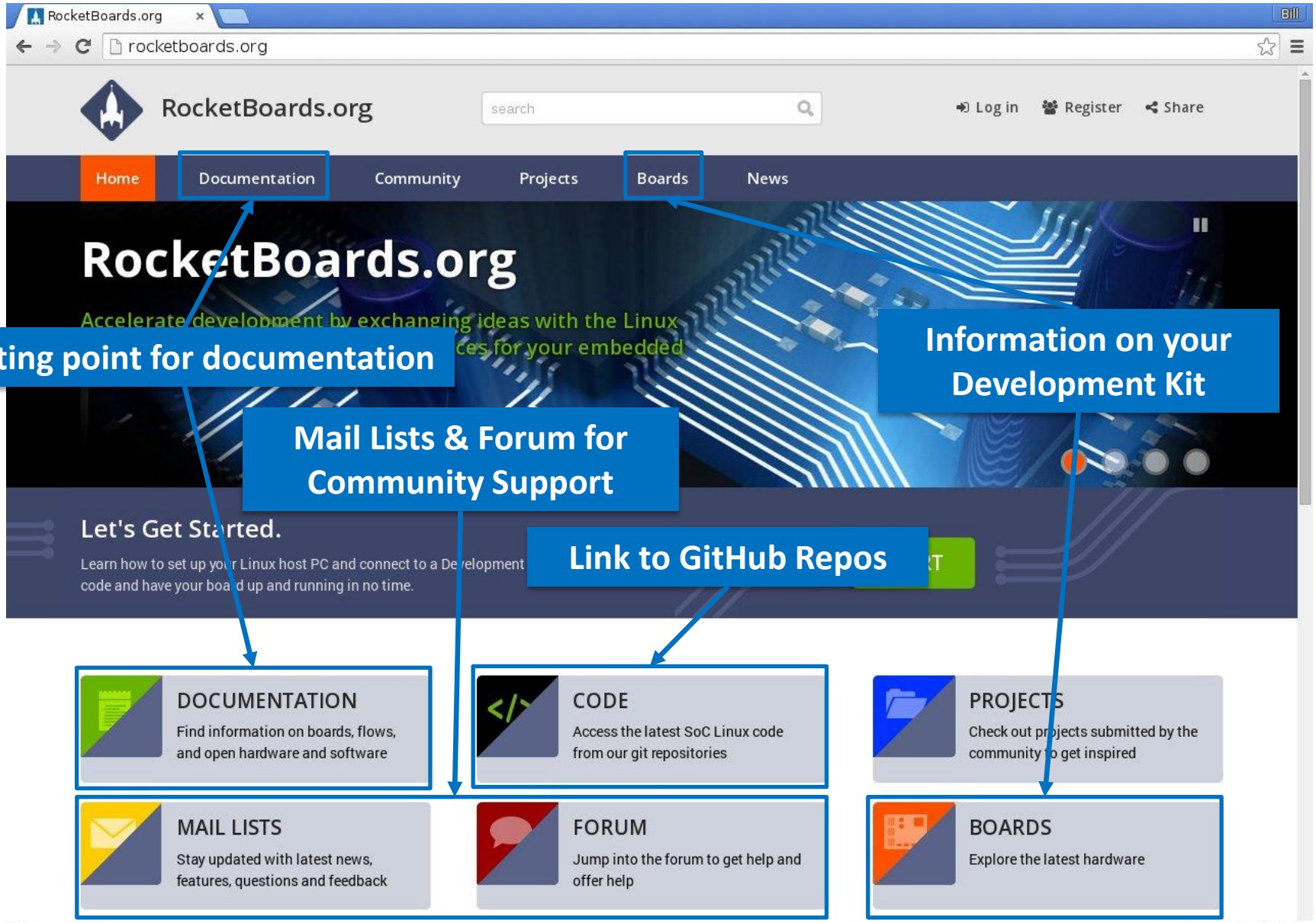


- The source for SoC FPGA Linux info
  - Golden System Reference Design (GSRD)
  - Updates on latest releases
  - Step-by-step getting started guides
- SoC FPGA Mailing List - RFI
  - Active community participation in answering SoC FPGA and Linux questions
- Example Projects, Applications, and Designs
  - From Altera and the SoC community
- Enables the SoC community to support Linux

RocketBoards.org

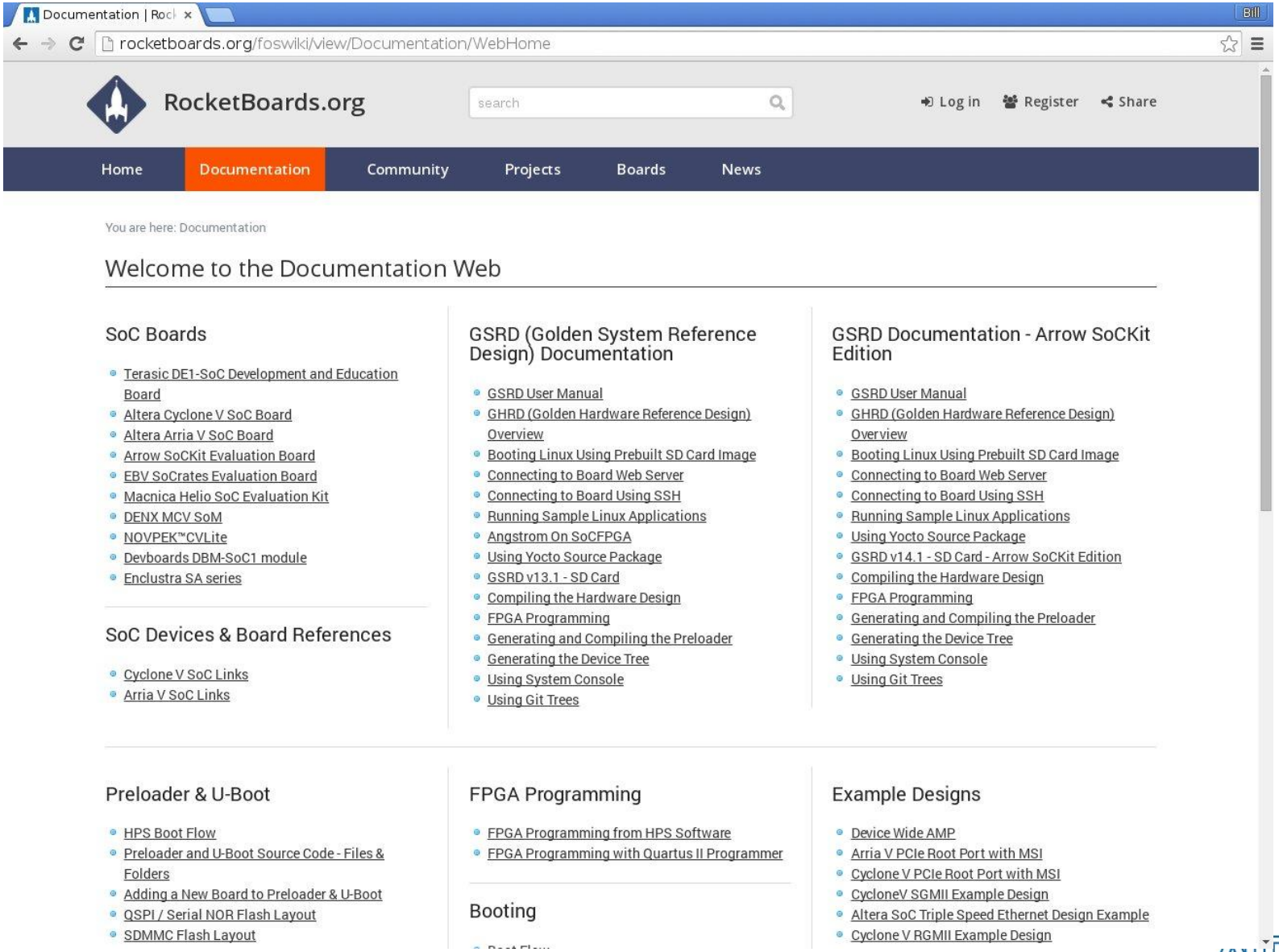
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RocketBoards.org Resources



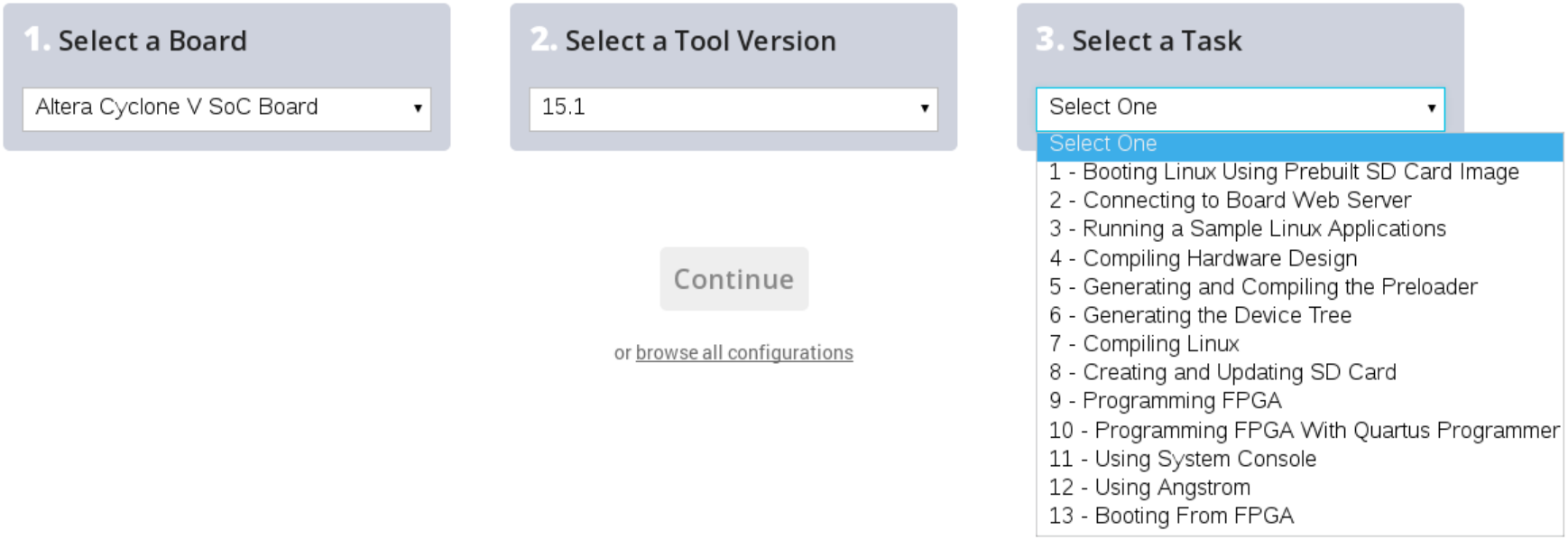
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RocketBoards.org Documentation



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RocketBoards.org – Useful Links



- Device Tree Generator User Guide
  - http://www.rocketboards.org/foswiki/Documentation/GSRD141DeviceTreeGenerator
- Programming FPGA from HPS
  - http://www.rocketboards.org/foswiki/Documentation/GSRD131ProgrammingFPGA
  - NOTE: the new FPGA manager core framework has recently been up streamed
- GSRD Releases
  - http://releases.rocketboards.org

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Several Ways to Learn!

- **Instructor-led training**
  - Face to face with an Altera expert Training Engineer
  - 20+ courses to choose from (8 hour classes)



- **Virtual classes (taught via WebEX)**
  - Can ask questions to Altera expert Training Engineer
  - Course content same as instructor-led classes (1/2 day sessions)



- **Online training (free and always available)**
  - 200+ topics available (~30 minutes in length)



- **Videos (free and always available)**
  - YouTube videos (~4 minutes each)



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SoC Classes Available



- Instructor-led or virtual classes
  - Designing with an ARM-based SoC
  - Developing Software for an ARM-based SoC
- Online classes
  - Hardware Design Flow for an ARM-based SoC
  - Software Design Flow for an ARM-based SoC
  - SoC Hardware Overview: the Microprocessor Unit
  - SoC Hardware Overview: Interconnect and Memory
  - SoC Hardware Overview: System Management, Debug, and General Purpose Peripherals
  - SoC Hardware Overview: Flash Controllers and Interface Protocols
  - SoC Bare-metal Programming and Hardware Libraries
  - Getting Started with Linux for Altera SoCs

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Essential SoC Software Tools Online Videos

- ARM DS-5 Altera Edition Toolchain
  - https://youtu.be/HV6NHr6gLx0
- DS-5 Altera Edition: Bare-metal Debug and Trace
  - https://youtu.be/u_xKybPhcHI
- DS-5 Altera Edition: FPGA-adaptive Linux Kernel Debug and Trace
  - https://youtu.be/lrR-SfVZd18
- Debugging Linux applications on the Altera SoC with ARM DS-5
  - https://youtu.be/ZcGQEjkYWOc
- FPGA-adaptive debug on the Altera SoC using ARM DS-5
  - https://youtu.be/2NBcUv2TxbI
- Streamline Profiling on Altera SoC FPGA. Part 1 - Setup
  - https://youtu.be/X-k9lmXQTio
- Streamline Profiling on Altera SoC FPGA. Part 2 - Running Streamline
  - https://youtu.be/Tzbd7qldKqY

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Essential SoC Hardware Documentation Resources

- Hard Processor System Technical Reference Manuals
  - Available in Device Handbooks:
    - https://www.altera.com/products/soc/portfolio/cyclone-v-soc/support.html
    - https://www.altera.com/products/soc/portfolio/arria-v-soc/support.html
    - https://www.altera.com/products/soc/portfolio/arria-10-soc/support.html
  - Contain Functional Descriptions Peripheral
  - Contain Control Register Address Map and Definitions
    - These are also available online at the links above in HTML and PDF formats
- HPS SoC Boot Guide
  - Cyclone V SoC & Arria V SoC: AN709 - HPS SoC Boot Guide
  - Arria 10 SoC: included in HPS TRM in Arria 10 Device Handbook
    - Arria 10 SoC secure booting: AN759 – Arria 10 SoC Secure Boot User Guide
- ARM Documentation Site
  - Documentation available for all ARM IP
    - Cortex-A9 & A53 MP Cores, FPU, NEON, GIC, ARM Peripherals, etc.
  - Requires free registration
  - Refer to HPS TRM for IP core names and revision information
  - http://infocenter.arm.com/help/index.jsp

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Essential SoC Software Documentation Resources

- Altera SoC Embedded Design Software (SoC EDS) Tools
  - User Guide:
    https://www.altera.com/content/dam/altera-www/global/en_US/pdfs/literature/ug/ug_soc_eds.pdf
    - Linux & Baremetal Software Development Tools Overview
    - HPS Preloader User Guide
    - HPS Flash Programmer User Guide
    - SD Card Boot Utility
  - Getting Started Guides: Preloader, Linux, Bare Metal, Debug, HW Library
    http://www.alterawiki.com/wiki/SoCEDSGettingStarted
  - SoC HPS Release Notes
  - SoC Abstraction Layer (SoCAL) API Reference
    - <SoC EDS install dir>/ip/altera/hps/altera_hps/doc/socal/html/index.html
  - Hardware Manager API Reference
    - <SoC EDS install dir>/ip/altera/hps/altera_hps/doc/hwmgr/html/index.html
  - GCC Documentation
    - <SoC EDS install dir>/ds-5/documents/gcc/getting_started.html
  - Bare Metal Compiler
    - <SoC EDS installation directory>/host_tools/mentor/gnu/arm/baremetal/share/doc/sourceryg++-arm-altera-eabi

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SoC Device Overview



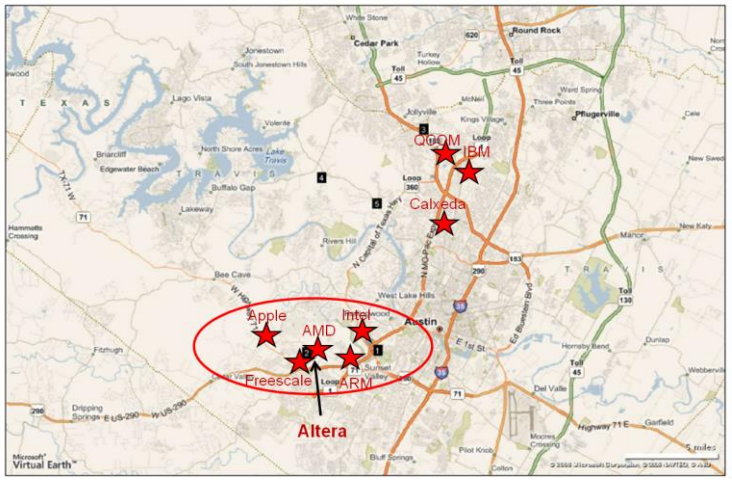
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Altera Investment in Embedded Technologies

- Altera established Austin Technology Center (ATC) in 2011
- Altera's primary embedded engineering center
- Austin provides access to one of the richest embedded processing talent bases in the world

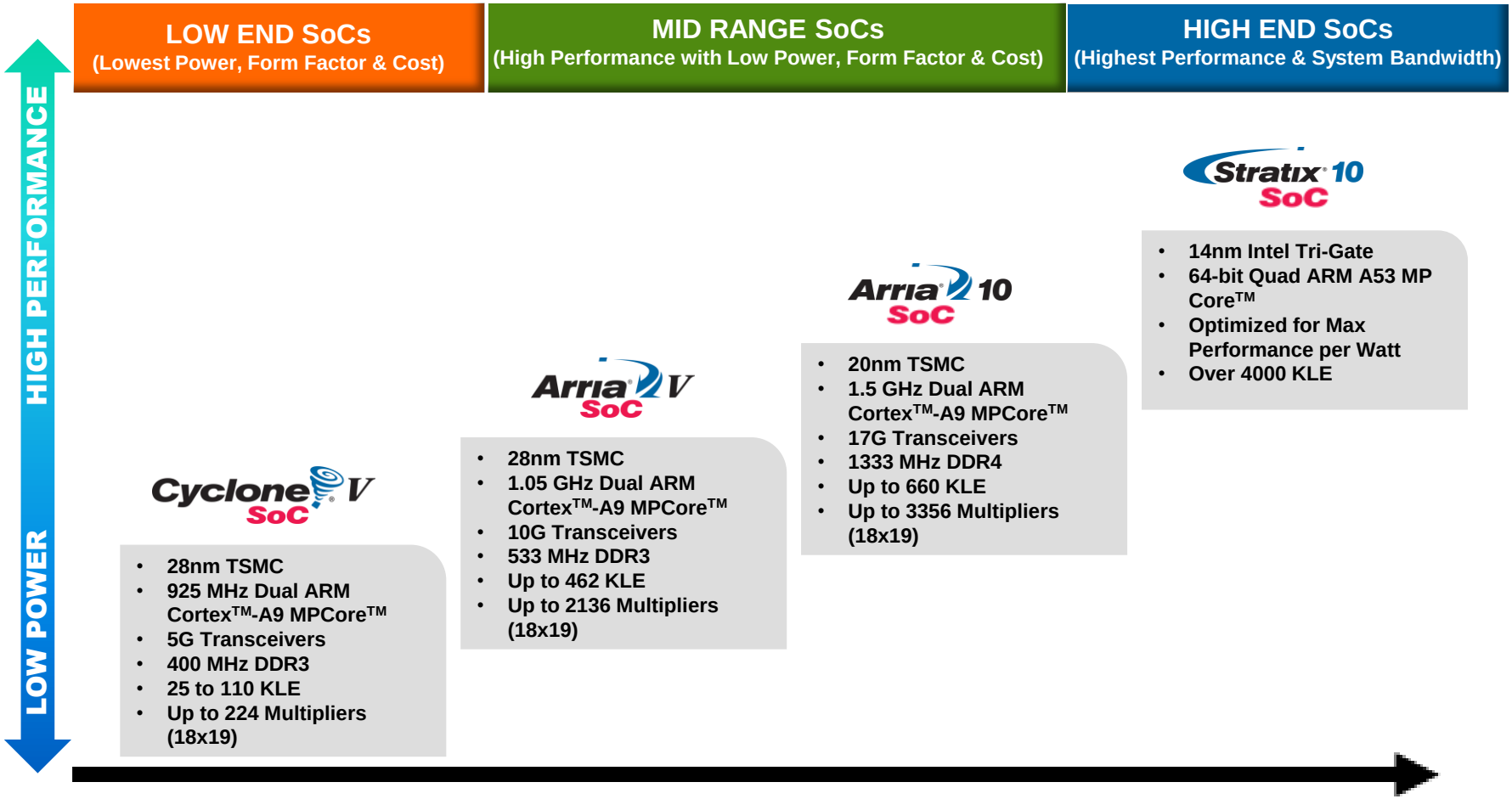






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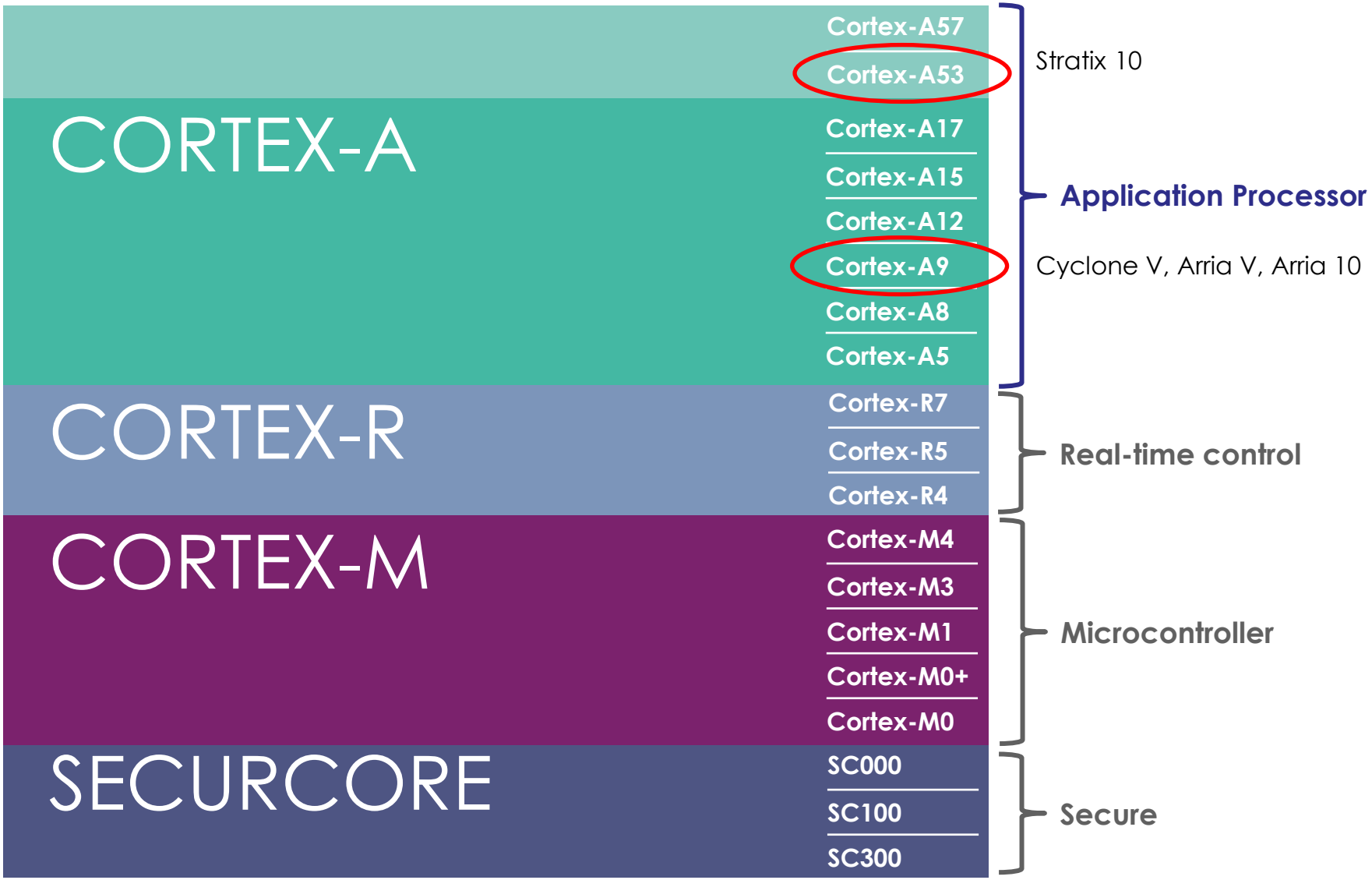
Altera SoC Product Portfolio



SoC devices available across entire product portfolio …

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ARM Public Processor Offering

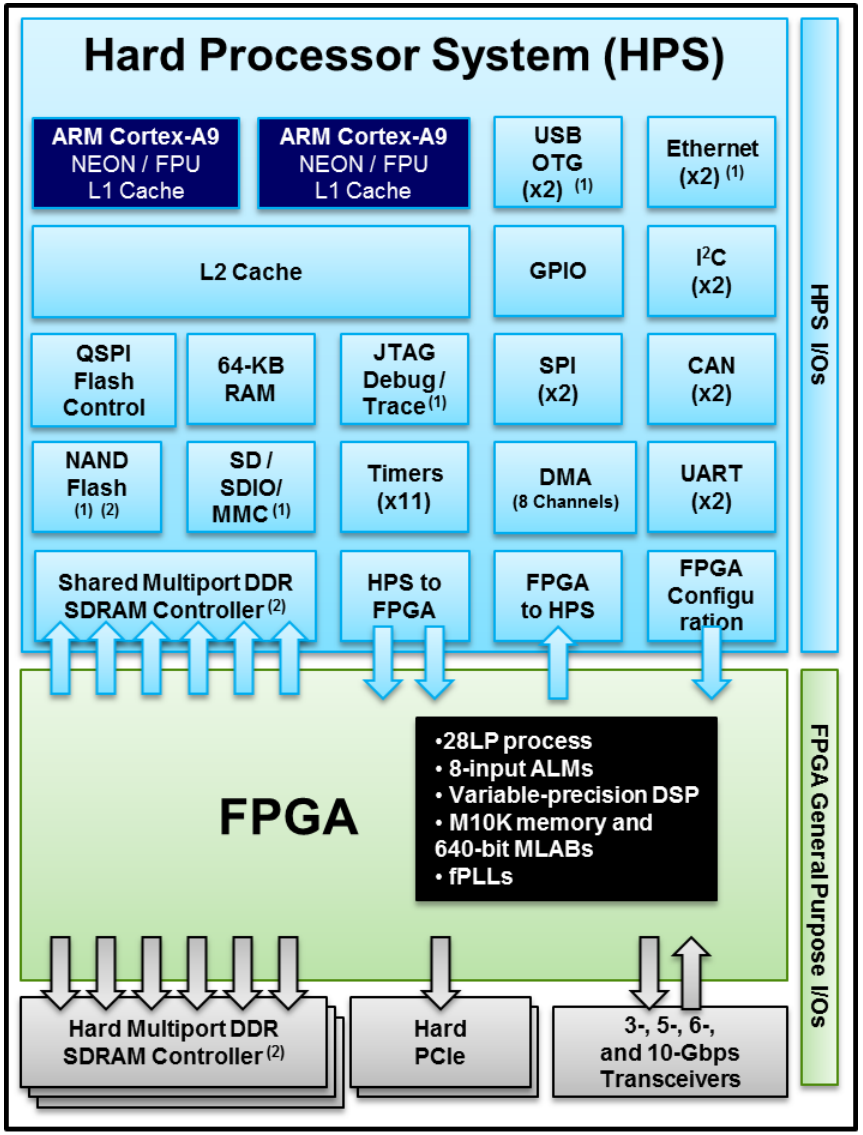


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28nm SoC System Architecture

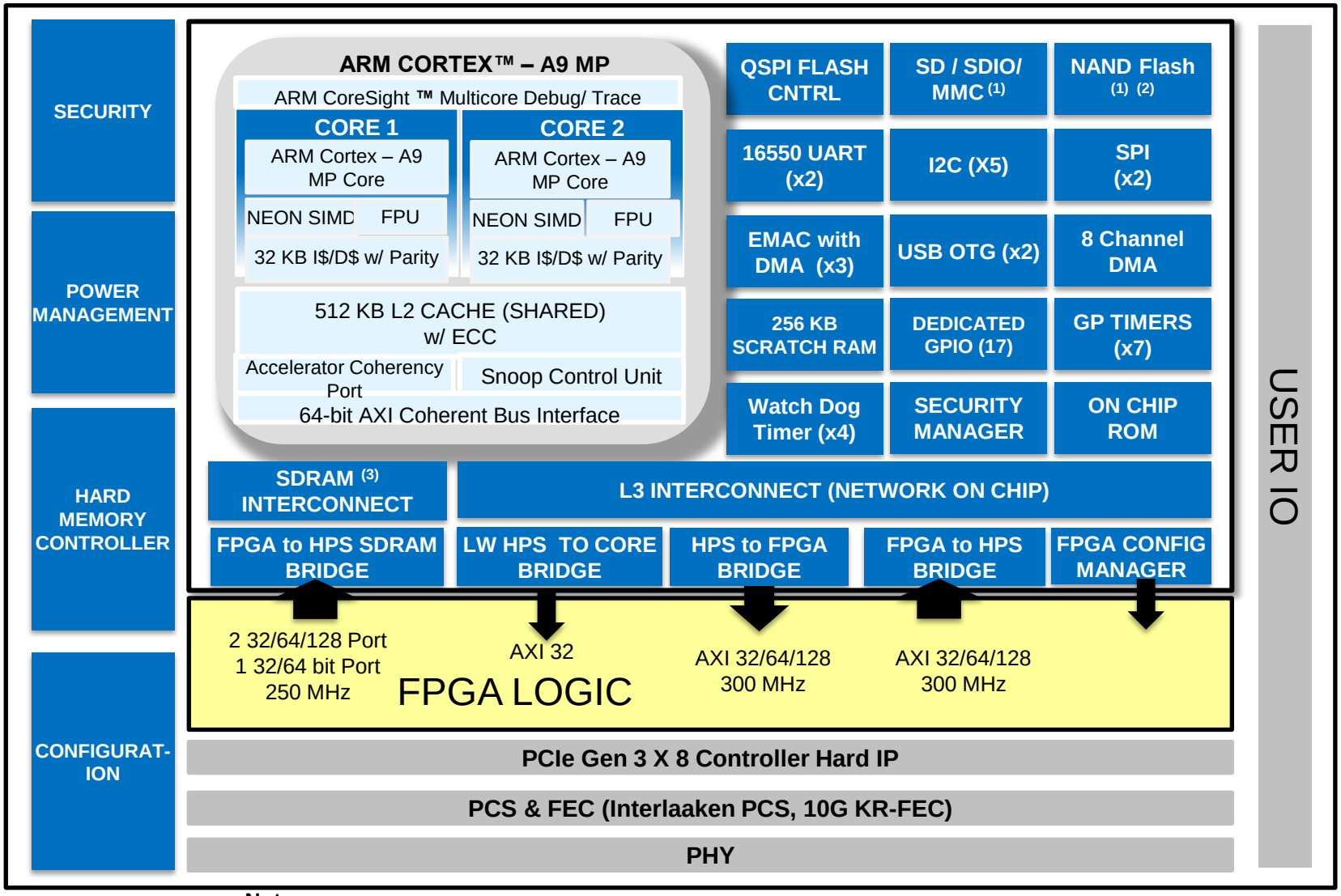
- Processor
  - Dual-core ARM® Cortex™-A9 MPCore™ processor
  - Up to 5,250 MIPS (1050 MHz per core maximum)
  - NEON coprocessor with double-precision FPU
  - 32-KB/32-KB L1 caches per core
  - 512-KB shared L2 cache
- Multiport SDRAM controller
  - DDR3, DDR3L, DDR2, LPDDR2
  - Integrated ECC support
- High-bandwidth on-chip interfaces
  - > 125-Gbps HPS-to-FPGA interface
  - > 125-Gbps FPGA-to-SDRAM interface
- Cost- and power-optimized FPGA fabric
  - Lowest power transceivers
  - Up to 1,600 GMACS, 300 GFLOPS
  - Up to 25Mb on-chip RAM
  - More hard intellectual property (IP): PCIe® and memory controllers



Notes:
(1) Integrated direct memory access (DMA)
(2) Integrated ECC

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Arria 10 HPS Block Diagram



Notes:

(1) Integrated direct memory access (DMA)

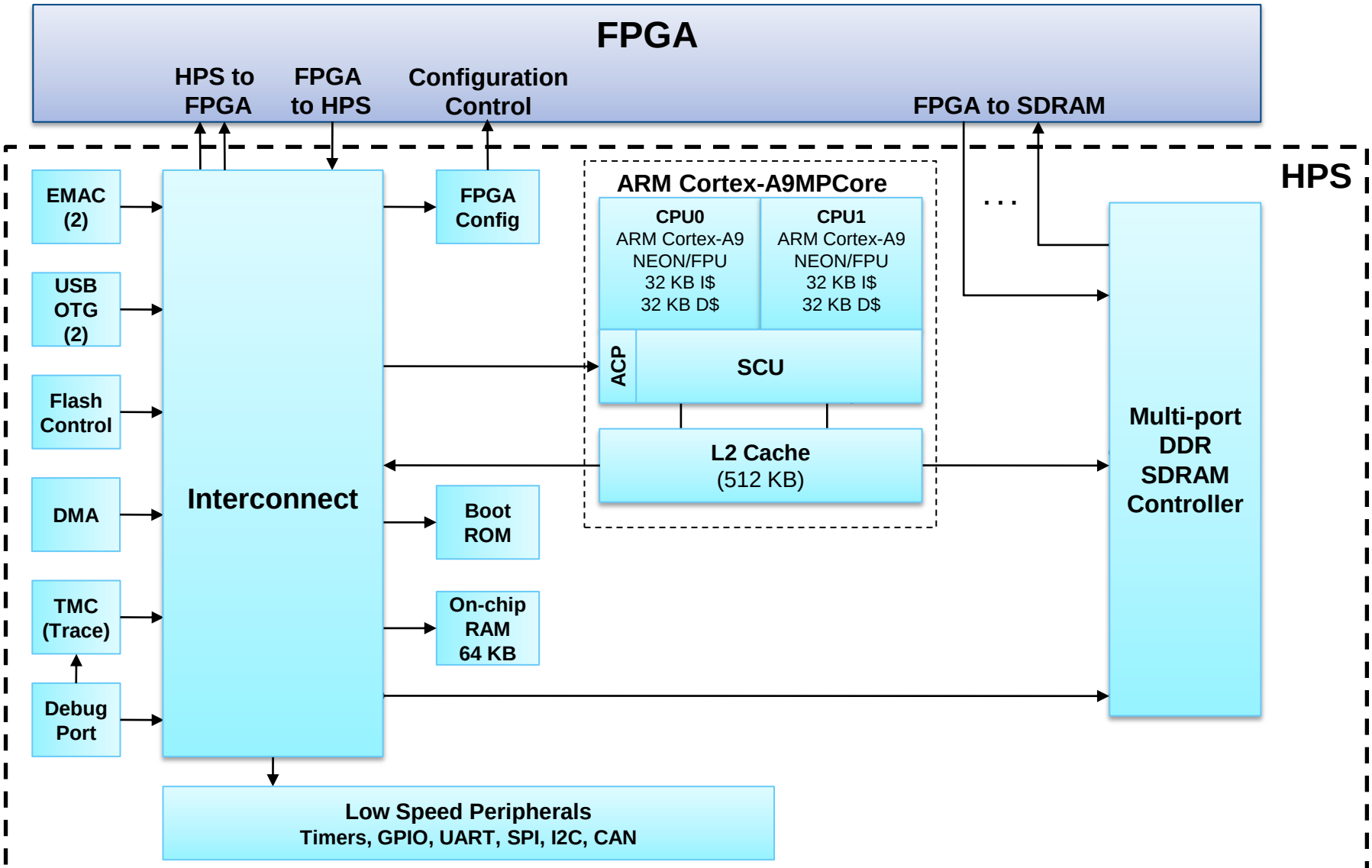
(2) Integrated ECC

(3) DDR3/4 SDRAM Support for HPS Memory

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High-Level Block Diagram



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A Comparison: Cyclone V SoC, Arria V SoC, Arria 10 SoC

Metric	Cyclone V SoC	Arria V SoC	Arria 10 SoC
Technology	28nm	28nm	20nm
Processor Performance	925 MHz	1.05 GHz	1.5 GHz
Total Power Dissipation	100%	100%	60% (40% Lower)
Max PCI Express Hard IP	Gen 2 x4	Gen 2 x8	Gen 3 x8
Memory Devices Supported	DDR2, DDR3, DDR3L, LPDDR2	DDR2, DDR3, DDR3L, LPDDR2	DDR4/3, LPDDR2/3, QDRIV, RLDRAM III, Hybrid Memory Cube*
Max. HPS DDR Data-Width	40-bit (32-bit + ECC)	40-bit (32-bit + ECC)	72-bit (64-bit + ECC)
EMAC Cores	EMAC x 2	EMAC x 2	EMAC x 3
NAND Device Supported	8-bit	8-bit	8-bit and 16-bit
SD/MMC devices supported	SD/SDIO/MMC	SD/SDIO/MMC	SD/SDIO/MMC 4.5 with eMMC
FPGA Logic Density Range (LEs)	25 - 110K	370 - 450K	160 - 660K
FPGA Core Performance	260 MHz	307 MHz	500 MHz

*** Listed memory protocols are supported by FPGA EMIF interface, the HPS EMIF interface supports a subset of these.**

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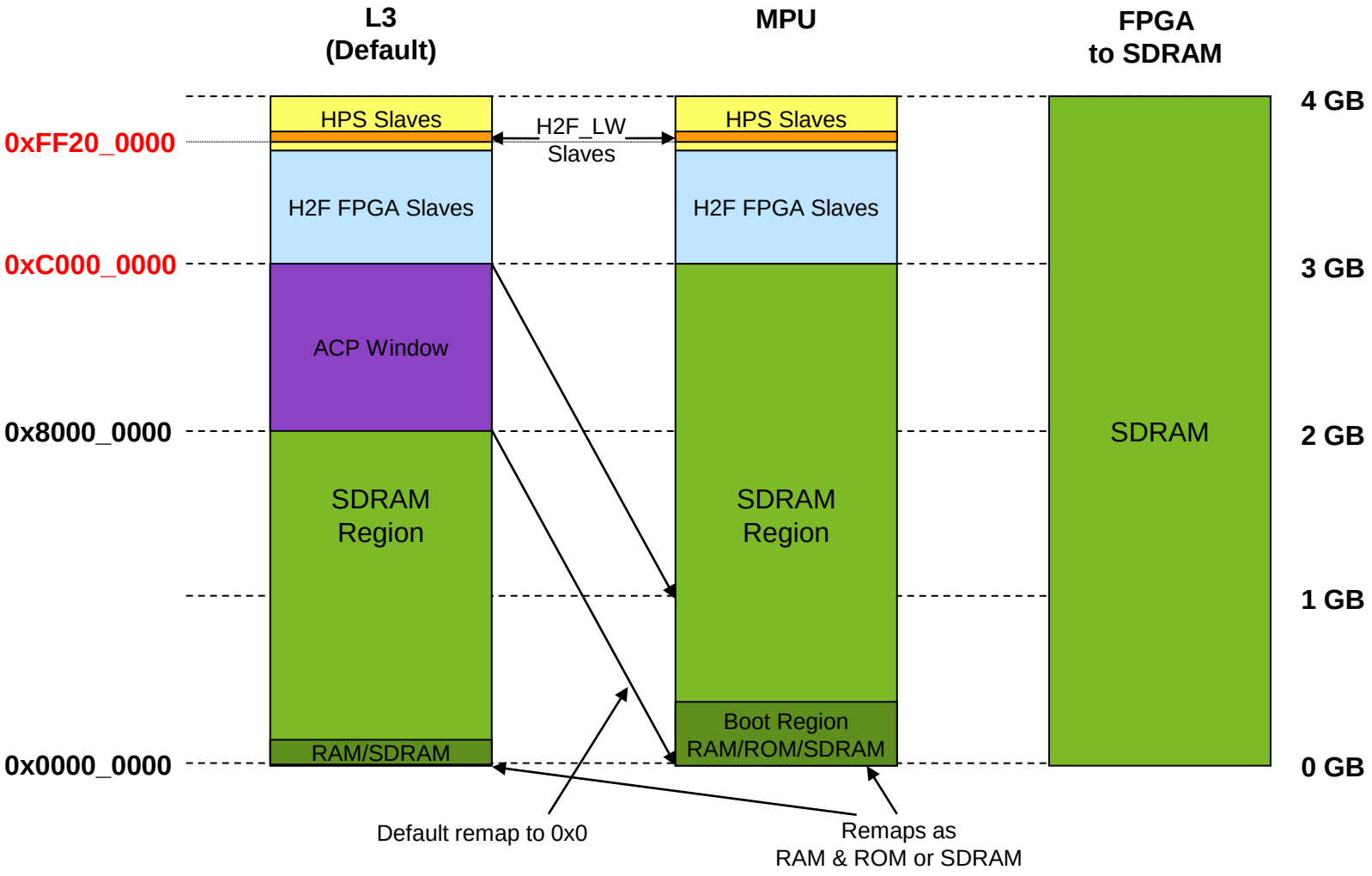
SoC Physical Address Map

Essential HW information for SW Developers



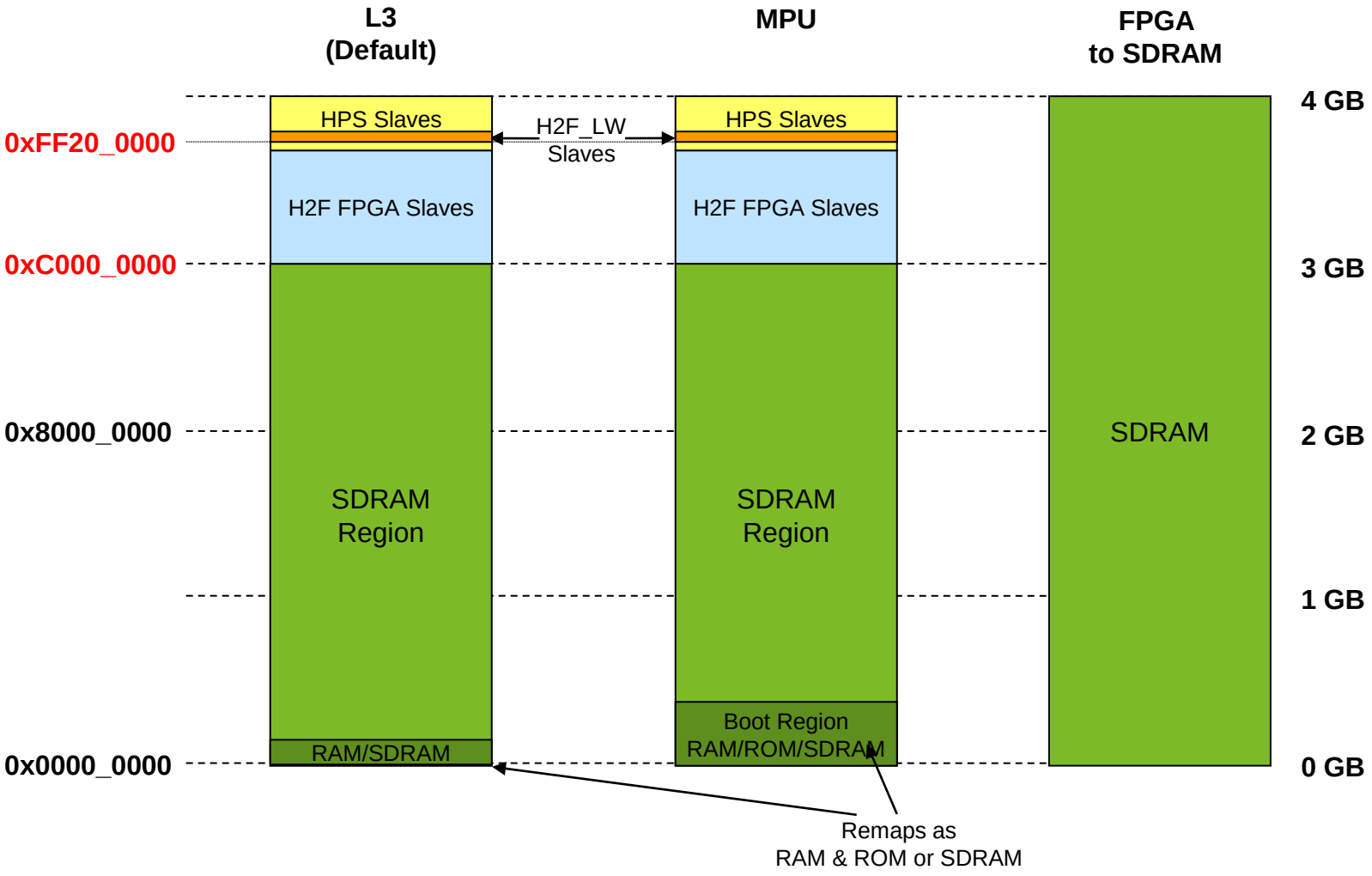
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Cyclone V & Arria V SoC HPS Physical Memory Map



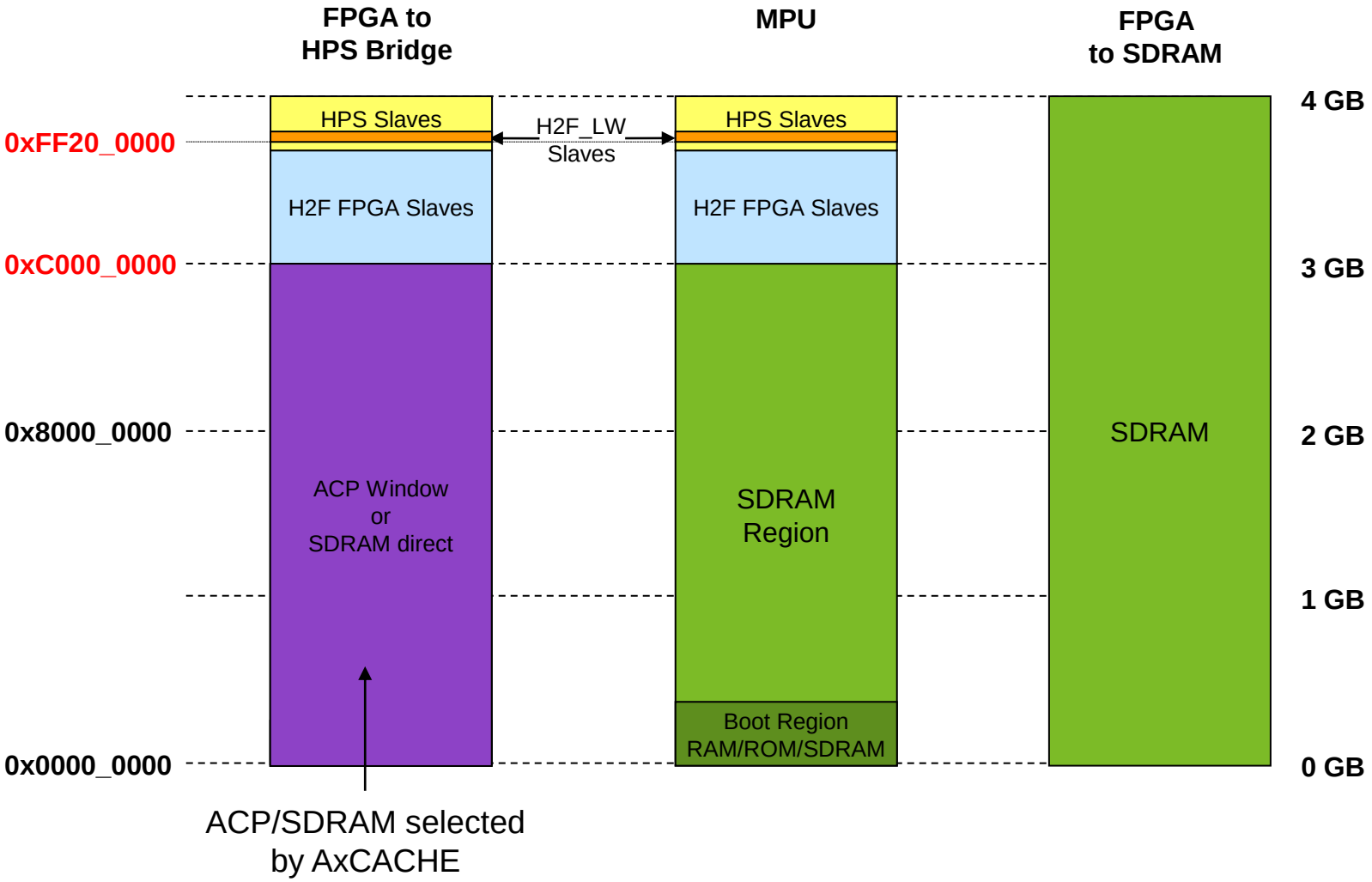
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Arria 10 SoC HPS Physical Memory Map



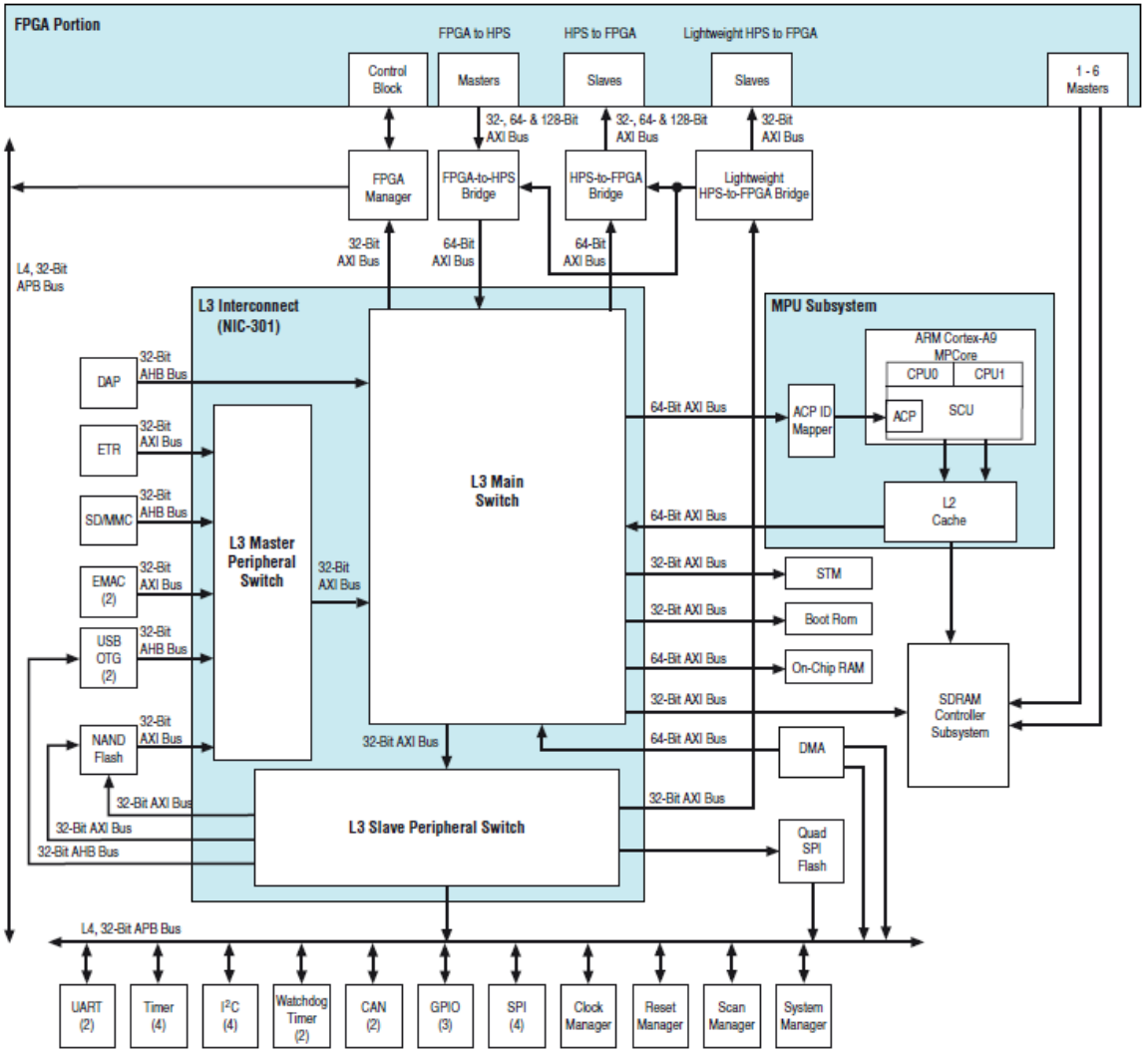
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Arria 10 SoC HPS Physical Memory Map



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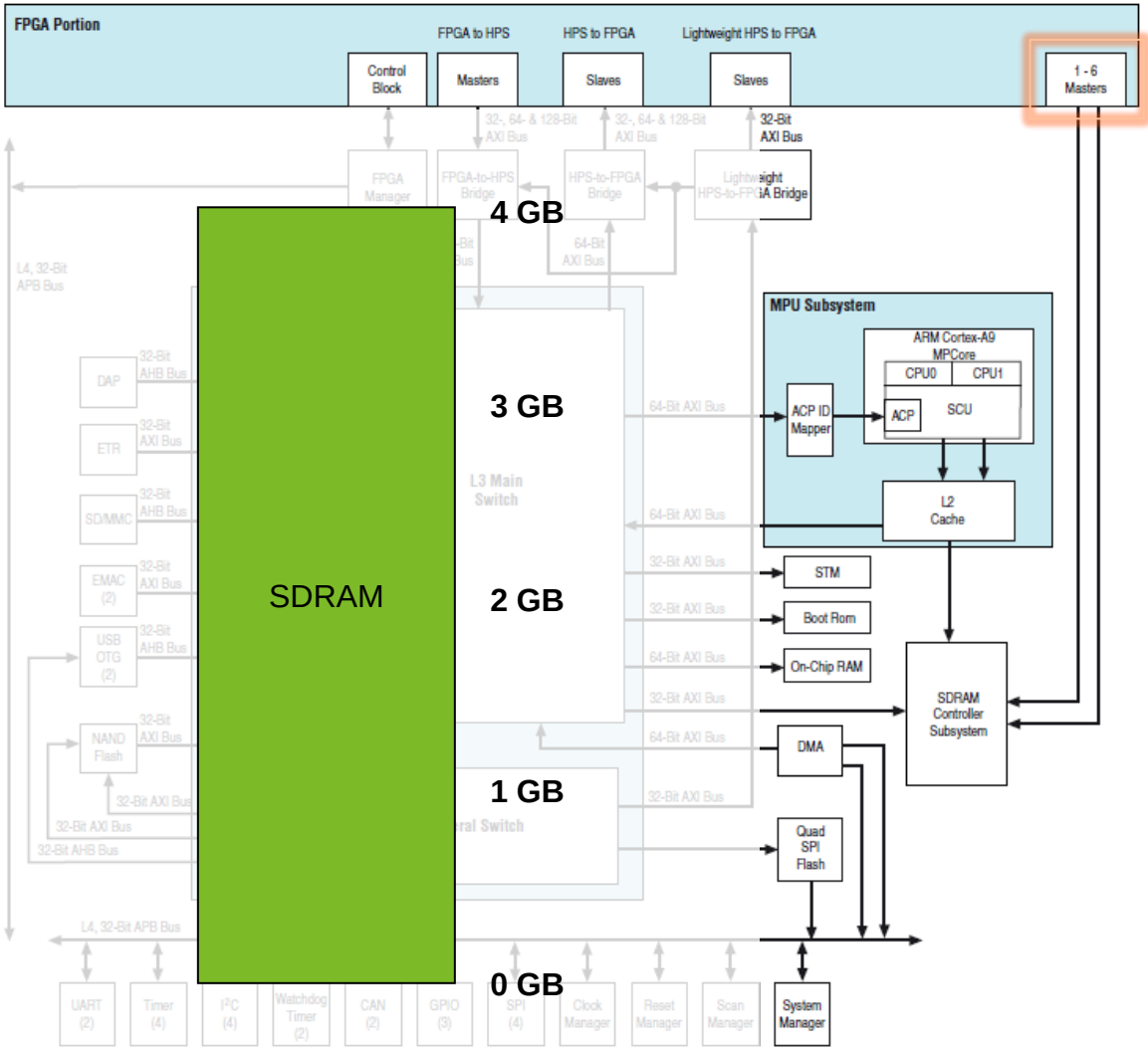
Physical Address Mapping



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Physical Address Mapping – FPGA to SDRAM

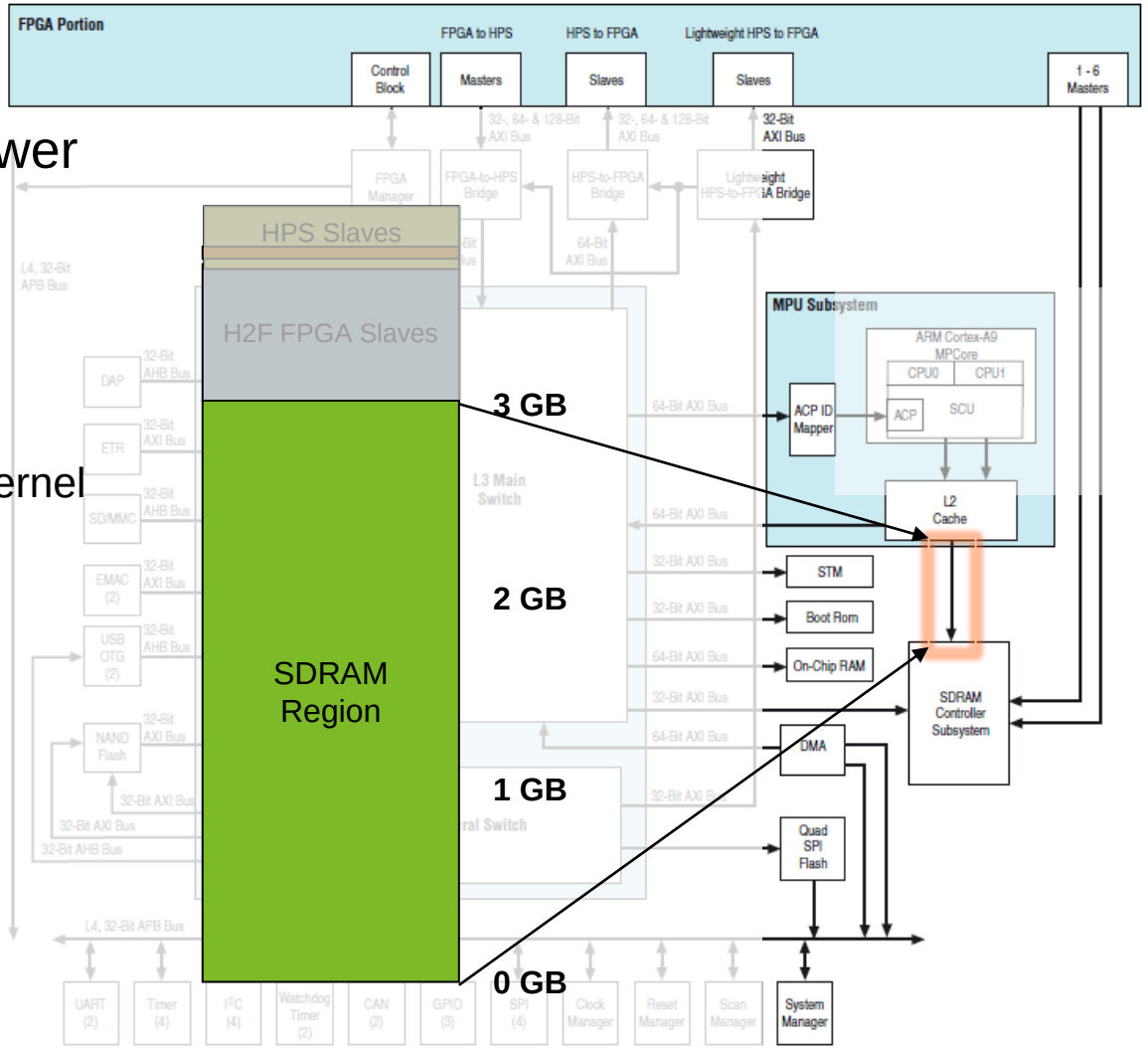
- FPGA Masters have access to full 4GB of SDRAM address space
  - Subject to MPFE MPU restrictions
  - No coherency
  - No virtual addressing



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Physical Address Mapping – MPU

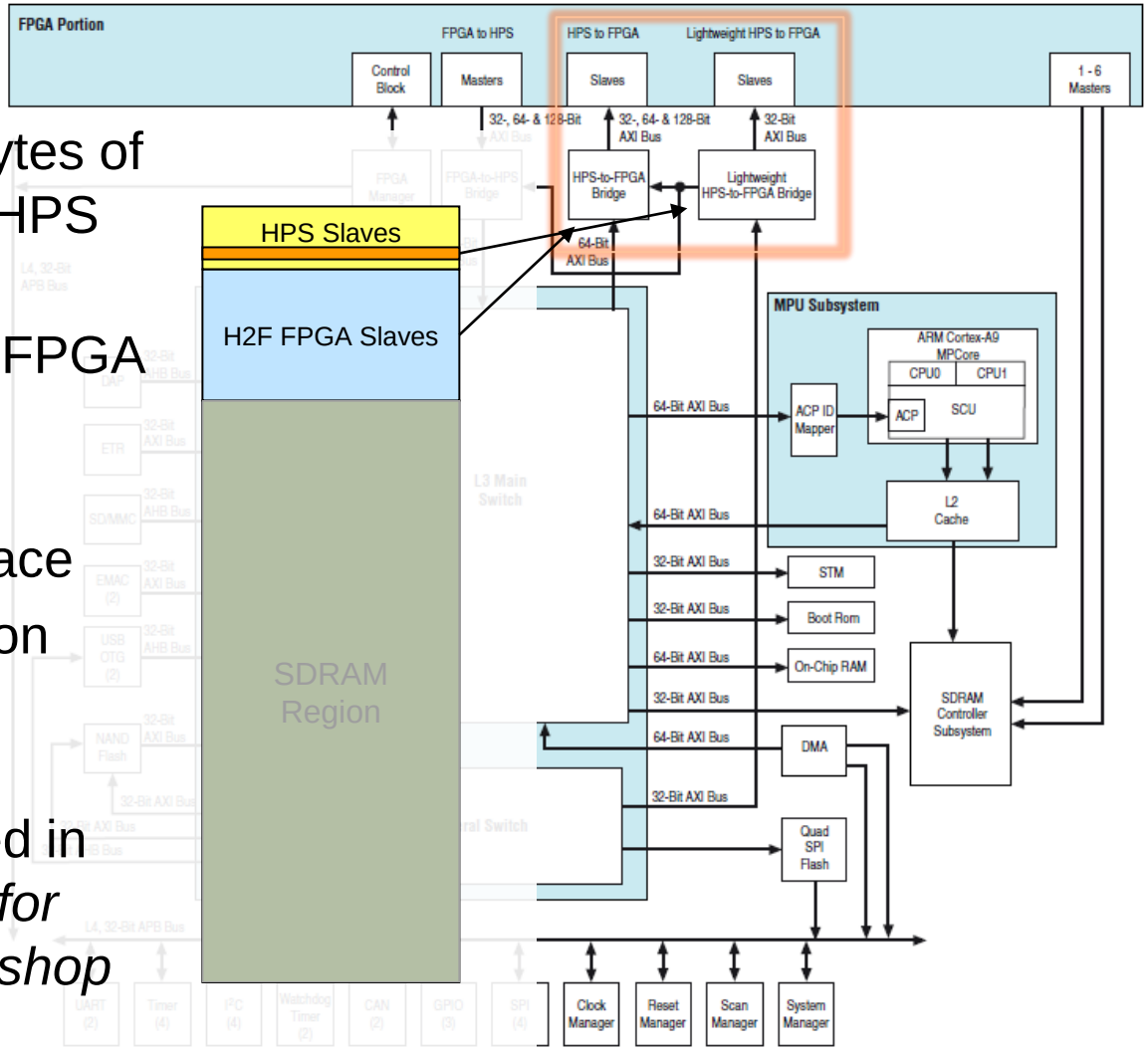
- MPU has access to the lower 3 GBytes of SDRAM
- Kernel manages and can allocate memory in this 3GByte space
  - Allocate for both user and kernel space
  - Allocatable on 4K Byte Boundaries (page size)



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Physical Address Mapping – MPU to FPGA

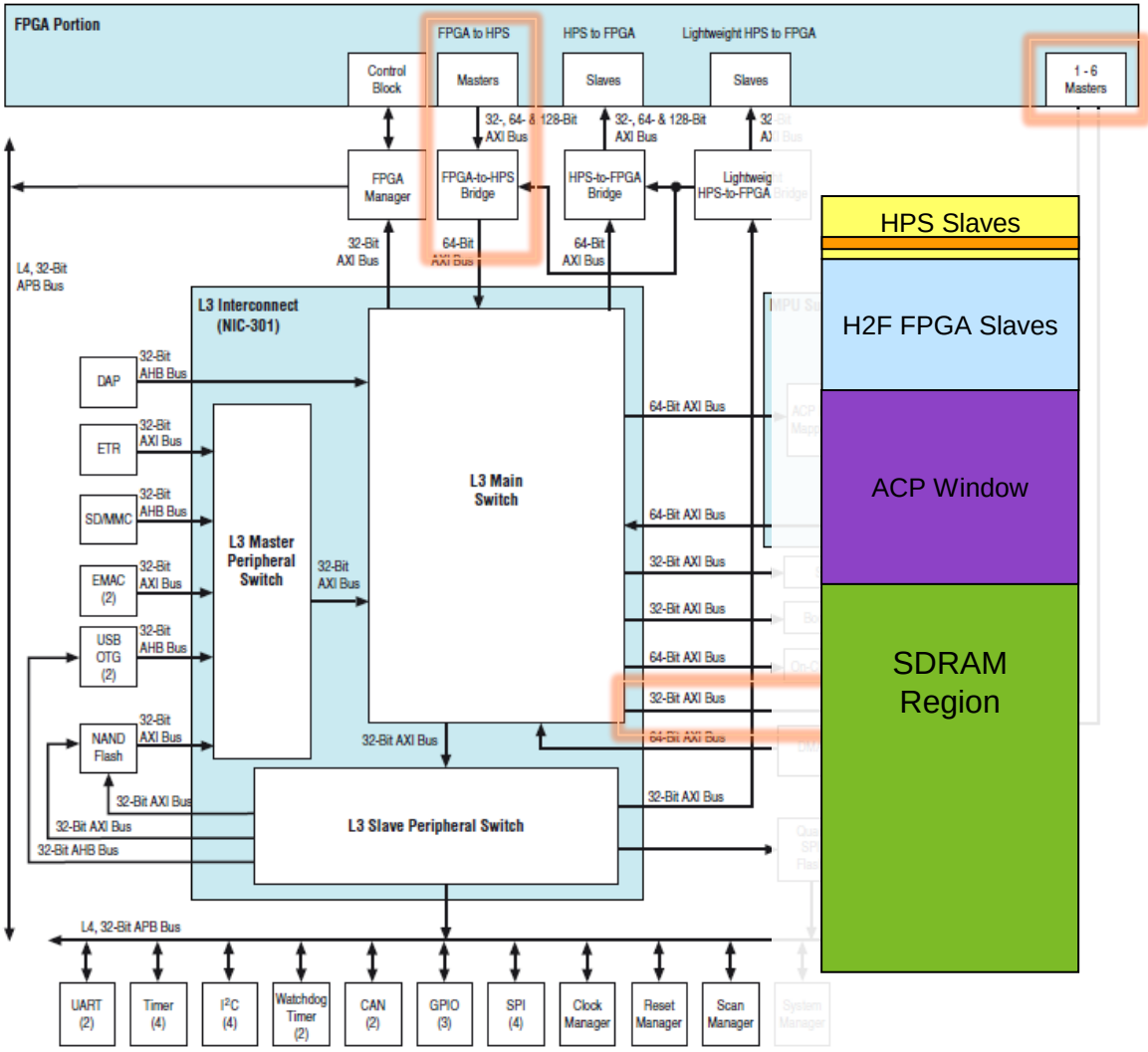
- MPU can access 960 MBytes of FPGA address space via HPS to FPGA Bridge
- MPU can access 2 MB of FPGA address space via HPS to FPGA Lightweight bridge
- Not allocatable in user space
- Space FPGA peripherals on Linux page size (4KB) boundaries
- Access methods discussed in *Developing Linux Drivers for Custom Peripherals Workshop*



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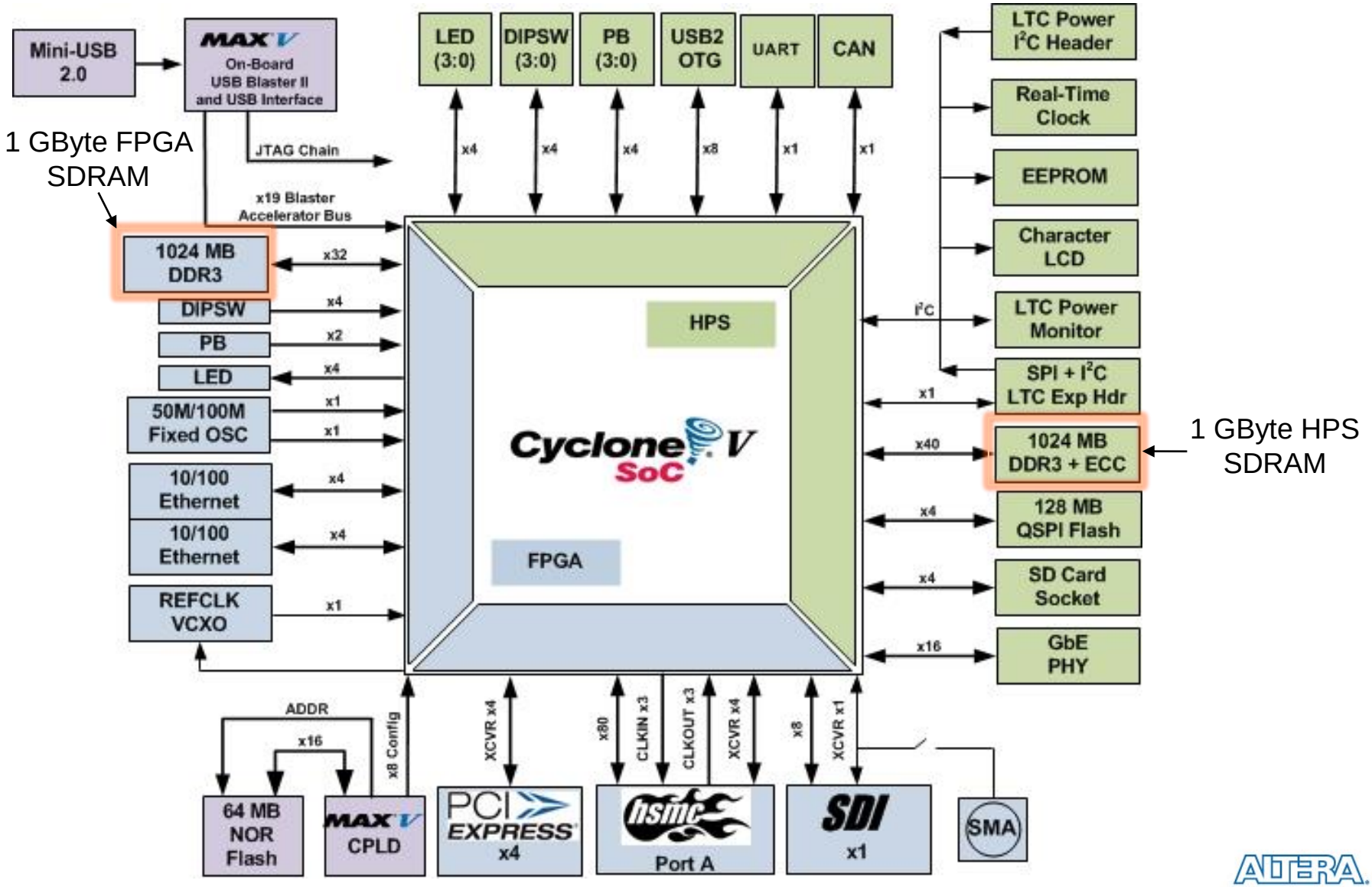
Physical Address Mapping – FPGA to HPS

- FPGA to HPS (F2H) masters see 4 GByte address space
- F2H bandwidth to SDRAM limited vs. FPGA to SDRAM bridge



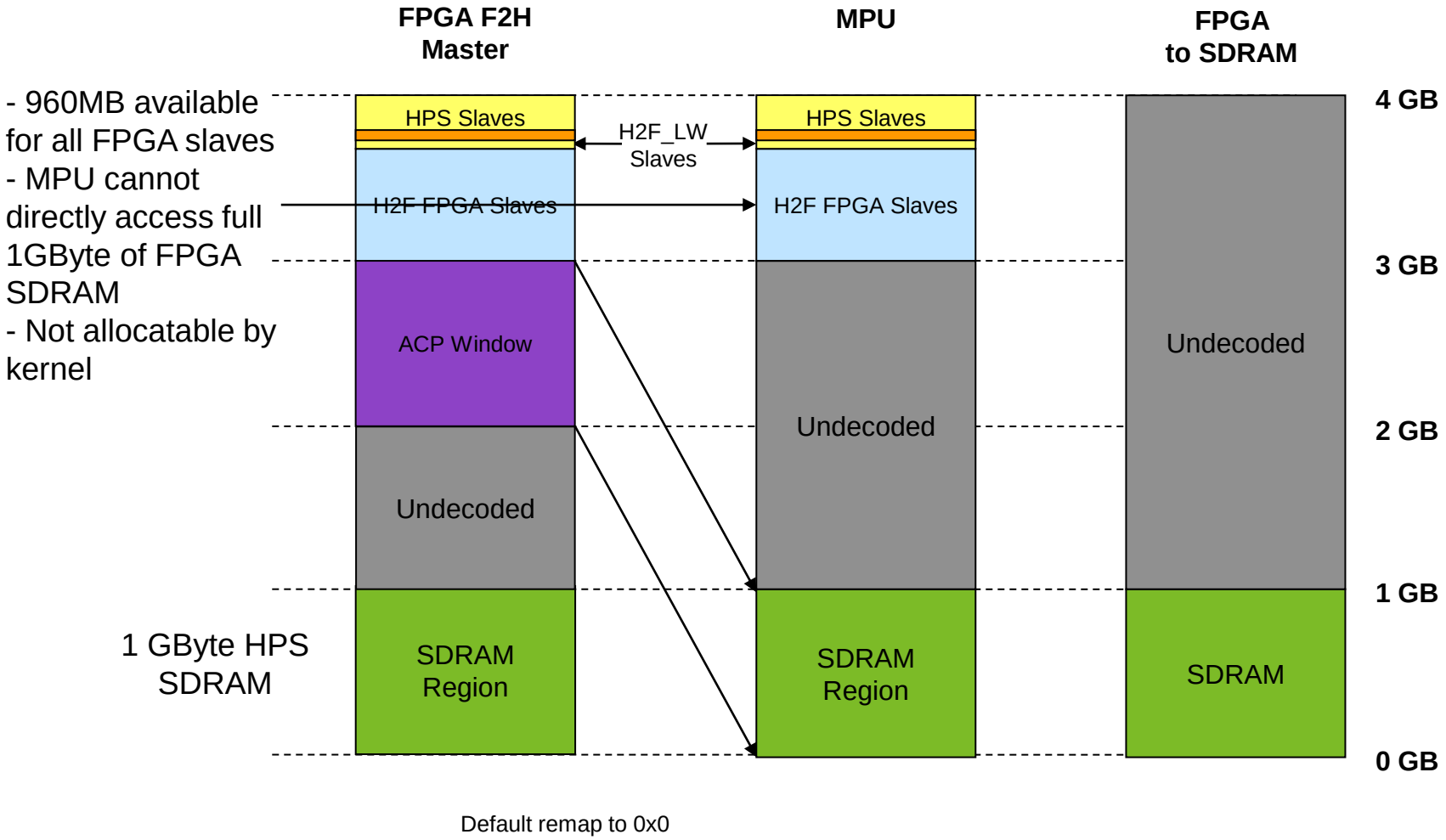
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Cyclone V SoC Memory Map Example - SDRAM



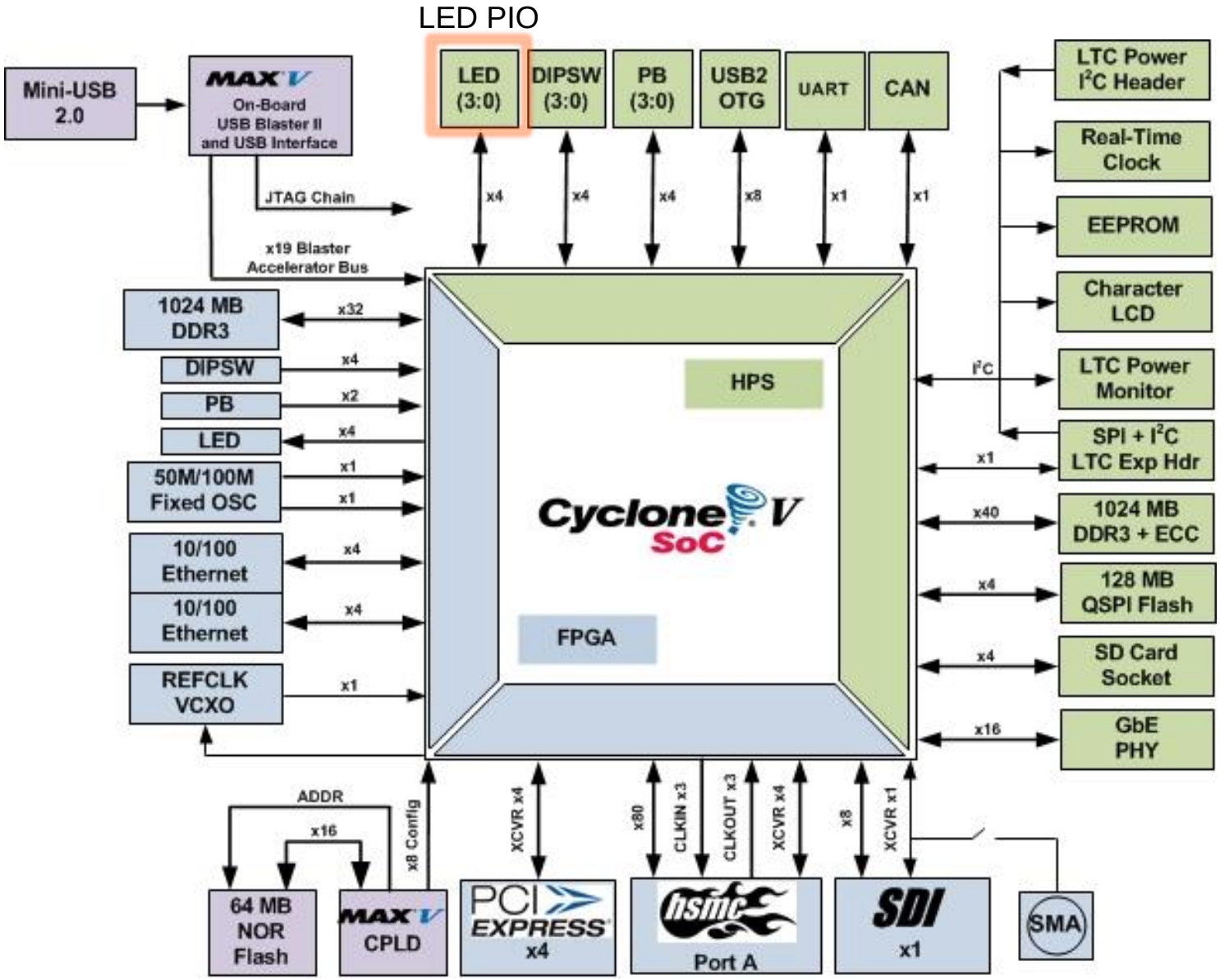
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Cyclone V SoC GSRD Memory Map Example - SDRAM



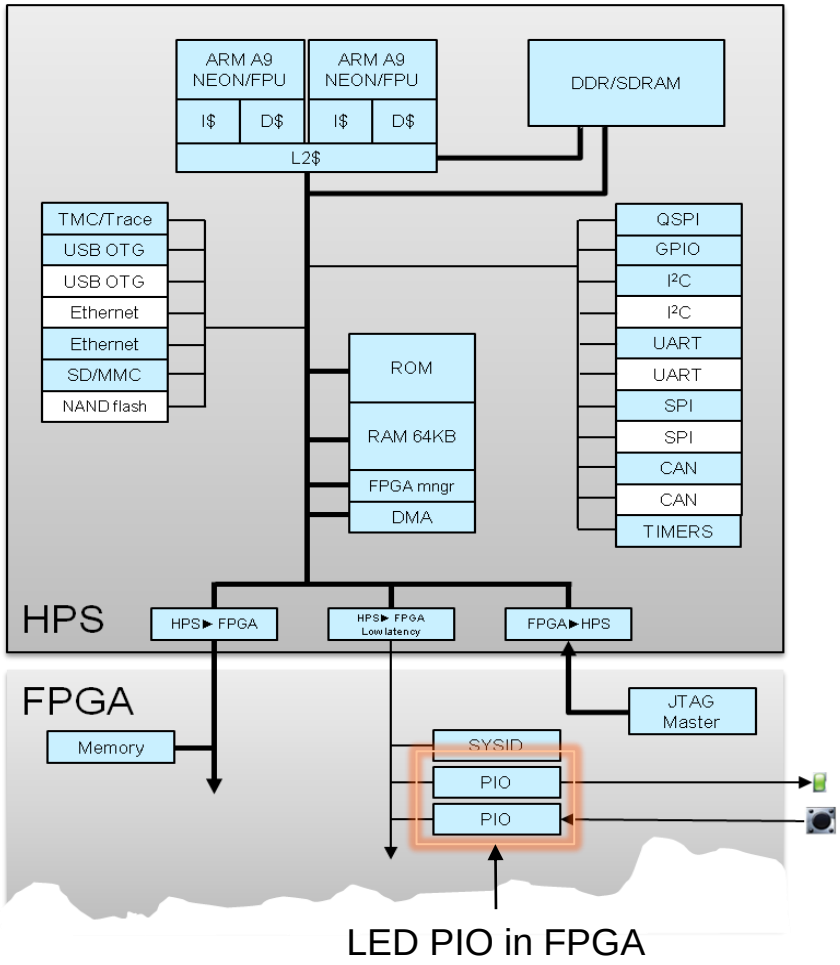
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Cyclone V SoC Memory Map Example - PIO



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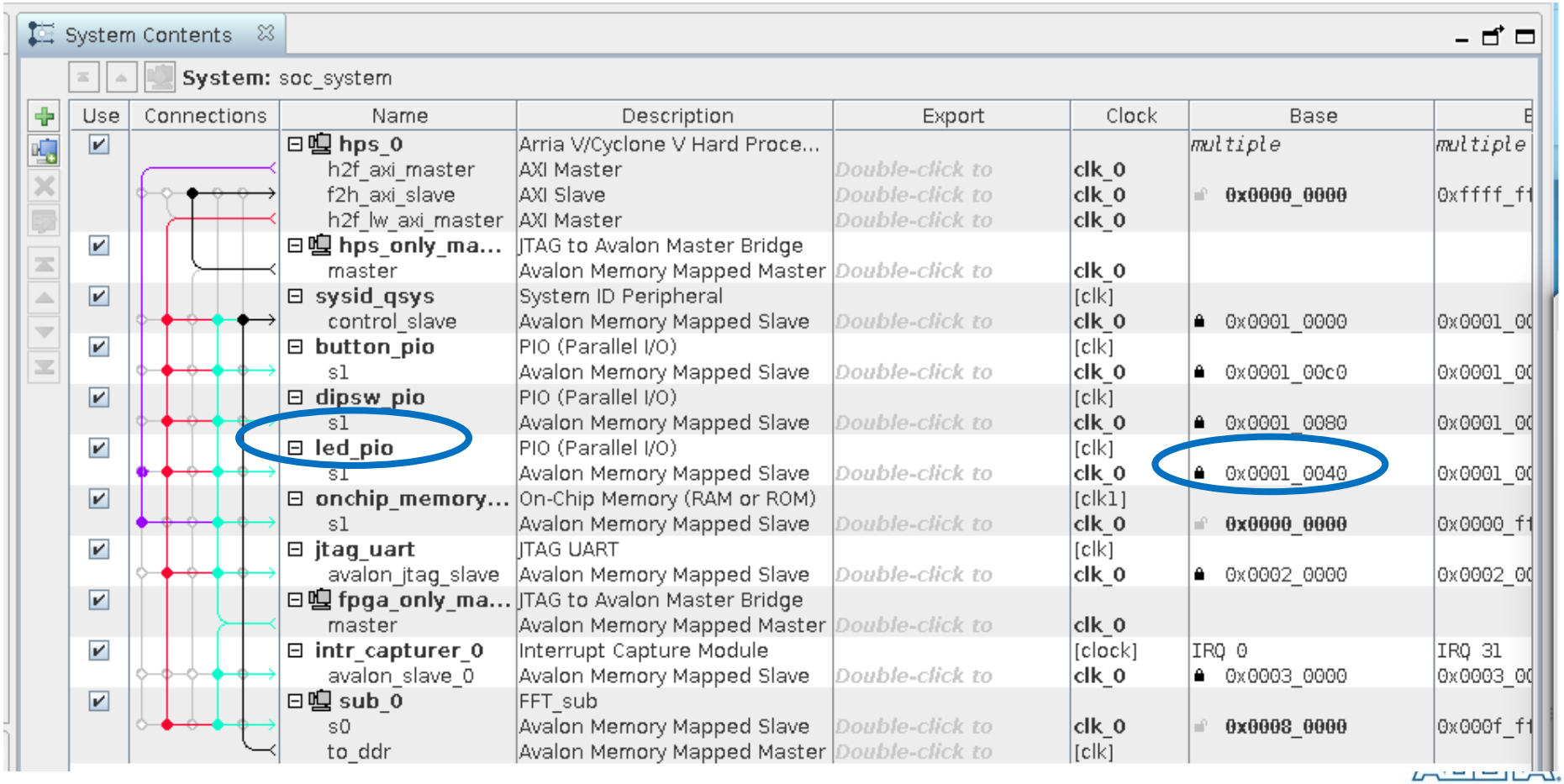
Cyclone V SoC Memory Map Example - PIO



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Cyclone V SoC GSRD Memory Map Example

- Look at led_pio at Address 0x0001_0040

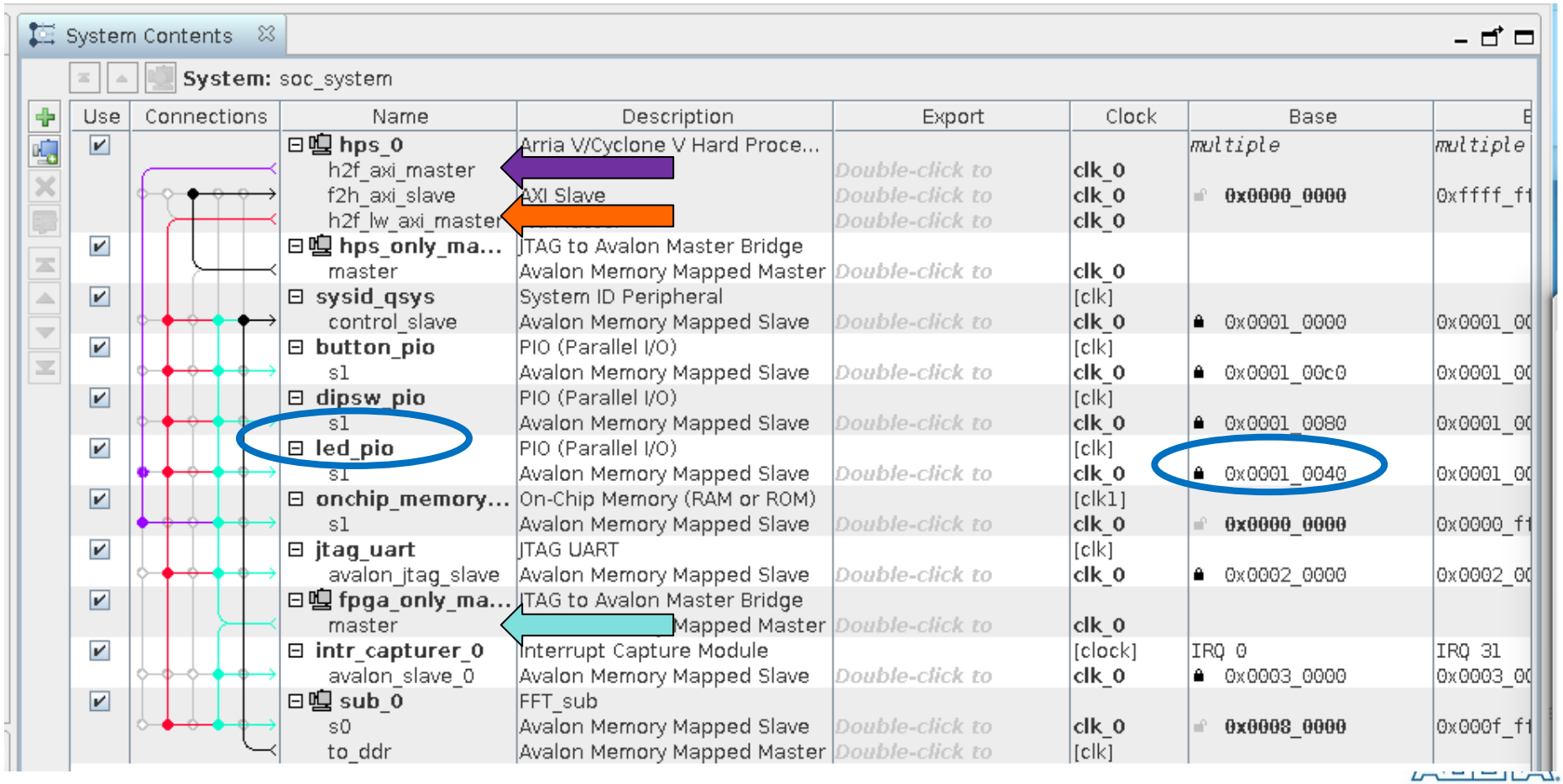


Use	Connections	Name	Description	Export	Clock	Base	E
☑		hps_0	Arria V/Cyclone V Hard Proce...			multiple	multiple
		h2f_axi_master	AXI Master	Double-click to	clk_0		
		f2h_axi_slave	AXI Slave	Double-click to	clk_0	0x0000_0000	0xffff_ff
		h2f_lw_axi_master	AXI Master	Double-click to	clk_0		
☑		hps_only_ma...	JTAG to Avalon Master Bridge				
		master	Avalon Memory Mapped Master	Double-click to	clk_0		
☑		sysid_qsys	System ID Peripheral		[clk]		
		control_slave	Avalon Memory Mapped Slave	Double-click to	clk_0	0x0001_0000	0x0001_00
☑		button_pio	PIO (Parallel I/O)		[clk]		
		s1	Avalon Memory Mapped Slave	Double-click to	clk_0	0x0001_00c0	0x0001_00
☑		dipsw_pio	PIO (Parallel I/O)		[clk]		
		s1	Avalon Memory Mapped Slave	Double-click to	clk_0	0x0001_0080	0x0001_00
☑		led_pio	PIO (Parallel I/O)		[clk]		
		s1	Avalon Memory Mapped Slave	Double-click to	clk_0	0x0001_0040	0x0001_00
☑		onchip_memory...	On-Chip Memory (RAM or ROM)		[clk1]		
		s1	Avalon Memory Mapped Slave	Double-click to	clk_0	0x0000_0000	0x0000_ff
☑		jtag_uart	JTAG UART		[clk]		
		avalon_jtag_slave	Avalon Memory Mapped Slave	Double-click to	clk_0	0x0002_0000	0x0002_00
☑		fpga_only_ma...	JTAG to Avalon Master Bridge				
		master	Avalon Memory Mapped Master	Double-click to	clk_0		
☑		intr_capturer_0	Interrupt Capture Module		[clock]	IRQ 0	IRQ 31
		avalon_slave_0	Avalon Memory Mapped Slave	Double-click to	clk_0	0x0003_0000	0x0003_00
☑		sub_0	FFT_sub				
		s0	Avalon Memory Mapped Slave	Double-click to	clk_0	0x0008_0000	0x000f_ff
		to_ddr	Avalon Memory Mapped Master	Double-click to	[clk]		

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Cyclone V SoC GSRD Memory Map Example

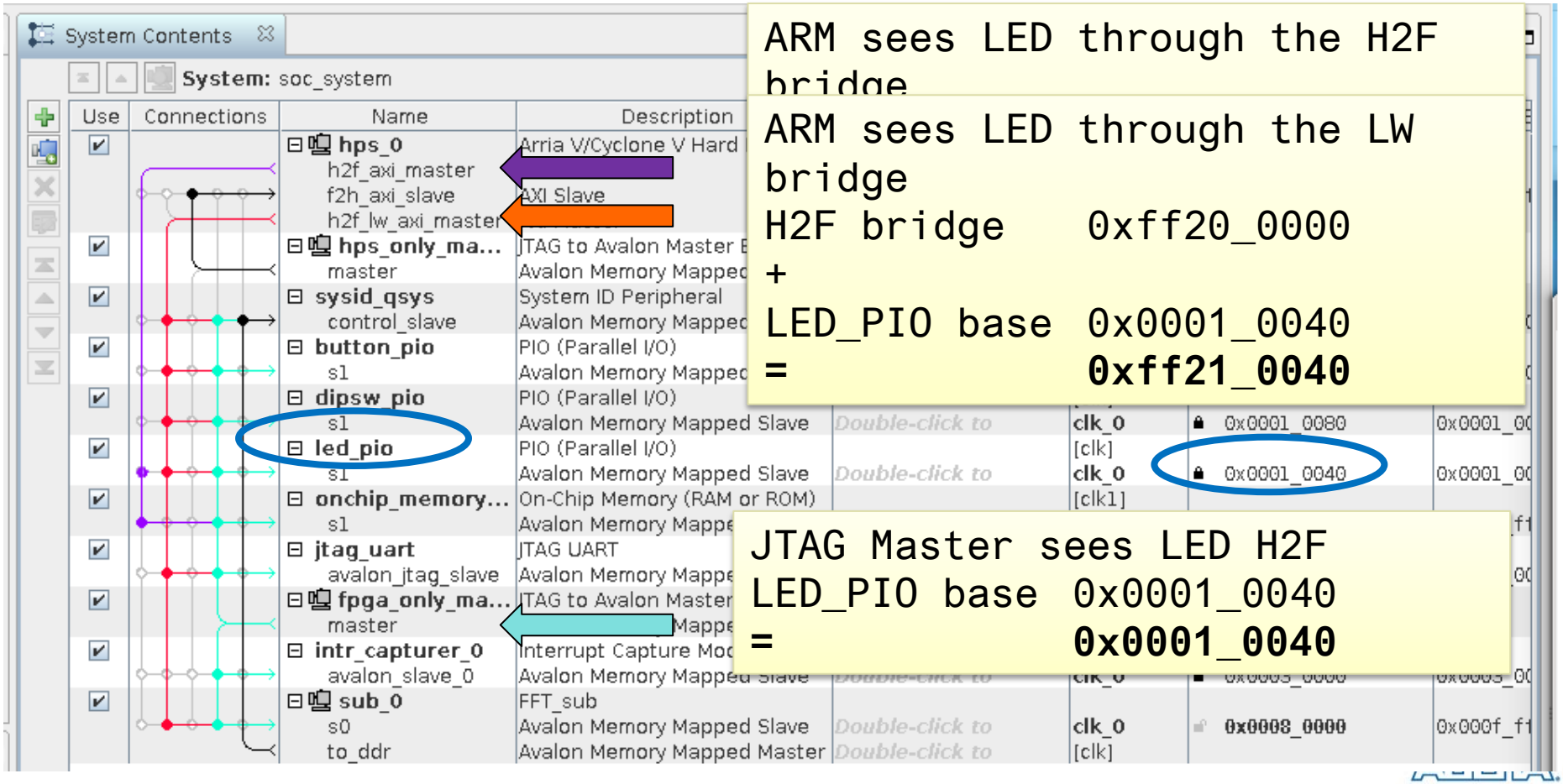
- It is connected to 3 Masters.



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Cyclone V SoC GSRD Memory Map Example

- Each master sees the slave at a different address
- These address are offsets from the HPS bridge address



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SoCFPGA Development Flow & Tools



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Why Does This Matter for Linux Development?

- Altera SoCs offer unique advantages
  - User specified peripheral set
  - Tightly-coupled MPU & FPGA fabric
  - One of a kind HW/SW debug capabilities
- Altera SoCs have unique requirements
  - Understanding of paths to exchange data between MPU and FPGA
  - Building custom BSP for user specified peripheral set
  - Correct handling and configuration of HPS/FPGA bridges
- Altera SoCs have tools to enable the power of a integrated MPU and FPGA

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Altera SoC Embedded Design Suite

FPGA Design Flow

Software Design Flow

QUARTUS® II

Qsys

Hardware Development

Software Development

- Quartus II design software
- Qsys system integration tool
- Standard RTL flow
- Altera and partner IP

Design

HW/SW Handoff

Design

- ARM Development Studio 5
- GNU toolchain
- OS/BSP: Linux, VxWorks
- Hardware Libraries
- Design Examples

- ModelSim, VCS, NCSim, etc.
- AMBA-AXI and Avalon bus functional models (BFMs)

Simulate

Simulate

- Virtual [illegible]

Software Development

- SignalTap™ II logic analyzer
- System Console

Debug

FPGA-Adaptive Debugging

Debug

- GNU, Lauterbach, DS5

- Quartus II Programmer
- In-system Update

Release

Release

- Flash Programmer



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So… what exactly is Qsys?

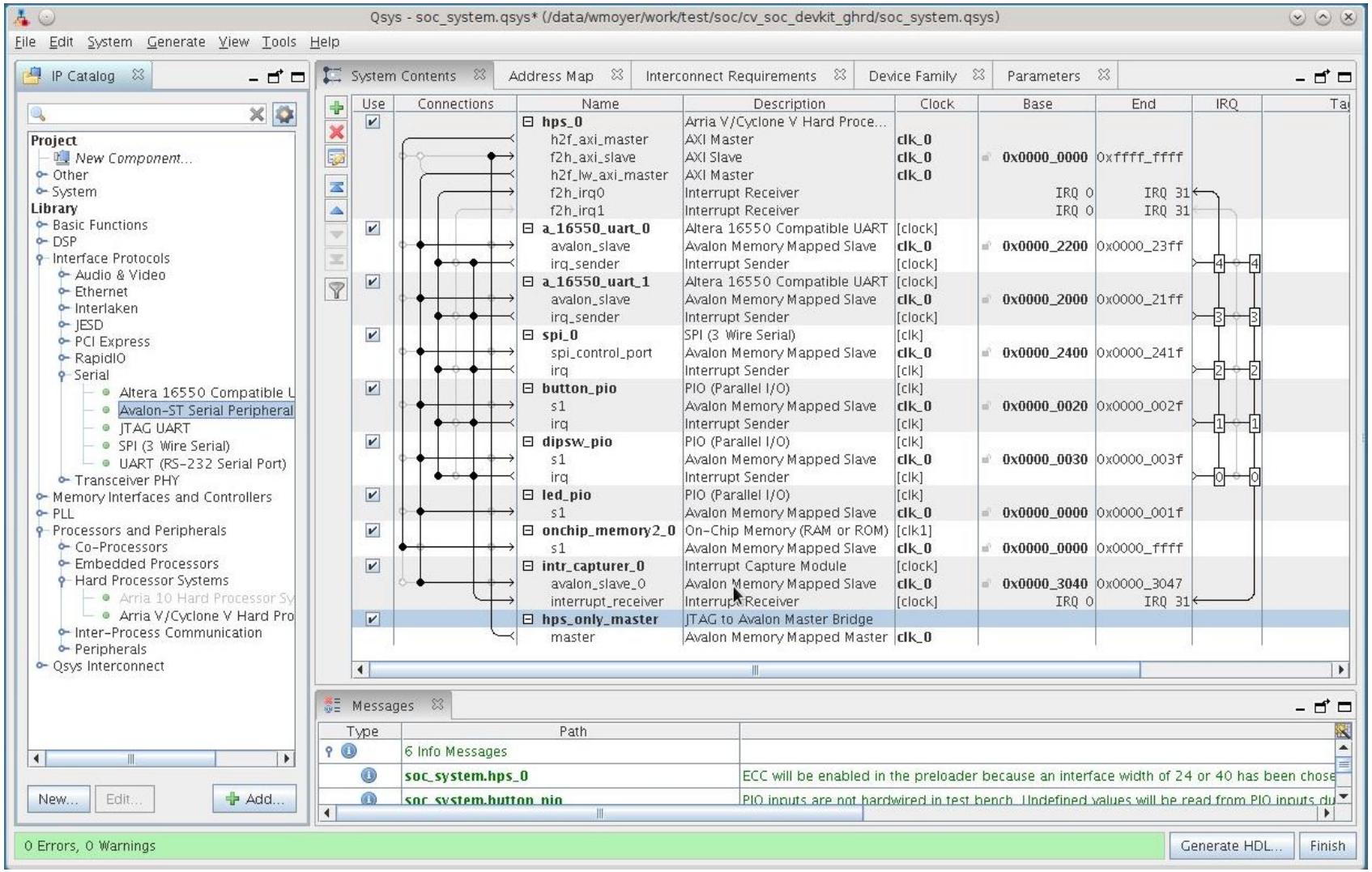
GUI based system integration tool for HW system design using IP blocks.

- Simplifies complex system development
- Raises the level of design abstraction
- Provides a standard platform:
  - IP integration
  - Custom IP authoring
  - IP verification
- Enables design re-use
- Scales easily to meet the needs of end product
- Reduces time to market



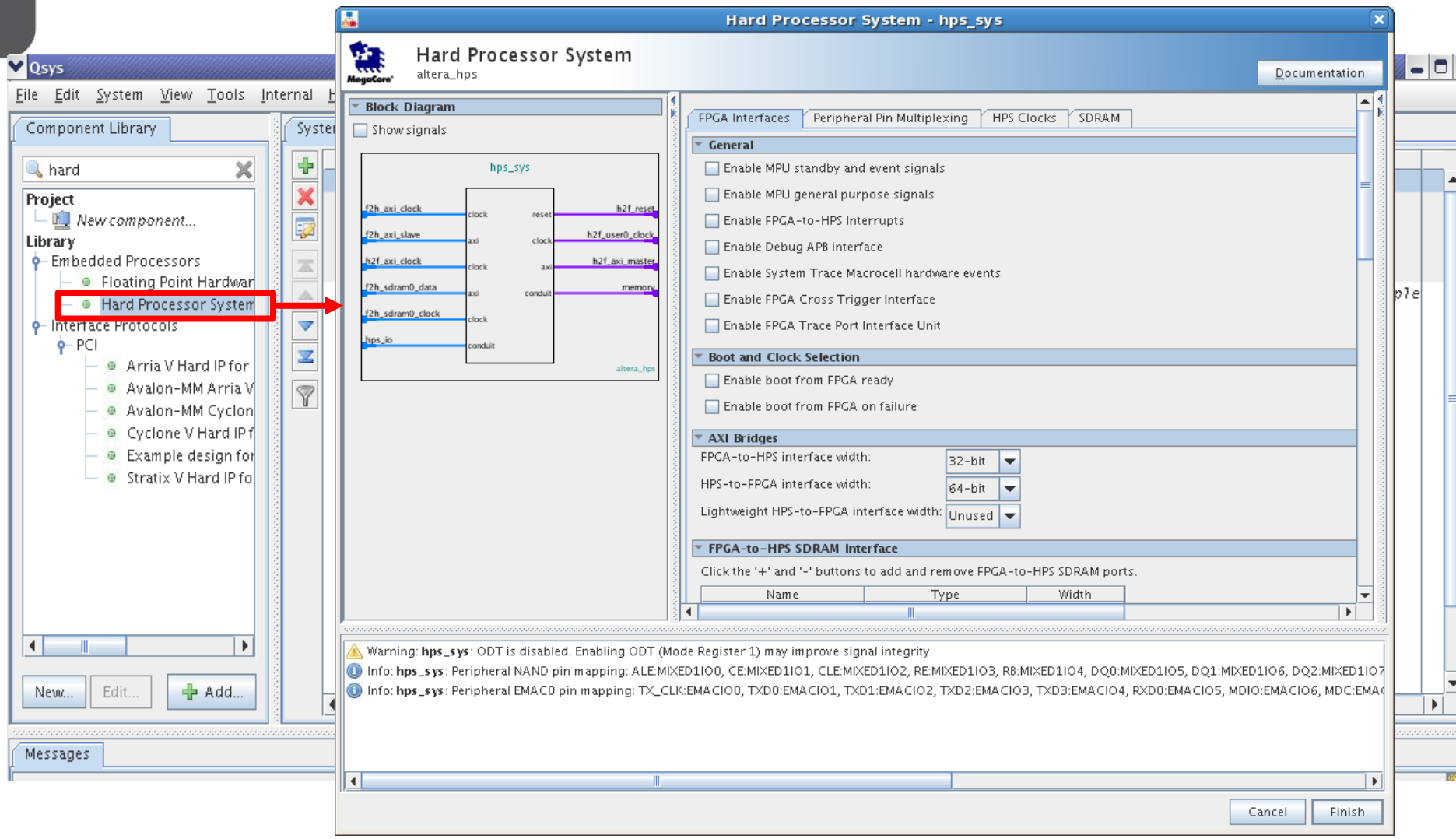
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Qsys System Integration Platform



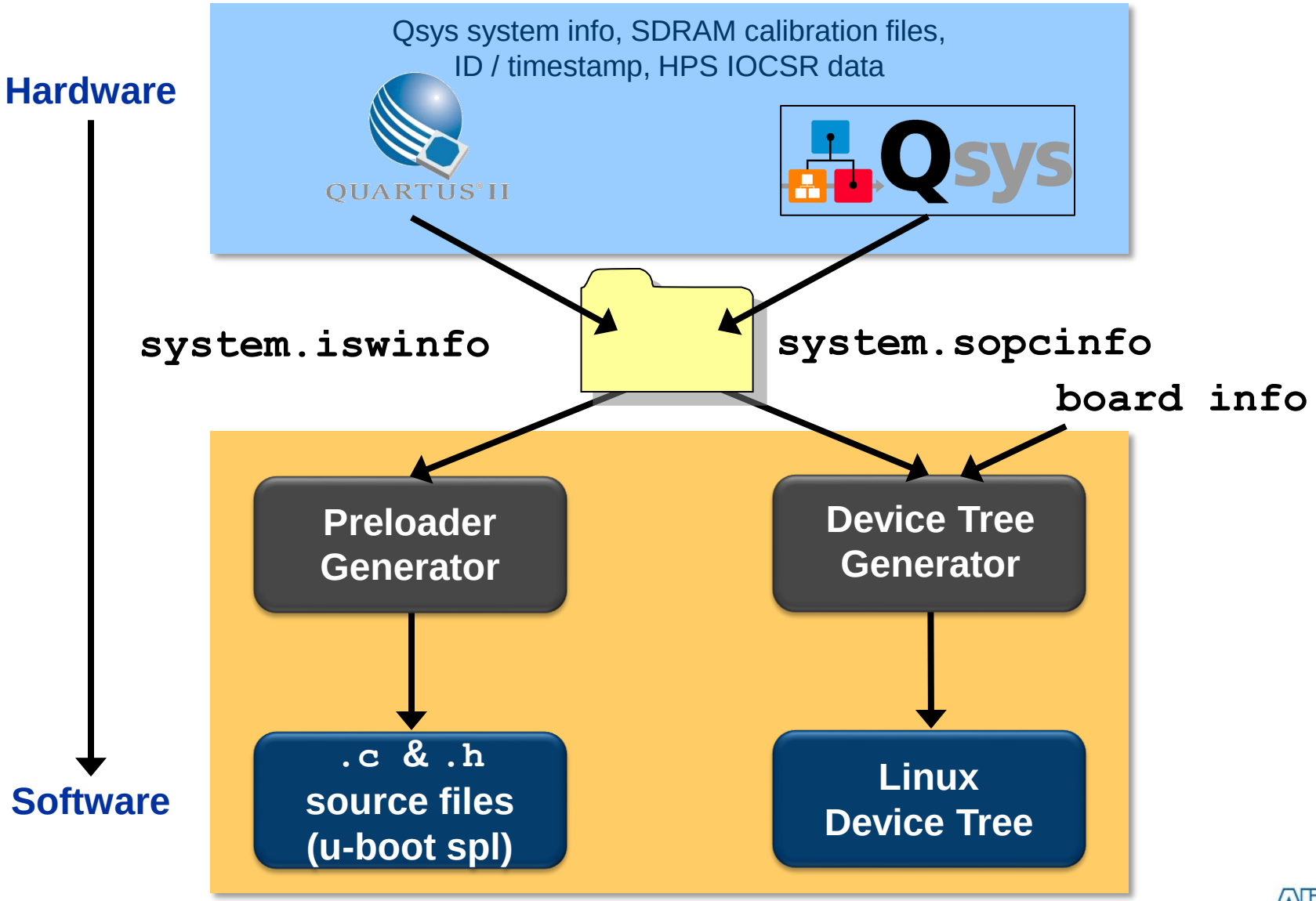
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Hard Processor System Configuration



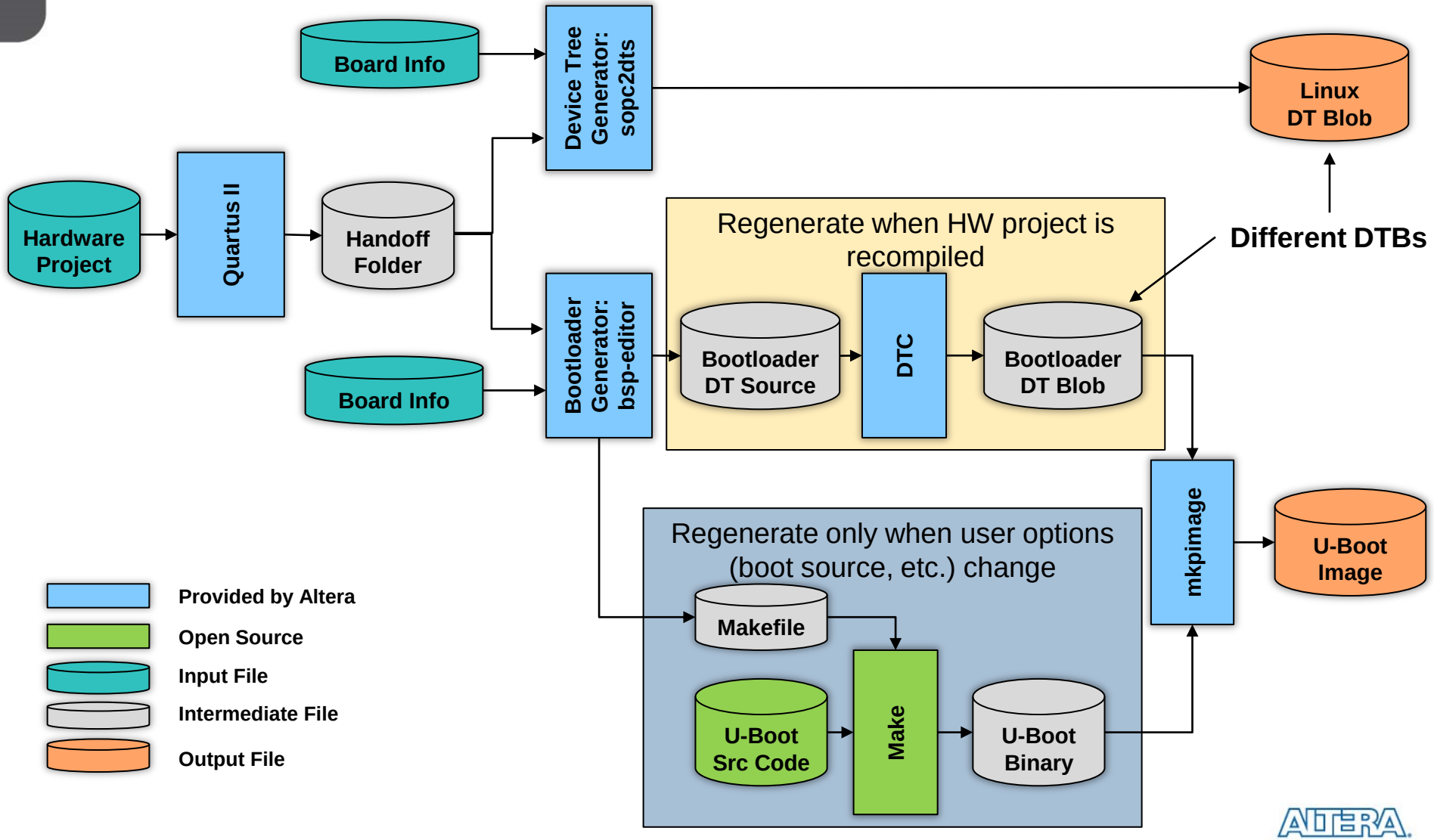
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Linux HW/SW Handoff – Cyclone V SoC and Arria V SoC



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Linux HW/SW Handoff – Arria 10 SoC



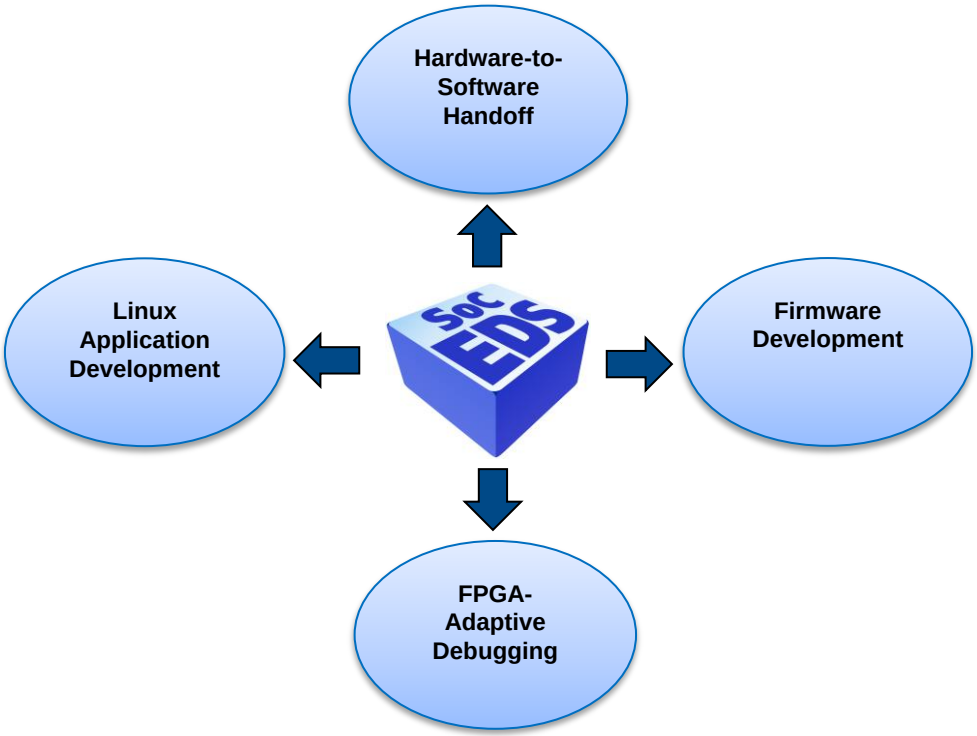
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Altera SoC Embedded Design Suite

Comprehensive Suite SW Dev Tools

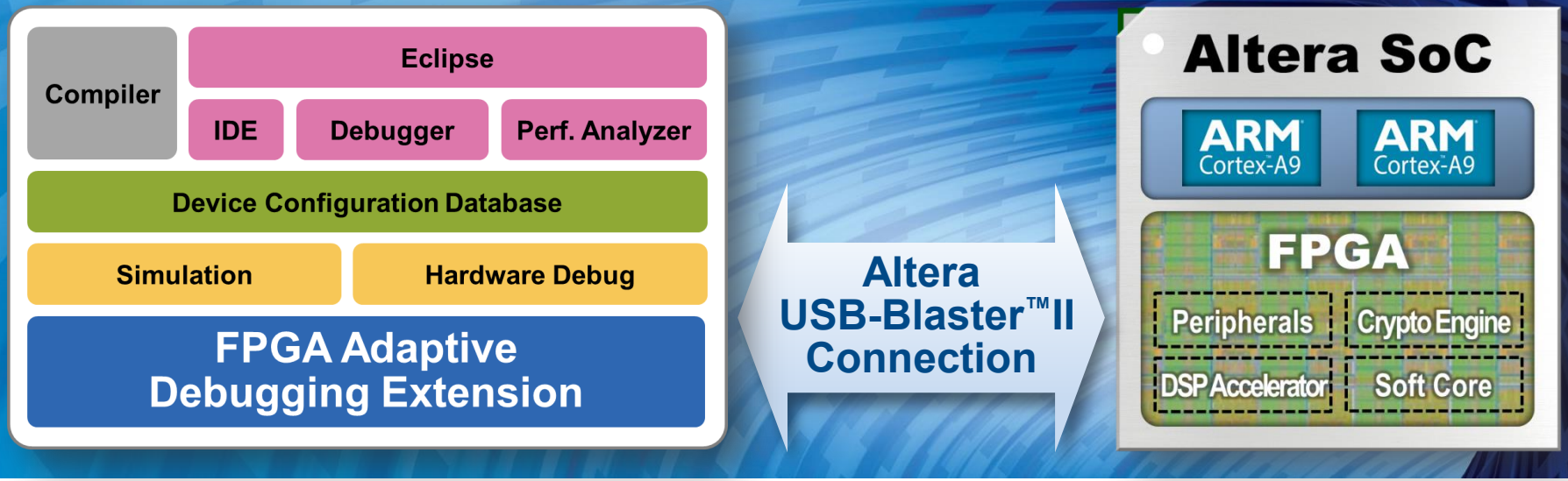
- Hardware / software handoff tools
  - Preloader & Device Tree Generators
- Bare-metal application development
  - SoC Hardware Libraries
  - Bare-metal compiler tools
- FPGA-adaptive debugging
  - ARM DS-5 Altera Edition Toolkit
- Linux application development
  - Yocto Linux build environment
  - Pre-built binaries for Linux / U-Boot
  - Work in conjunction with the Community Portal
- Design examples



✓ ***Free Web Edition***
✓ ***Subscription Edition***
✓ ***Free 30-day Eval***

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Industry First: FPGA-Adaptive Debugging



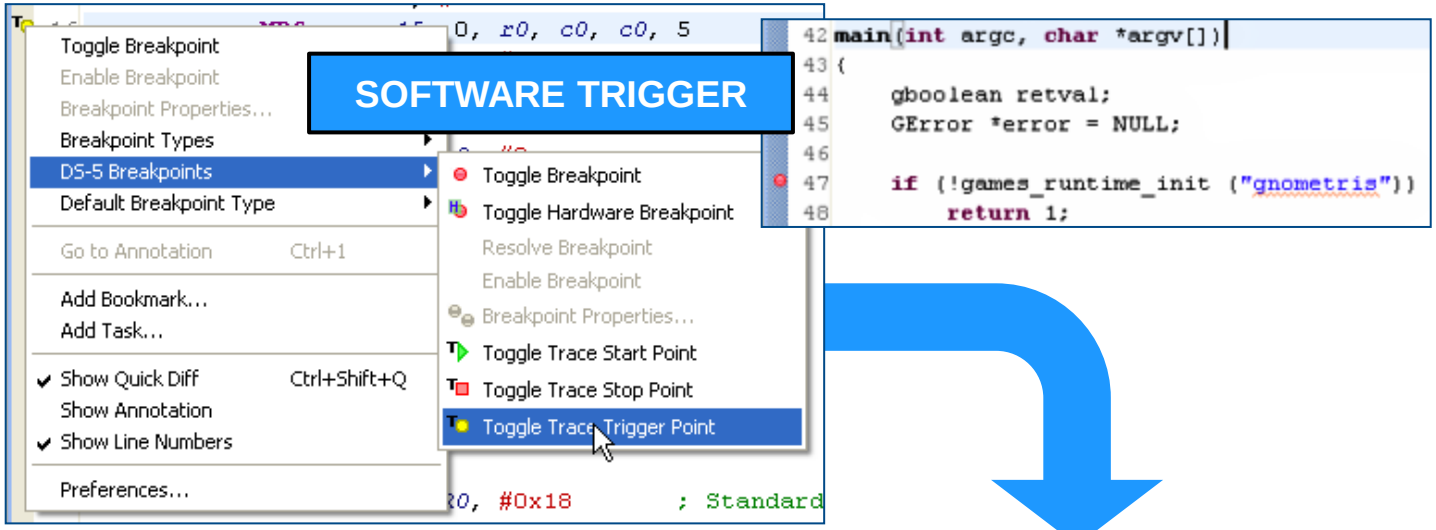
ARM® Development Studio 5 (DS-5™) Altera® Edition Toolkit

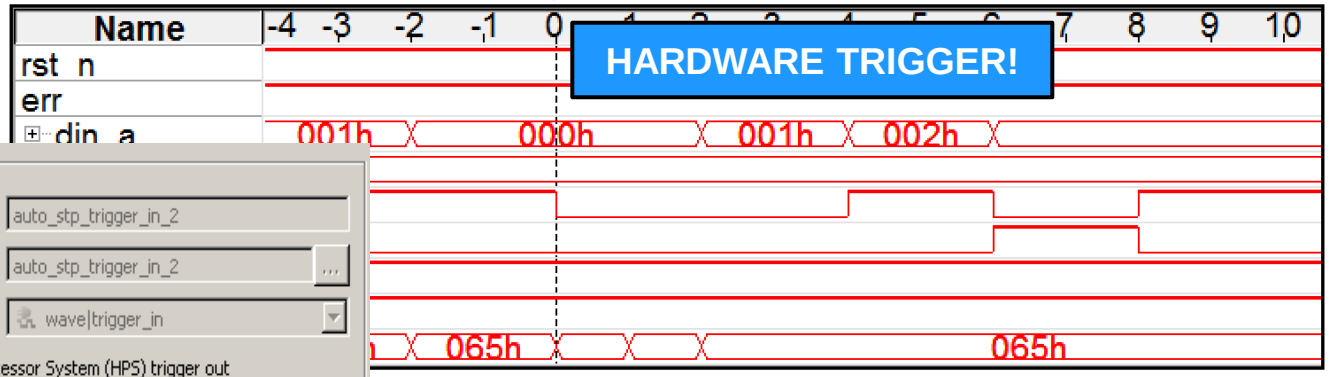
- Removes debugging barrier between CPUs and FPGA
- Exclusive OEM agreement between Altera and ARM
- Result of innovation in silicon, software, and business model
- Supports FPGA-Adaptive Linux kernel, driver & application debug

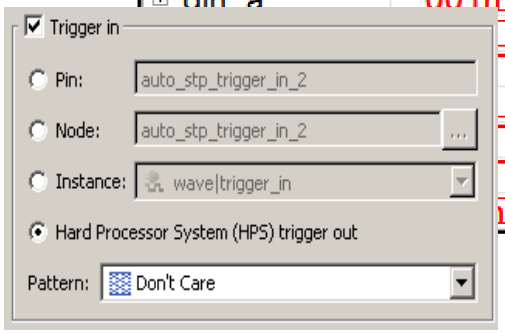
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Cross-Domain Debug 1

- Trigger from software world to FPGA world





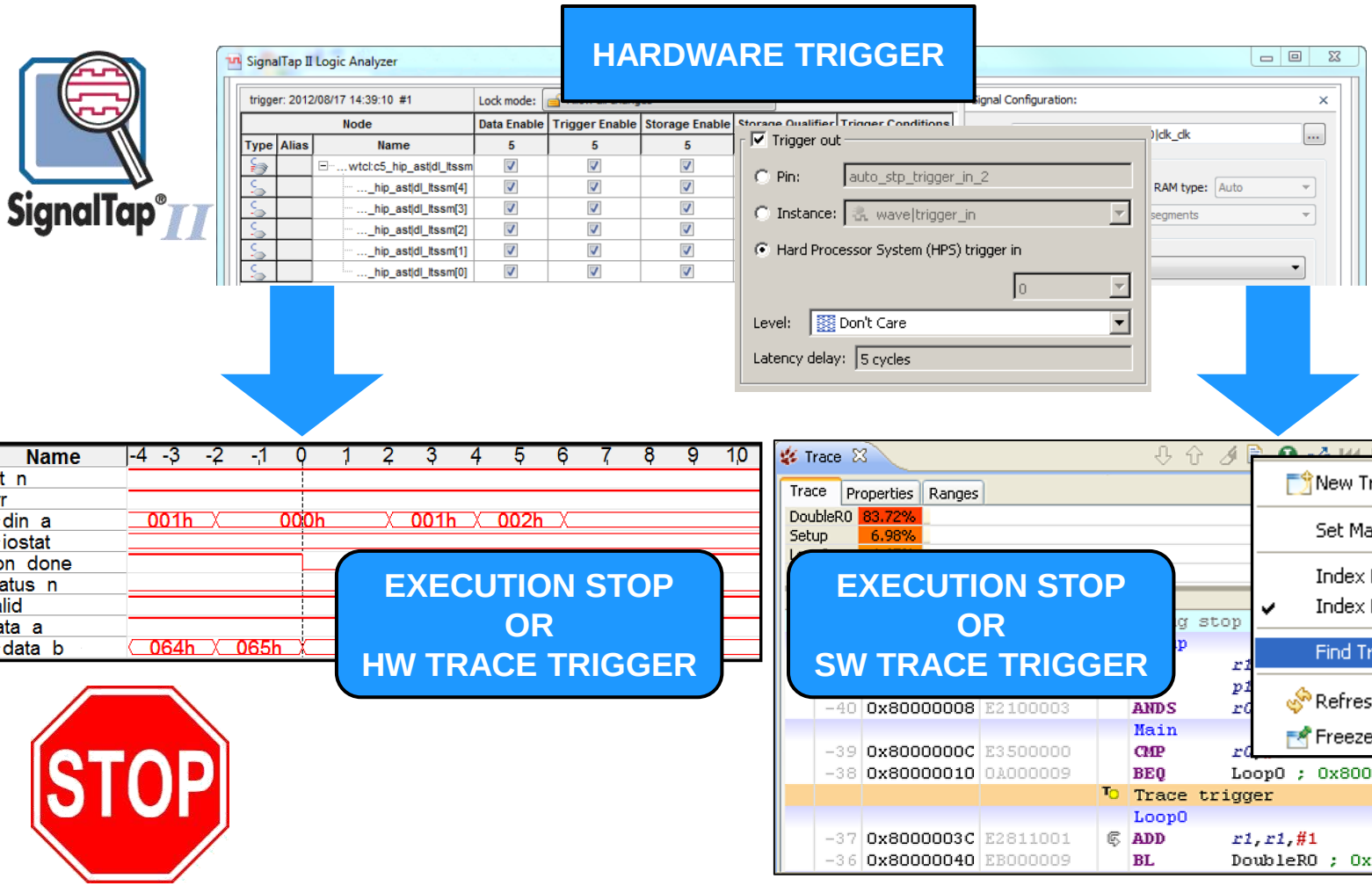


SignalTap II

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Cross-Domain Debug 2

- Trigger from FPGA world to software world



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Correlate HW and SW Events

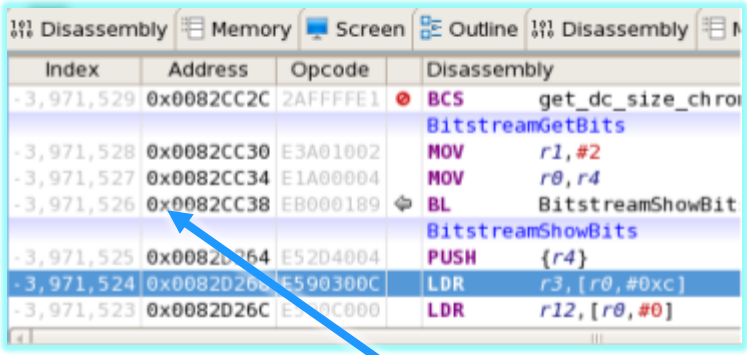


- Debug event trigger point set from either:

  SignalTap™ II Logic Analyzer
  or
  DS-5 debugger

- Captured trace can then be analyzed using timestamp-correlated events

ARM® DS-5™ Toolkit



Timestamp Correlated

SignalTap II Logic Analyzer



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SoC EDS Editions Summary

Component	Key Feature	Web Edition	Subscription Edition	30-Day Evaluation
Hardware/Software Handoff Tools	Preloader Image Generator	x	x	x
	Flash Image Creator	x	x	x
	Device Tree Generator (Linux)	x	x	x
ARM DS-5 Altera Edition	Eclipse IDE	x	x	x
	Debugging over Ethernet (Linux)	x	x	x
	Debugging over USB-Blaster II JTAG		x	x
	Automatic FPGA Register Views		x	x
	Hardware Cross-triggering		x	x
	CPU/FPGA Event Correlation		x	x
Compiler Tool Chains	Linaro Tool Chain (Linux)	x	x	x
	CodeBench Lite EABI (Bare-metal)	X	x	x
Hardware Libraries	Bare-metal programming Support	x	x	x
SoC Programming Examples	Golden System Reference Design	x	x	x

Everything needed for Linux development is free & open source

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Altera SoC Linux Overview

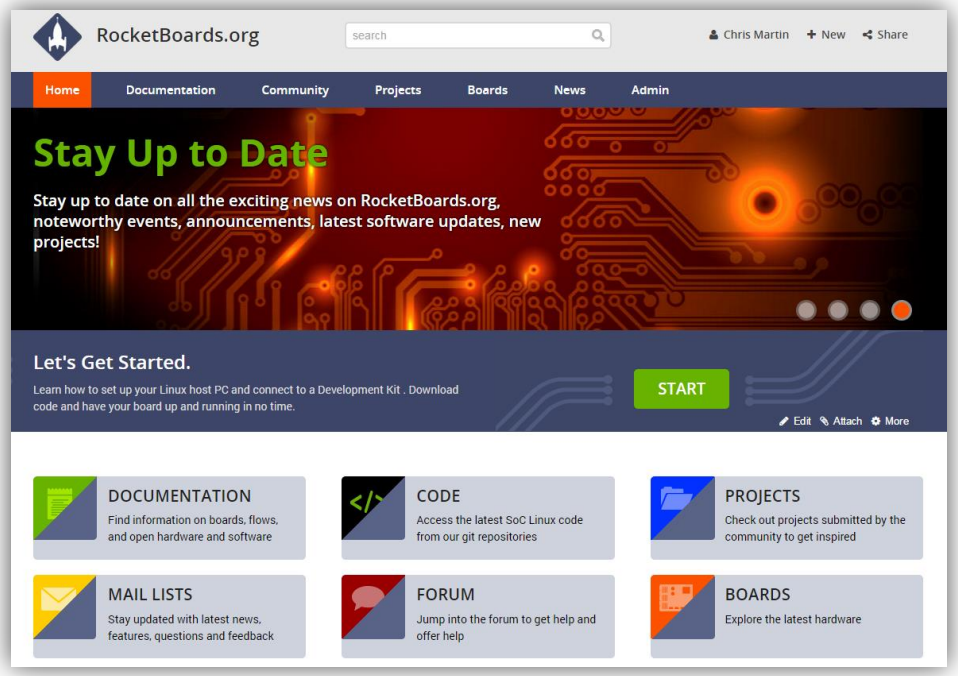


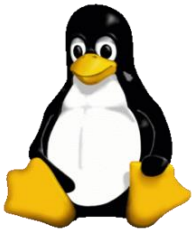
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Linux for Altera SoCs

- High Quality Linux Support
- Modern release strategy
- Multiple Kernel Versions
- Community Enablement





LONG TERM SUPPORT INITIATIVE

Ångström

git

Linaro

yocto PROJECT

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Linux Strategy

- Kernel
  - Same kernel source tree for all SoC's and NIOS II
    - Same kernel binary for all 32bit SoCFPGA
    - Same kernel binary for all 64bit SoCFPGA
  - Device tree support (SoCs and NIOS II)
  - Upstream and maintain to kernel.org
- U-Boot
  - Cyclone V & Arria V SoC currently 2013.01.01
    - Updating to mainline release at Altera 16.0 release
  - Arria 10 SoC currently 2014.10
    - Updating to mainline release at Altera 16.1 release
  - Same U-Boot source tree for all SoC's and NIOS II
- Toolchain
  - Standard, un-patched Linaro Toolchain:
    gcc-linaro-arm-linux-gnueabihf-4.9-2014.09

Nios® II



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Linux Strategy - Build Systems

- Offer support for Angstrom for SoC
  - Embedded Linux distribution, Yocto Project configuration, package manager (OPKG)
  - Angstrom v2015.12 / Yocto 2.0
    - via kraj/meta-altera opensource meta-altera layer
    - Linaro GCC 4.9 for all kernels
    - Linaro GCC 5.2 for 4.1 LTSI kernel
- Yocto Project Support for SoC
  - SoCFPGA layer (meta-altera) upstreamed to Angstrom
  - Yocto Project v2.0 ready
- Buildroot for both SoC and Nios II
  - Roll your own from relevant source



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Build System Resources

- Angstrom flow for SoC

  http://rocketboards.org/foswiki/view/Documentation/AngstromOnSoCFPGA_1

- Yocto flow for SoC

  http://www.rocketboards.org/foswiki/Documentation/YoctoDoraBuildWithMetaAltera

  - Source Poky from the Yocto Project website

    git://git.yoctoproject.org/poky.git

- Buildroot flow for Nios II

  http://rocketboards.org/foswiki/view/Documentation/NiosIILinuxUserManual

- Buildroot flow for SoC

  http://www.rocketboards.org/foswiki/Documentation/BuildrootForSoCFPGA

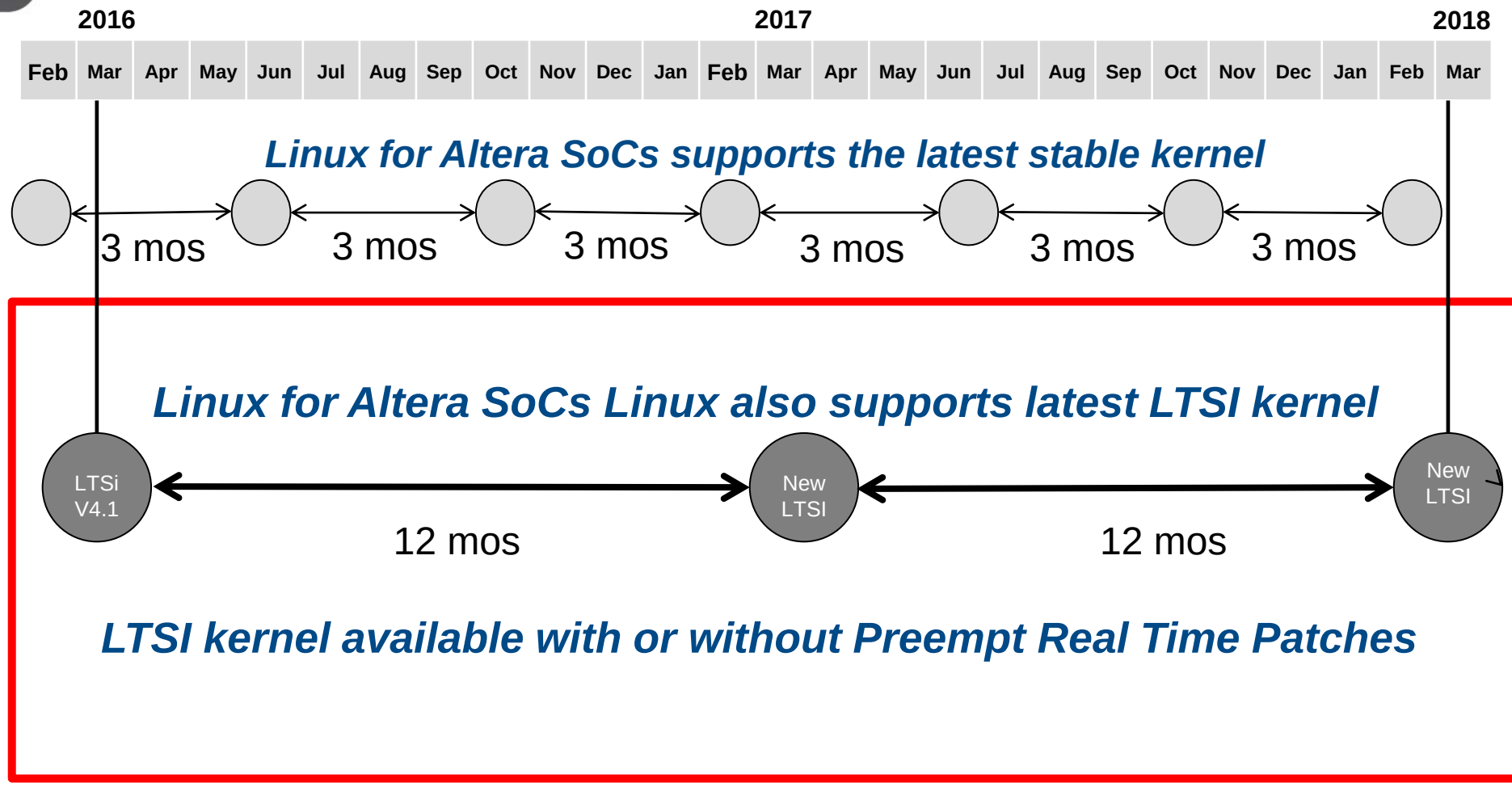
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Angstrom/Yocto Information Resources

- Yocto Project
  https://www.yoctoproject.org/documentation
- Angstrom Distribution
  http://www.angstrom-distribution.org/
- Open Embedded
  http://www.openembedded.org/wiki/Main_Page
- Building Linux w/ Yocto Linux Foundation class
  LFD405 – Building Embedded Linux with the Yocto Project

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Altera SOC Linux Provides Customers Kernel Choices



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Status of the Linux kernel for SoC FPGA

- Current versions
  - Latest stable: refer to linux-socfpga git repo tags: rel_socfpga-x.x…
    - We keep up with Linus Torvalds releases
    - Latest mainline kernel is currently 4.4
  - LTSI v4.1
    - Maintained until next LTSI is released
  - Real-Time: 4.1-ltsi-rt
    - LTSI kernel with PREEMP_RT patches
- All branches are kept in sync
  - Bug fixes
  - New features
  - No changes of API in the LTSI branches

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Wind River Linux

- Wind River Linux version 8
- Linux SMP Kernel version 4.1 (LTSI)
  - Real Time patches & Carrier Grade Linux available
- Yocto 2.0 project user space
- Bitbake build system
- WR Workbench Tools
- GNU toolchain 5.2
- Available now from Wind River
- Technical Support:
  - www.windriver.com
  - support@windriver.com

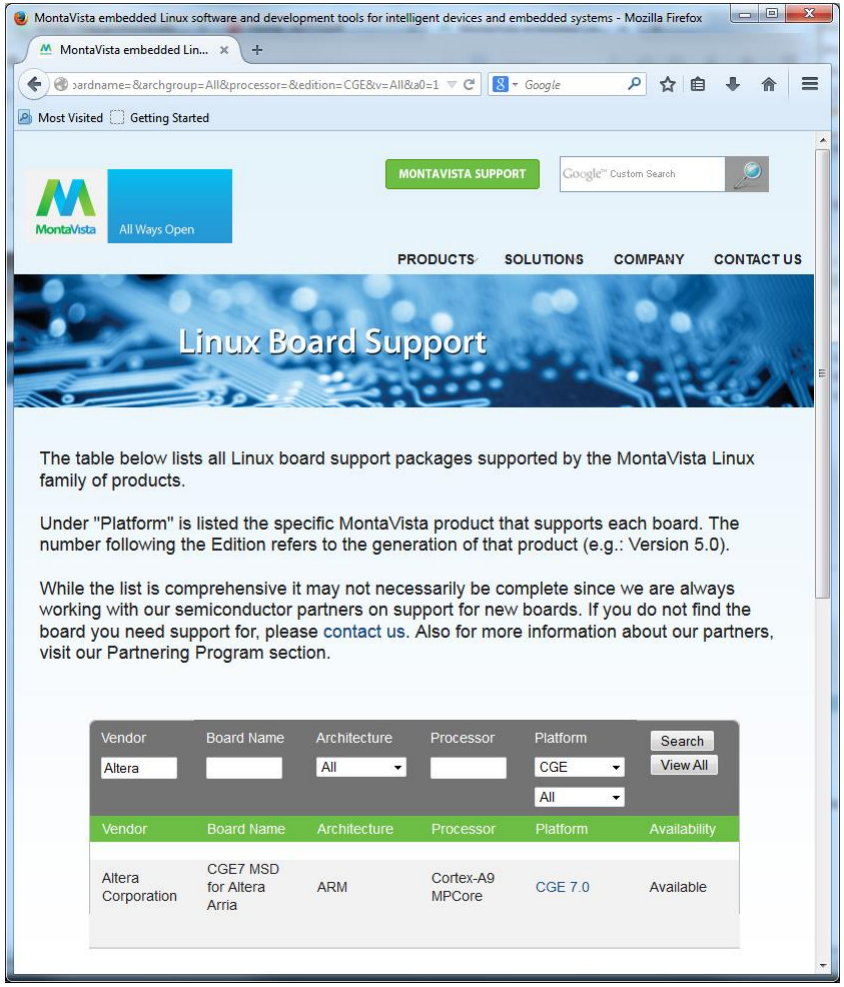


WIND RIVER

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Monta Vista Linux

- Monta Vista Linux CGE7
  - Carrier Grade Edition
- Linux SMP Kernel version 3.10 (LTSI)
- Yocto project user space
- Available from Monta Vista
- Technical Support:
  - www.mvista.com



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Latest Stable Kernel vs. LTSI Kernel

Latest Stable Kernel

- Access to latest kernel features
- New features and drivers often released only to the latest kernel version
- Significant investment in kernel maintenance:
  - Back porting features, bug fixes, device support, & new drivers or…
  - Constant upgrades to latest stable kernel

LTSI Kernel

- LTSI kernel versions supported for 2 years
- Critical bug fixes, priority features, & new device or driver support back-ported to LTSI kernel by the community
- Reduces investment in kernel maintenance
  - Features, bug fixes, device support, & new drivers ported by the community

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LTSI – Economic value

Economic Value of LTSI?

- Cost of Back porting security and bug fixes are 3M$ per year per version
- Cost of maintaining in-house patch is 288K$ in case of LTSI 3.4



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Linux Code Quality

- Altera's internal development process is similar to the community's
  - Code/Peer reviews
  - Code style checked
  - Copyright, licenses, etc checked
  - All checks enforced
- Daily Builds
  - Automated builds run daily
  - Complete system: boot loader, kernel, Angstrom
  - SD card image produced
- Daily Tests
  - Linaro's LAVA is used
  - All kernel branches tested

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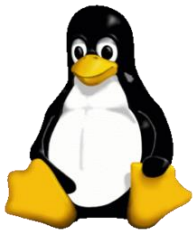
Linux Code Tests

- Objectives
  - Daily test of the supported Linux kernel branches
  - Provide feedback to developers
- Infrastructure
  - Linaro's LAVA is used
    - Linaro Automated Validation Architecture
    - Runs our unit tests and log results
  - Tests start automatically after each build is complete

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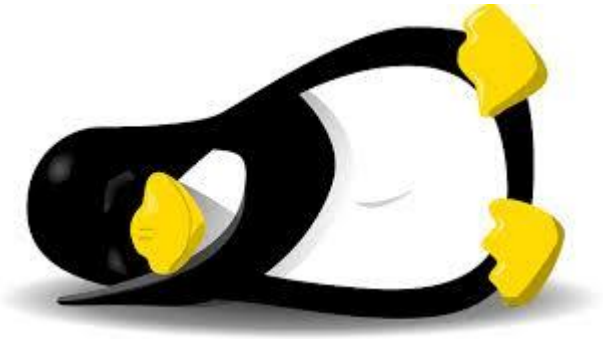
Altera SoC Linux Support Model

- Rocketboards.org
  - SoC & Nios II Linux documentation
  - SoC & Nios II SoC Linux reference & example designs
- Rocketboards.org RFI & Linux Community
  - Kernel/RFS/u-boot questions
  - SoC/Nios II subsystem and driver questions
- Altera.com and Rocketboards.org
  - SoCEDS & Quartus/QSys documentation and questions
  - SoC Preloader questions
  - SoC HPS implementation specific questions
  - Use myAltera for service requests
- Support from Altera is focused on SoCFPGA and NiosII Linux Board Support Package
- Altera enables Linux community development on SoCFPGA & Nios II



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Break



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Components of the SoC FPGA Linux BSP



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Building a Custom Embedded Linux Distribution

How do I get from here….



…to here?







- Altera provides a Linux BSP…

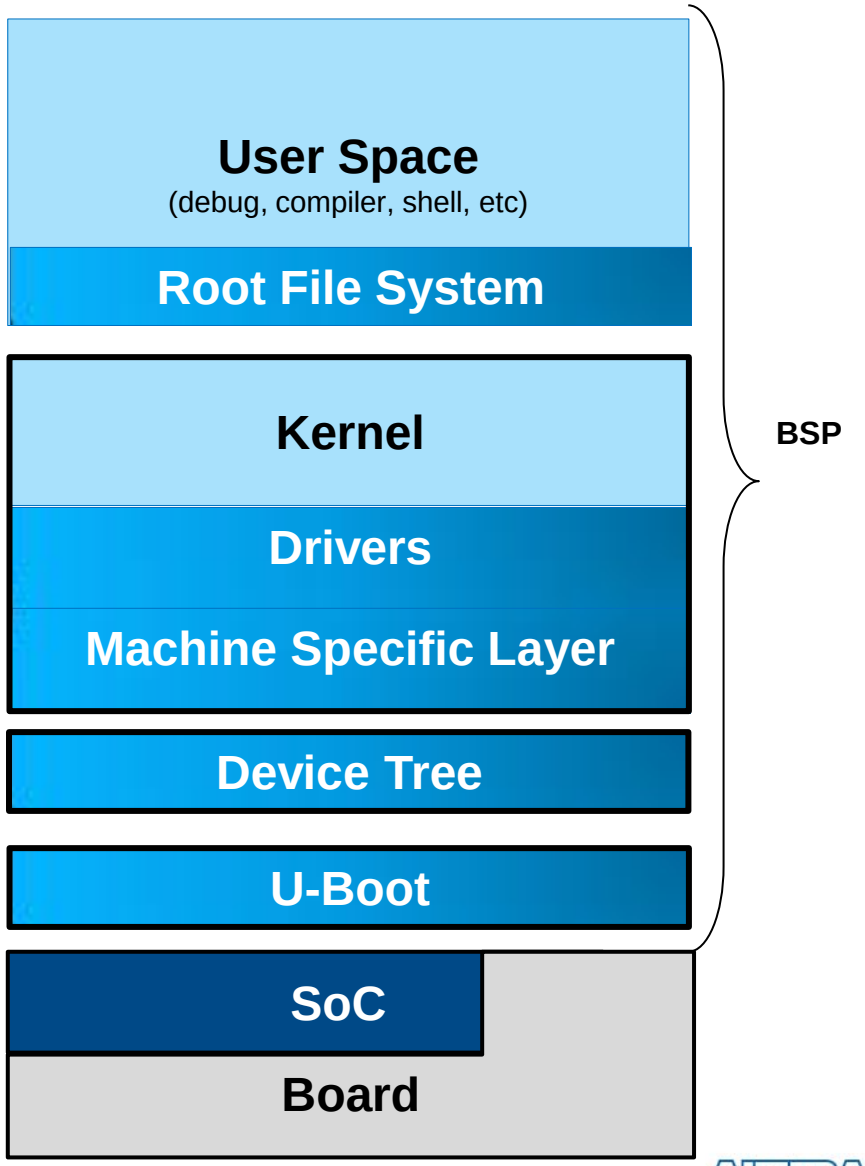
  …not a Linux distribution.
- The BSP enables the creation of a custom distribution

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SoC Linux Board Support Package

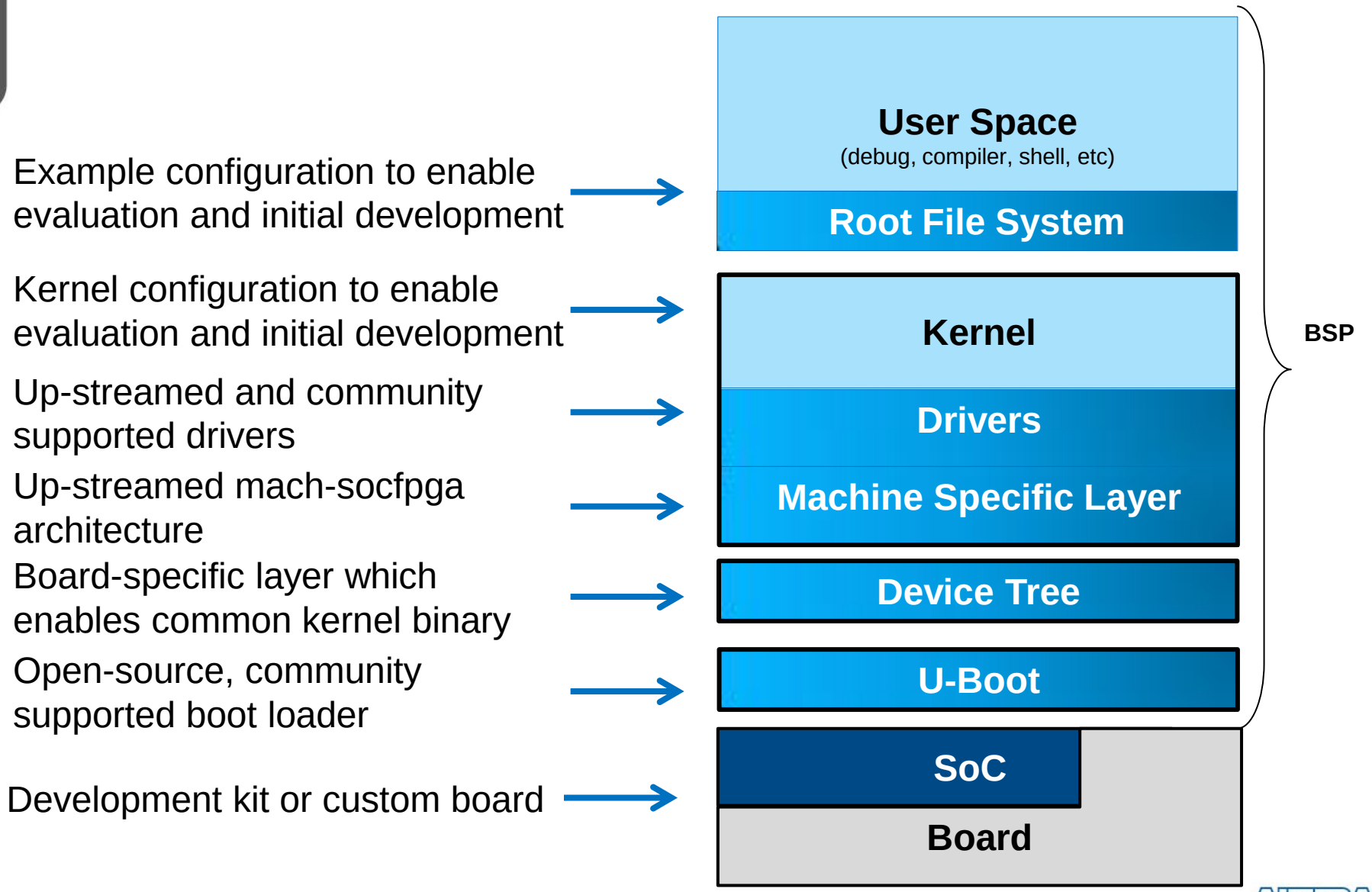
- U-Boot
- Device Tree
- SoC Machine Specific Layer – mach-socfpga
- Drivers for SoC and board components
- Kernel
- Root File System

SoC Linux BSP release provides all of the components in fully non-proprietary source code form



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SoC Linux Board Support Package



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Cyclone V SoC Development Kit



- Everything you need to begin Linux development
  - SoC Development Kit
  - Golden System Reference Design
    - A Linux distribution for the dev. kit
- Features:
  - 1 user license for ARM DS-5 Altera Edition Toolkit
  - Ethernet, USB, CAN, UART
  - DDR3 (HPS and FPGA), SDCard, QSPI
  - PCIe (rootport & endpoint)
  - Expansion header
  - Much more . . .

www.altera.com/products/devkits/altera/kit-cyclone-v-soc.html

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Linux GSRD for Development Kits



- Boot Linux from an SD card
  - Updated images on:

    http://releases.rocketboards.org/
    - Choose a release date folder, the “gsrd” folder, then the “bin” folder. Ex:

      http://releases.rocketboards.org/release/2014.12/gsrd/bin
  - Latest LTSI kernel
  - Angstrom Linux distribution for SoC
    - Package manager to load packages from Angstrom’s on-line package feed
    - Add whatever tools are needed for evaluation: gstreamer, usb-utils, etc…
- GSRD contents
  - Complete HW reference design w/ FPGA programming file
  - Bootable SD card image & component binaries
  - Tagged for rebuilding in angstrom-socfpga git repository

    ACDSX.X_REL_GSRD_PR

A complete SD card ships with the board

- **Take it out and stick it in a drawer**
- **Based on out-of date 3.9 kernel**

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Multiple Dev. Kit Options w/ Linux BSP

Kit	Vendor	Family
Arria 10 SoC Dev. Kit	Altera	**Arria 10 SoC**
Arria V SoC Dev. Kit	Altera	**Arria V SoC**
Cyclone V SoC Dev. Kit	Altera	**Cyclone V SoC**
Atlas Board	Altera	**Cyclone V SoC**
SoCKit	Arrow	**Cyclone V SoC**
Helio SoC Eval. Platform	Macnica	**Cyclone V SoC**
SoCrates	EBV Elektronik	**Cyclone V SoC**

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SoC Linux Up-streaming & Driver Support

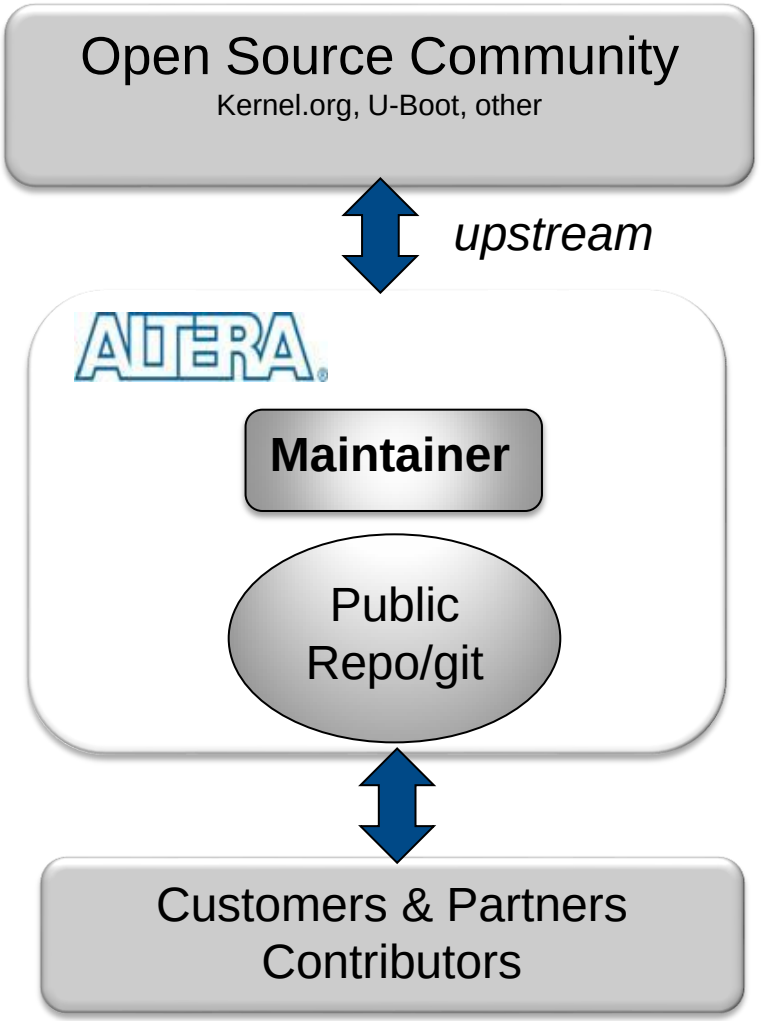


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Maintaining and Up-streaming

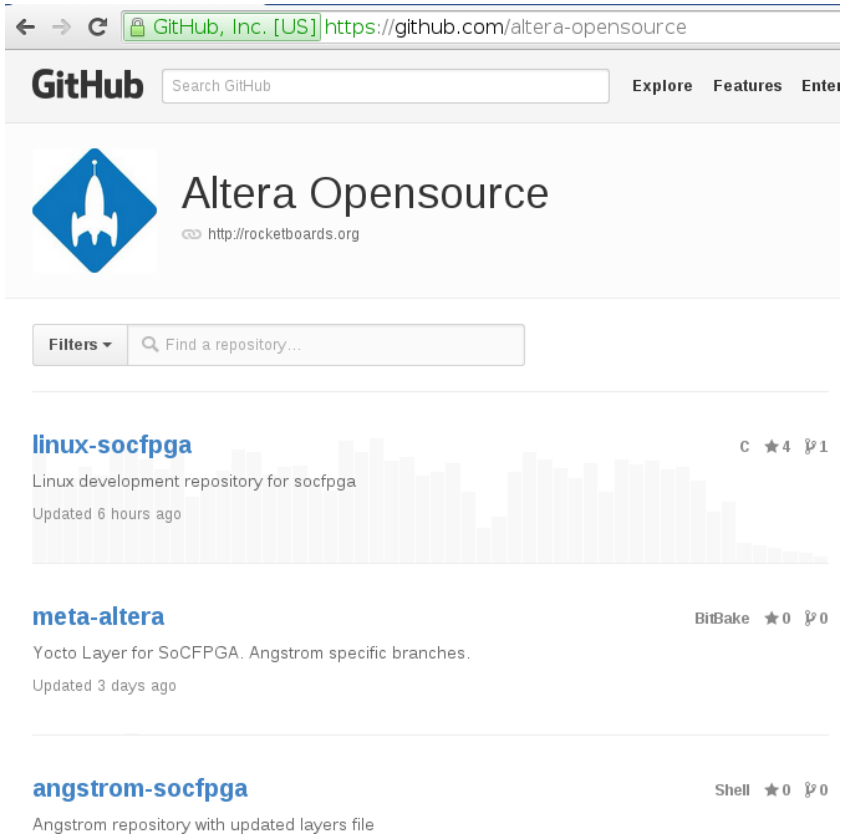
- Altera awarded maintainership for the 'SoC FPGA' architecture
  - Kernel (arch/arm/mach-socfpga)
  - U-Boot (altera/socfpga_cyclone5)
- Being a maintainer means
  - We upstream the SoC related code
  - We control the changes against the SoC code requested by the community
  - See kernel.org and git.denx.de
- Other community contributions
  - Device Tree Generator: sopc2dts
  - Yocto meta-altera layer



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SoCFPGA Linux Code Repositories on GitHub

- Public git repos for SoCFPGA
  https://github.com/altera-opensource



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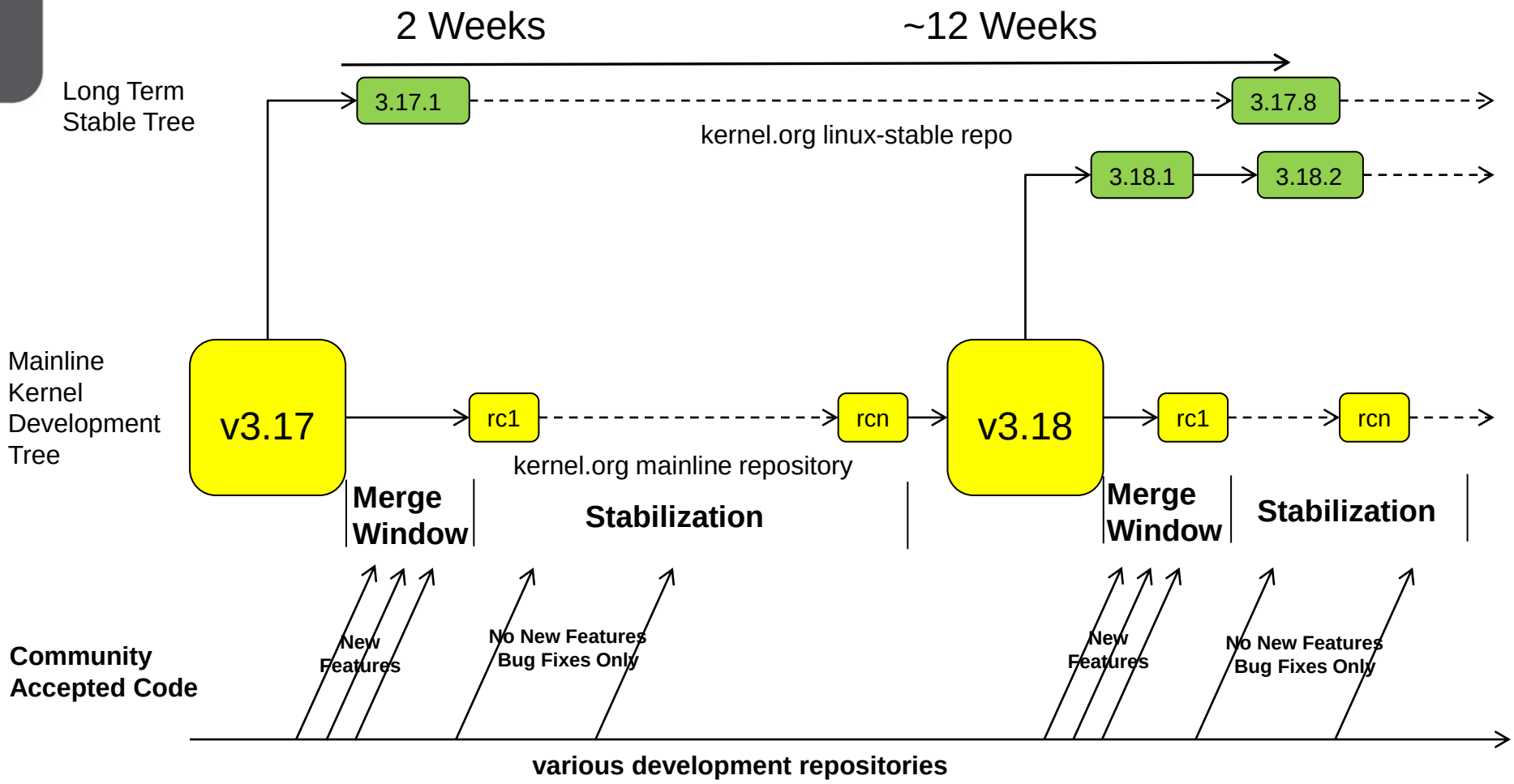
SoCFPGA Linux Code Repositories on GitHub

Repository	Description
linux-socfpga.git/	SoCFPGA Linux development repository • Mirrors kernel.org linux repo releases • Downstream branches for socfpga specific patches and updates
meta-altera.git/	Repository for Yocto recipes for SoCFPGA • Starting point for custom Yocto recipes
angstrom-socfpga.git/	Setup scripts for SoCFPGA Angstrom distribution
uboot-socfpga.git/	SoCFPGA u-boot development repository
sopc2dts.git/	Device Tree Generator (sopc2dts) repository
linux-refdesigns.git/	SW source code for Linux reference designs

**** Sourced from: github.com/altera-opensource/***

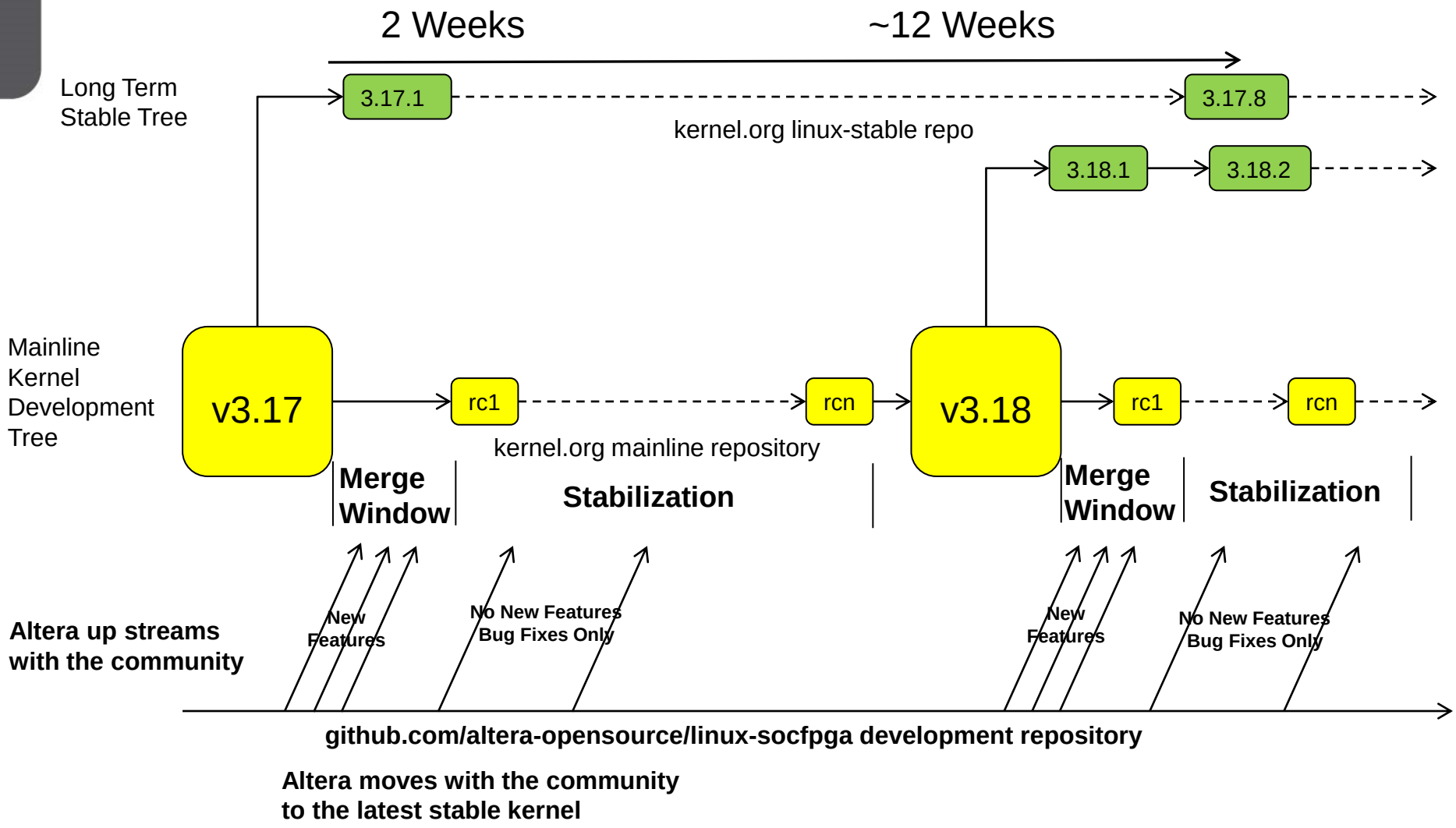
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Kernel Release Cycle – Merge & Bug Fix



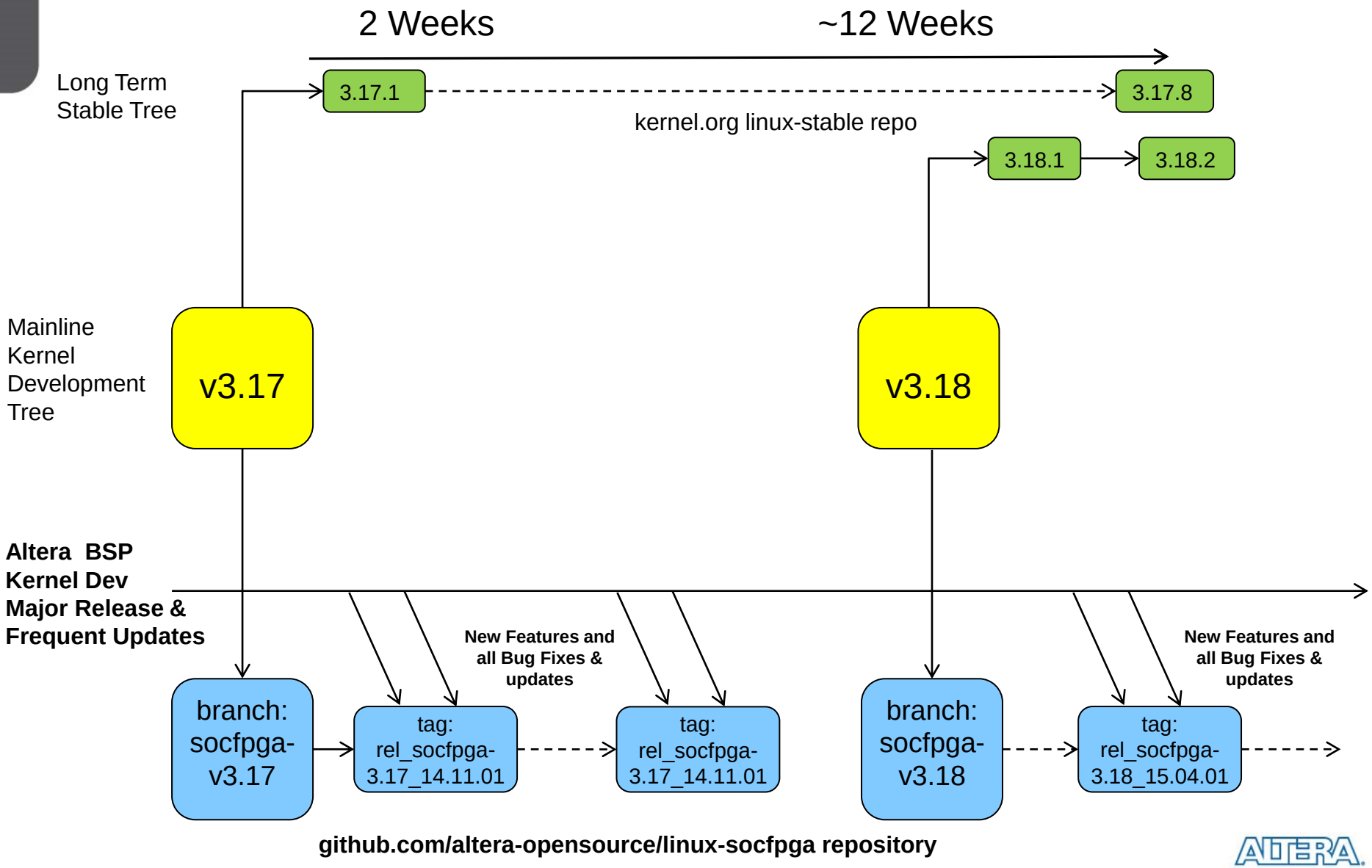
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Kernel Release Cycle – Merge & Bug Fix



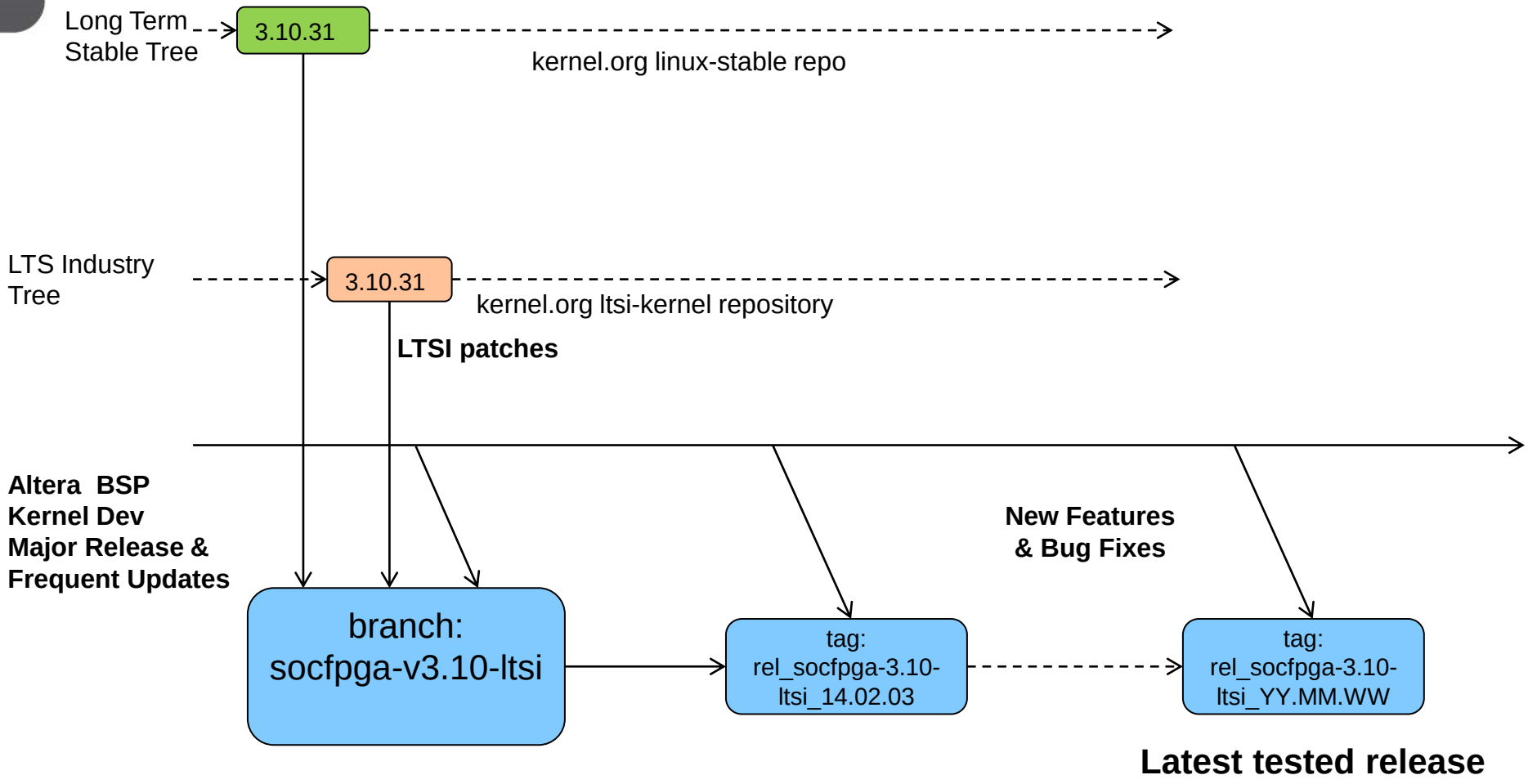
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Altera BSP Kernel Development



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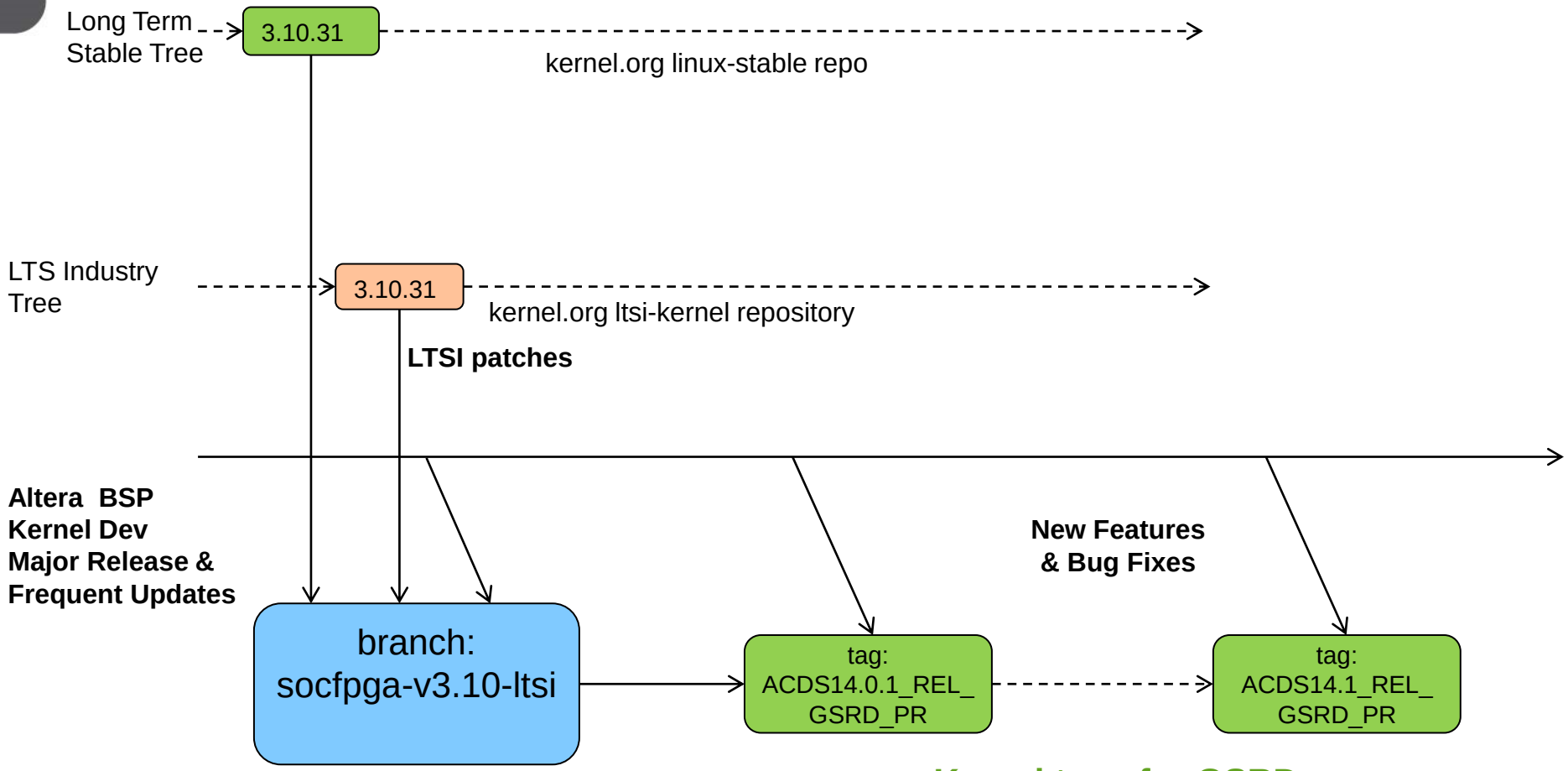
Altera LTSI Kernel Development



github.com/altera-opensource/linux-socfpga repository

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Altera LTSI Kernel Development



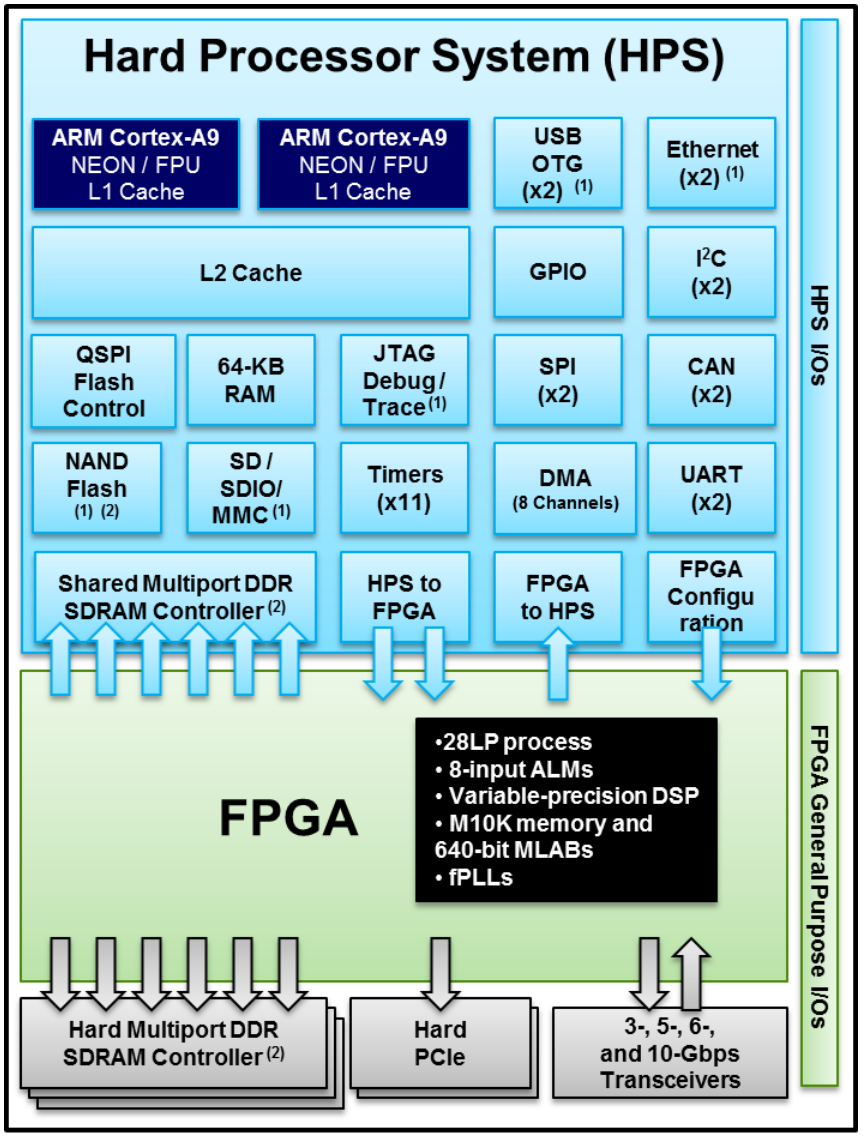
Kernel tags for GSRD releases on Rocketboards

github.com/altera-opensource/linux-socfpga repository

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SoC Hard IP Driver Support

- Altera SoC HPS built mainly with off the shelf Hard IP components
  - ARM
  - Synopsys DesignWare
  - Cadence
- Wide-spread usage and community support result in high-quality drivers
- Altera is actively contributing features, updates, and fixes to the community



Notes:
(1) Integrated direct memory access (DMA)
(2) Integrated ECC

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Linux Driver Support for HPS Peripherals

Driver	Kernel Source Tree Location	Maintainer
SPI	drivers/spi/spidev.c drivers/spi/spi-dw.c drivers/spi/spi-dw-mmio.c	**Community**
CAN	drivers/net/can/c_can/c_can_platform.c	**Community**
Ethernet	drivers/net/ethernet/stmicro/stmmac/stmmac_platform.c	**Community**
NAND	mtd/nand/denali_dt.c	**Community**
I2C	drivers/i2c/busses/i2c-designware-platdrv.c	**Community**
USB	drivers/usb/dwc2/	**Community**
USB PHY	usb/phy/phy-generic.c	**Community**
SDMMC	drivers/mmc/host/dw_mmc-pltfm.c	**Community**
Timer	drivers/clocksource/dw_apb_timer_of.c	**Community**

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Linux Driver Support for HPS Peripherals

Driver	Kernel Source Tree Location	Maintainer
Watchdog	drivers/watchdog/dw_wdt.c	**Community**
PL330 DMA	dma/pl330.c	**Community**
GIC	drivers/irqchip/irq-gic.c	**Community**
GPIO	drivers/gpio/gpio-dwapb.c	**Community**
Timer	drivers/clocksource/dw_apb_timer_of.c	**Community**
UART	drivers/tty/serial/8250/8250_dw.c	**Community**
QSPI	spi/spi-cadence-qspi.c	**Altera**
Clock Manager	drivers/clk/socfpga/clk.c	**Altera**
FPGA Manager	drivers/fpga/fpga-mgrs/altera.c	**Altera**
FPGA Bridges	drivers/misc/fpga-bridge/	**Altera**
EDAC (ECC)	drivers/edac/altera_*	**Altera**

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Linux Driver Support for soft Peripherals

Driver	Kernel Source Tree Location	Maintainer
TSE Ethernet	drivers/net/ethernet/altera	**Altera**
PCIe Root Port	www.rocketboards.org www.rocketboards.org – w/o MSI	**Altera**
Frame Buffer	video/altvipfb.c	**Community**
Avalon SPI	spi/spi-altera.c	**Community**
Avalon UART	tty/serial/altera_uart.c	**Community**
JTAG UART	tty/serial/altera_jtaguart.c	**Community**
QSYS Sys ID	misc/altera_sysid.c	**Altera**
Mailbox	drivers/mailbox/mailbox-altera.c	**Altera**
Altera 16550 UART	drivers/tty/serial/8250/8250_core.c	**Community**
Avalon PIO	drivers/gpio/gpio-altera.c	**Altera**

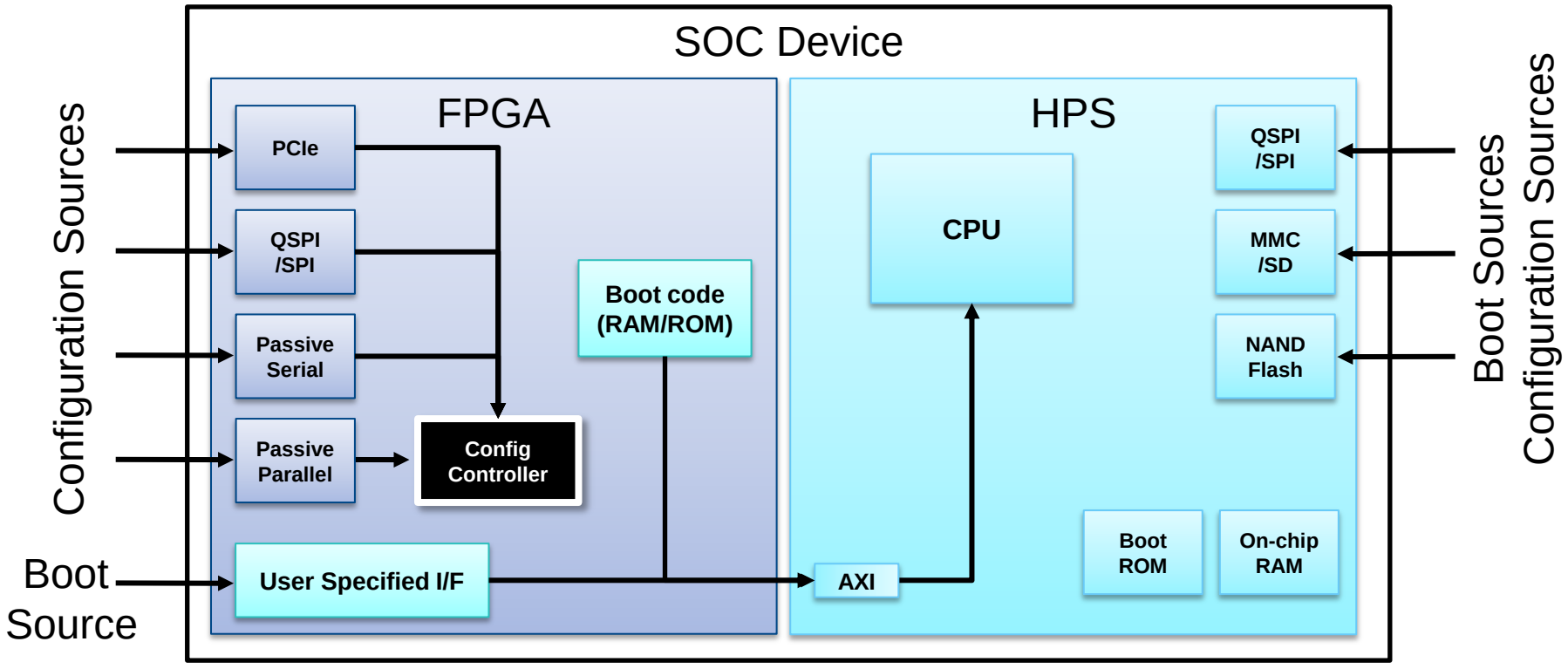
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Altera SoC Linux Boot Flow



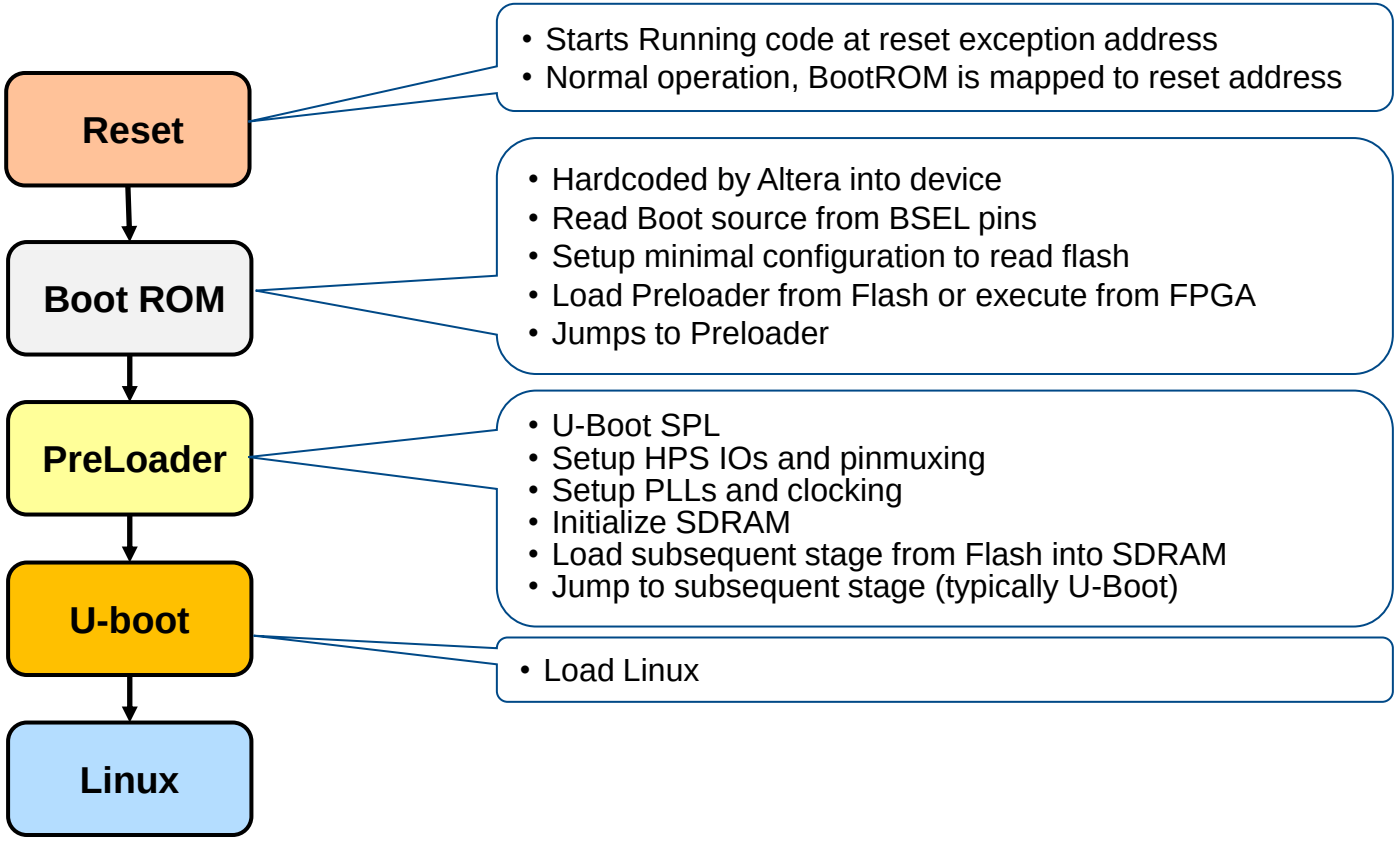
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Altera SoC FPGA Configuration Options



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SoCFPGA Linux Boot Flow – Cyclone V & Arria V SoC



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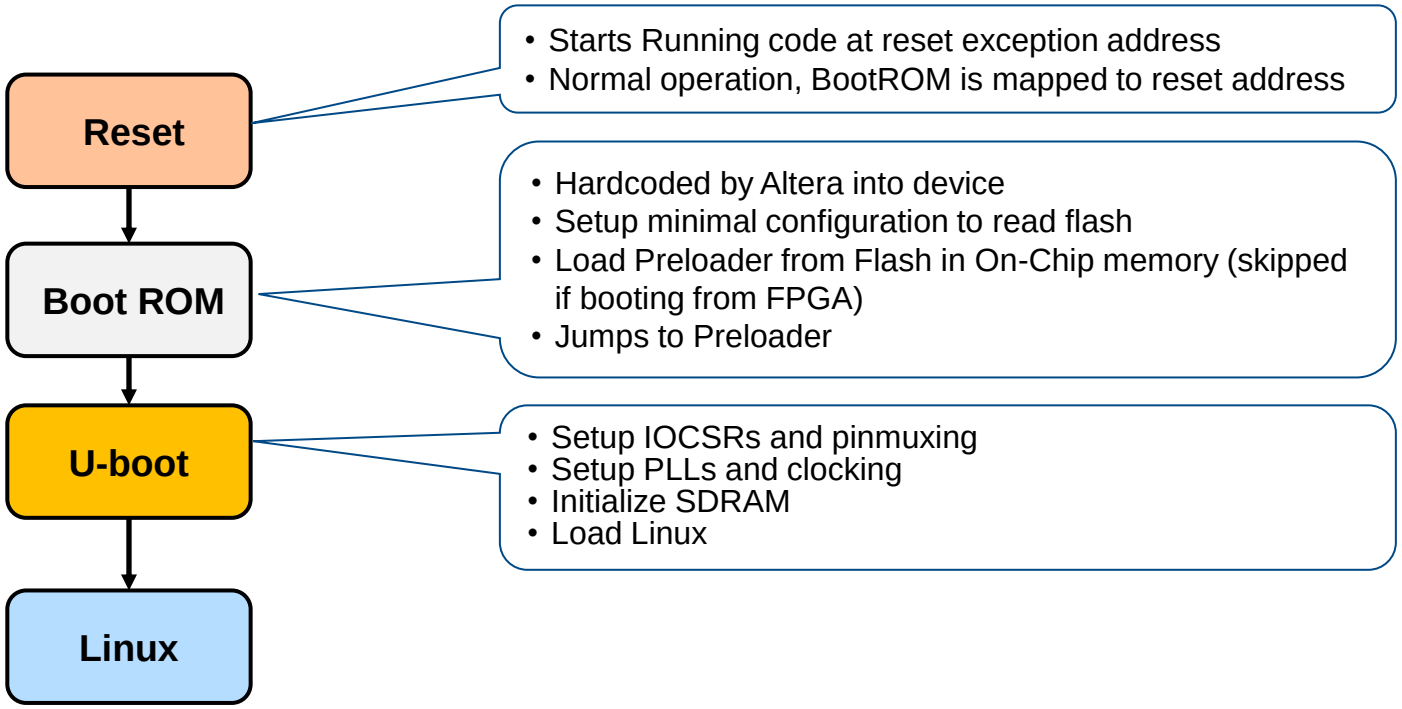
Preloader Overview – Cyclone V & Arria V SoC

- Loaded by Boot ROM
  - From flash and executed from on-chip RAM
  - or run directly from FPGA
- Uses U-boot Secondary Program Loader (SPL)
  - Open source, GPL Licensed
- Loads U-boot into RAM and jumps to U-boot
- Always regenerate and recompile Preloader when
  - QSys system or HPS configuration changes
  - Quartus/QSys version changes

  Not regenerating and recompiling the Preloader is the single most common source of SoC SW problems!
- Covered in detail in SoCEDS User Guide and *Designing Software for ARM-Base SoC* training class

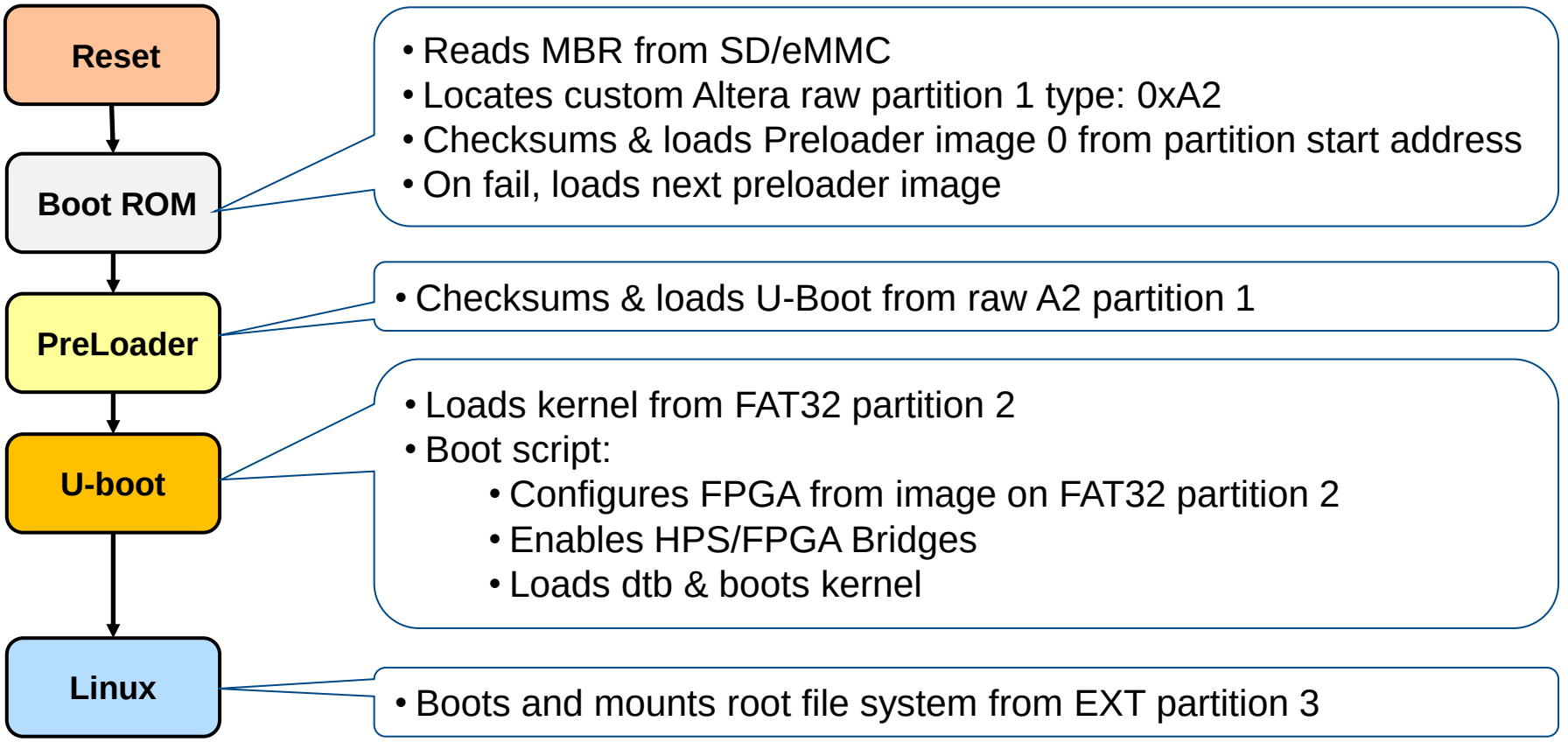
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SoCFPGA Linux Boot Flow – Arria 10



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Booting from SD/eMMC – GSRD Flow - Simplified

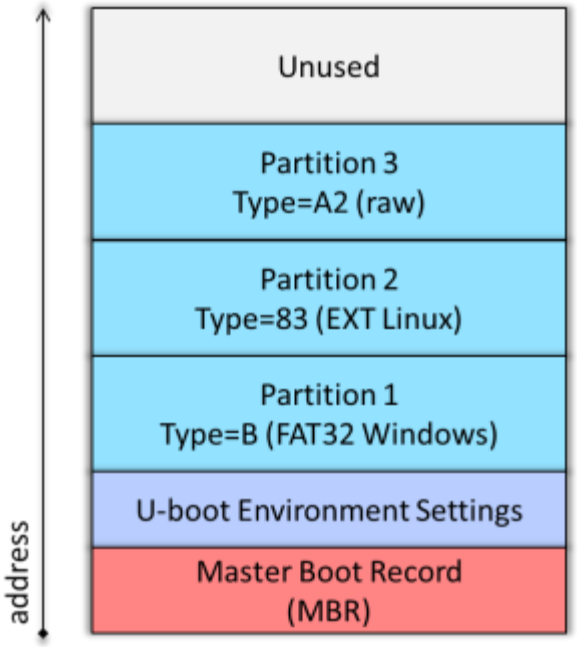


Provides ONE example of an SD/eMMC Linux boot flow

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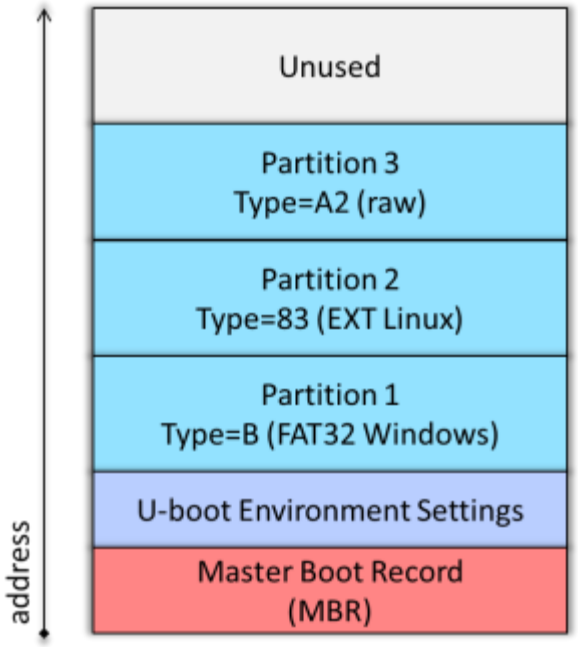
GSRD SD Card Image for Cyclone V & Arria V



Location	File Name	Desciption
Partition 1 (FAT32)	socfpga.dtb	Device Tree Blob
	soc_system.rbf	FPGA configuration file
	u-boot.scr	U-Boot script: configures FPGA and loads kernel
	zImage	Compressed Linux kernel image file
Partition 2 (EXT3)	Various	Linux root file system
Partition 3 (A2 raw)	n/a	Preloader image(s)
	n/a	U-Boot image

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GSRD SD Card Image for Arria 10



Location	File Name	Desciption
Partition 1 (FAT32)	socfpga.dtb	Device Tree Blob
	soc_system.rbf	FPGA configuration file
	u-boot.scr	U-Boot script: not currently used
	zImage	Compressed Linux kernel image file
Partition 2 (EXT3)	Various	Linux root file system
Partition 3 (A2 raw)	n/a	U-Boot image

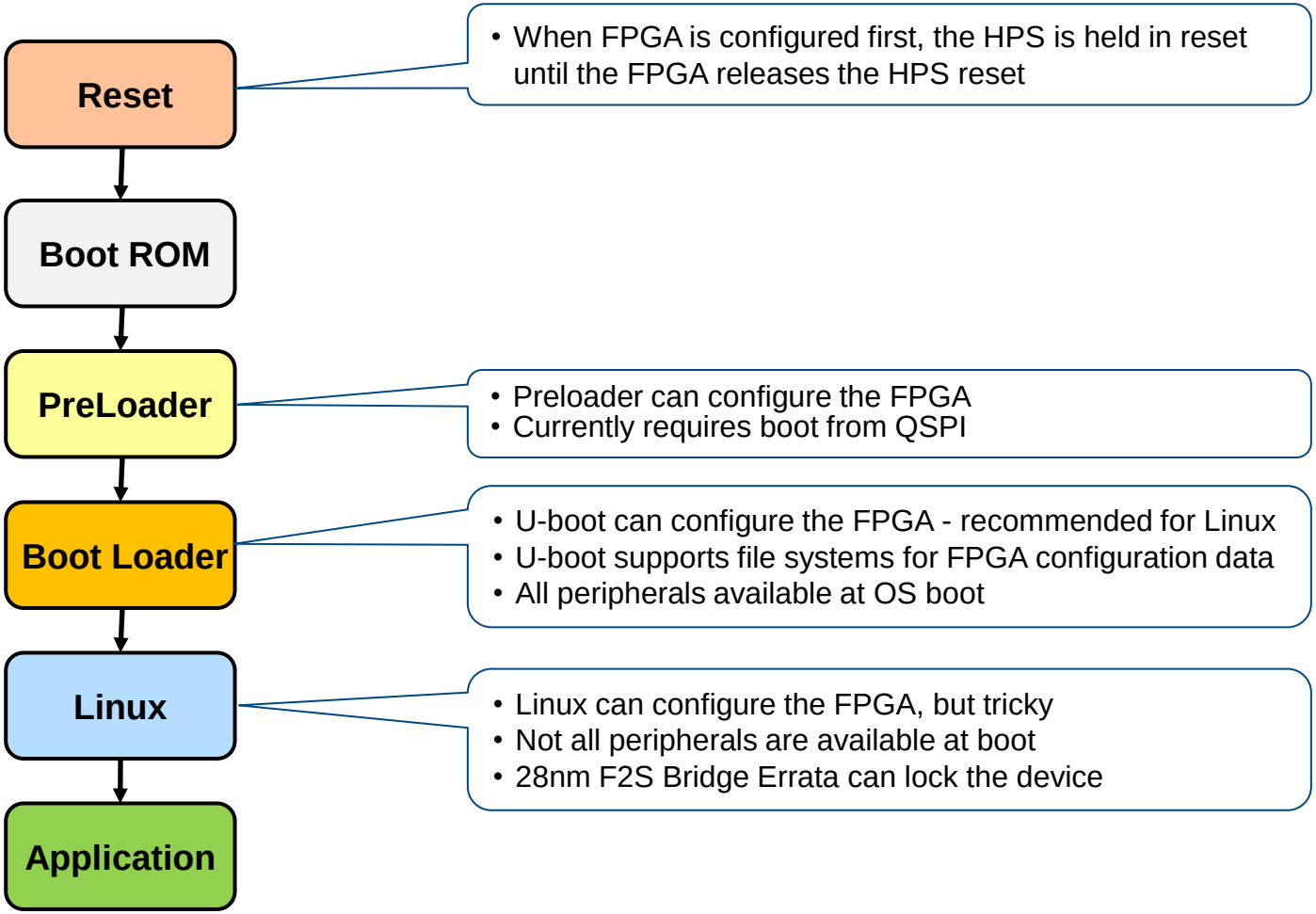
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Creating SD Card Images

- Create using Altera provided script
  - See “tools” folder under GSRD release folders
    http://releases.rocketboards.org/
  - Builds complete SD card image which can be directly copied
- Use pre-built images
  - In “bin” folder under GSRD release folders
    http://releases.rocketboards.org/
  - <SoCEDS install directory>/examples/software
  - Can be directly copied to SD card
- Described in “Creating and Updating SD Card” section of GSRD User Manual
  http://www.rocketboards.org/foswiki/Documentation/GSRD

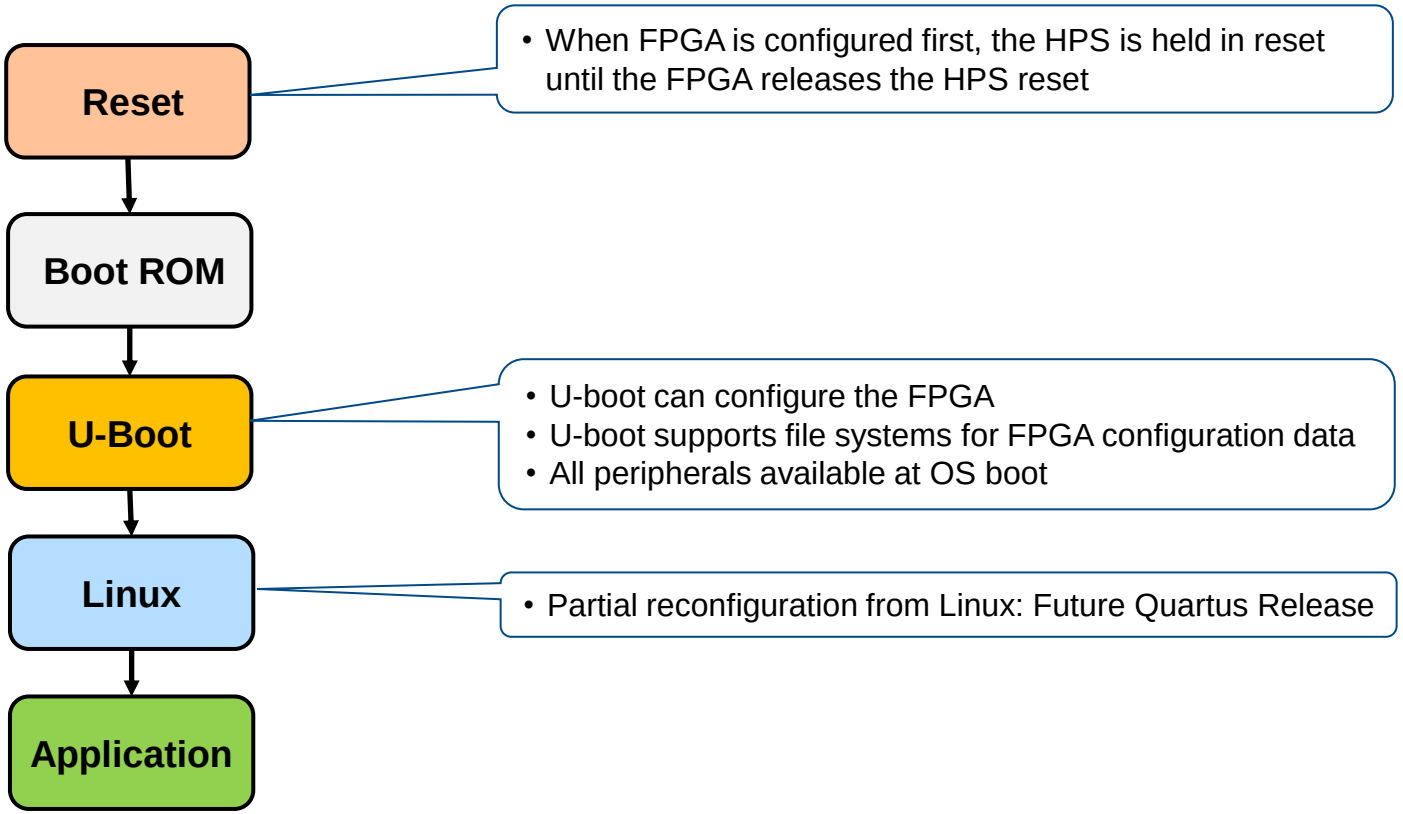
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FPGA Configuration – Cyclone V & Arria V SoC



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FPGA Configuration – Arria 10 SoC



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Das U-Boot Bootloader



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What is u-boot?

- Common embedded bootloader
  - Command line interface w/ decent help and lots of hardware support
  - Capable board bring up tool
  - Driver support for a wide variety of essential peripherals
- Loads the Device Tree and modifies Device Tree configuration at run-time
- Loads the kernel and passes boot arguments
  - From local file system, over network, or over a serial link
- Open-source & GPL licensed
  - http://www.denx.de/wiki/U-Boot

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U-Boot for SoCFPGA

- Sourced from GitHub Altera Opensource or SoCEDS

  https://github.com/altera-opensource/u-boot-socfpga.git

  <SoC EDS install dir>/examples/hardware/<*_ghrd>/…/uboot-socfpga
- Supported u-boot versions:
  - Cyclone V & Arria V SoC currently 2013.01.01
    - Updating to mainline release at Altera 16.0 release
  - Arria 10 SoC currently 2014.10
    - Updating to mainline release at Altera 16.1 release
- u-boot-socfpga branch & tag convention similar to linux-socfpga

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U-Boot for SoCFPGA

- SoCFPGA-specific u-boot documentation

  <u-boot-socfpga repo>/doc/README.SOCFPGA

- u-boot controls SDRAM size passed to kernel
  - u-boot SDRAM sizing algorithm overwrites device tree SDRAM *memory* node entry
  - Device tree entry overrides boot arguments
- SoCFPGA u-boot environment variables
  - Pass HPS/FPGA bridge status and configuration info from preloader
  - Enable FPGA programming from preloader

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SoCFPGA HPS Peripherals Supported in U-boot

- Peripheral support in u-boot-socfpga 2013.01.01

MPU Subsystem

Cache/MMU
- Cache Mgmt
- MMU Mgmt

Interrupt Ctrl

FPGA Manager

Clock Manager

Reset Manager

System Manager

SDRAM Ctrl

Interrupt Ctrl

Pin I/O Cnf Mgmt

ECC Mgmt

Ethernet MAC

Flash Memory
- QSPI
- NAND
- SD/MMC

GPIO

DMA

Serial
- UART

Timers
- Watchdog
- General Purpose

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Useful U-boot Commands and Variables

- Write memory location

```
mw <address> <size> <optional count>
mw 0xC0000000 0x10 0x6
```

- Read memory location

```
md<optional .b .w .l> <address> <optional size>
md.w 0xC0000000 0x2
```

- Control u-boot auto-boot
  - Boot without delay:

```
setenv bootdelay 0
```

  - Disable auto-boot (stop at u-boot command line):

```
setenv bootdelay -1
```

- Save environment variables to flash

```
saveenv
```


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Linux Device Tree for SoC FPGA



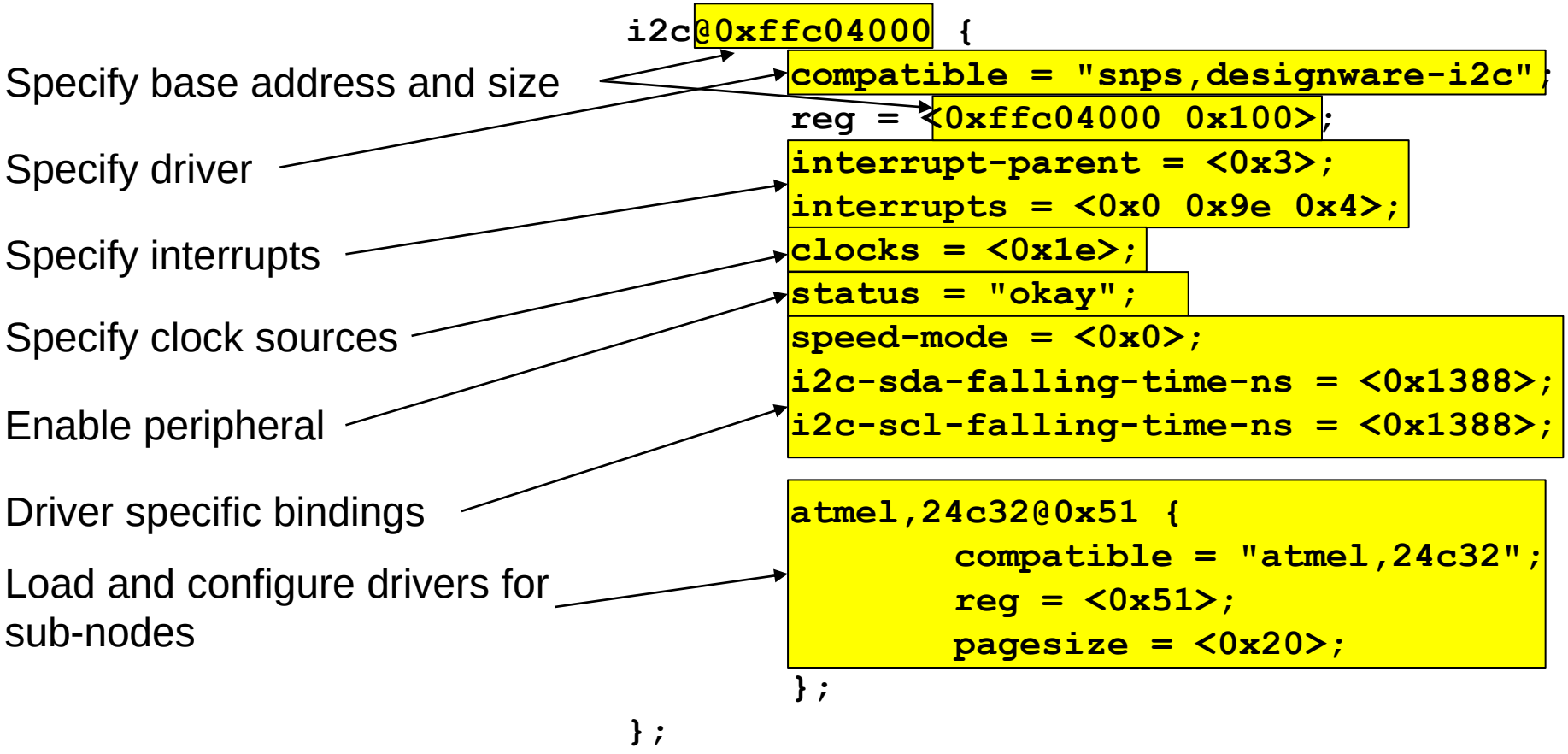
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What's a Linux Device Tree?

- A tree-like data structure for describing hardware in embedded systems
- Enables device drivers to be linked to Linux kernel at run-time
  - No Linux kernel recompile required
  - Drivers loaded dynamically after loading Device Tree
- Driver-specific Device Tree bindings are documented in the kernel documentation
  - Documentation/devicetree/bindings
- See Device Tree Generator User Guide
  - See link in GSRD User Guide

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SoC FPGA Device Tree Bindings Example

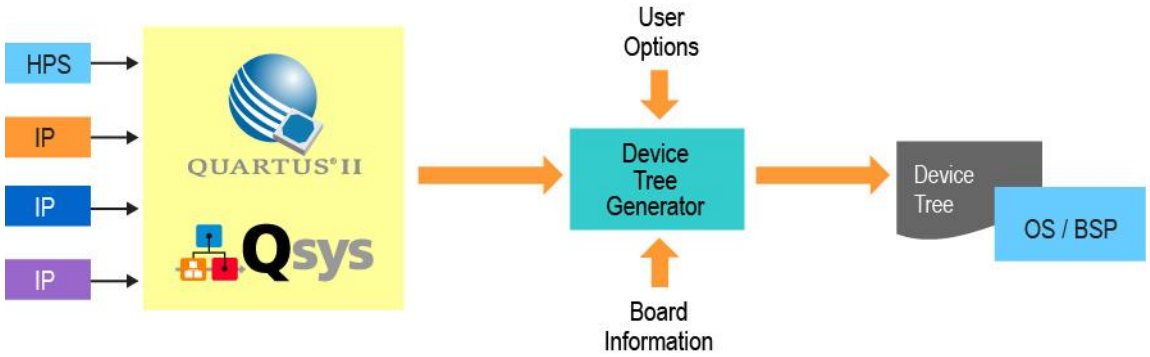


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Device Trees for a Configurable Peripheral Set

- Typically developers build Linux for fixed form chips
- How do you build Linux for an FPGA fabric that changes?

You use the *Device Tree Generator*

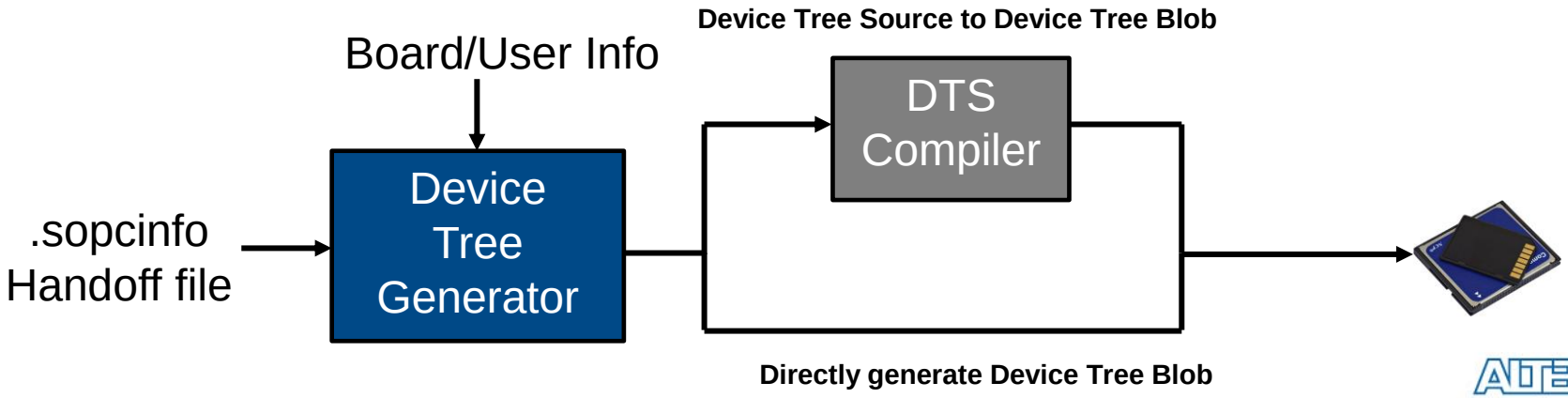


- Device tree information can be added to custom Qsys peripherals and custom drivers
- Device Tree Source can be edited by hand

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How to use Altera SoC Device Tree Generator

- .sopcinfo file describes HPS and FPGA system
- Board info file describes external devices on board
- Device Tree Generator creates a plain text representation of the device tree – called the Device Tree Source (DTS)
  - Device Tree Generator: SoCEDS sopc2dts tool
  - sopc2dts distributed as part or SoCEDS or on RBO GIT repo
- Compile the text into a binary representation called the Device Tree Blob (DTB)
- Optionally directly generate DTB from sopc2dts



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Board Info File

- XML file required as an input to sop2dts
- Specifies information of which QSys isn't aware
  - SPI or I2C timing or external peripherals and their properties
  - External flash (QSPI, SPI, or NAND) properties (organization, specs, etc)
- Board specific Ethernet & PHY information
- Allows the developer to disable peripherals which may be enabled in preloader or FPGA HW

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Take Home Lab



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Workshop 2 Lab Overview

- Goal:
  - Familiarize you with SoC FPGA Linux components and where to obtain them
- Overview
  - You will build the SoC-specific pieces of a Cyclone V or Arria V SoC Linux distribution and run it on your dev. kit
  - This flow builds each component discretely without a build system. It does not use the optional Yocto or Angstrom based build systems.
  - It configures the distribution in a way which works for this lab, which may differ from your actual system requirements

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What You'll Need

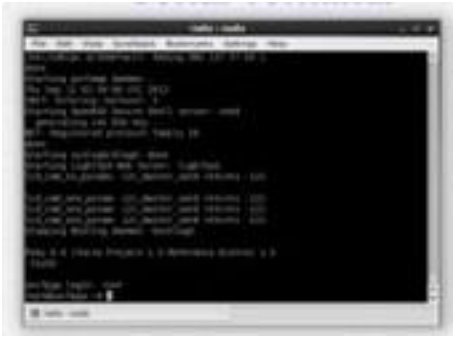
- A supported Dev. Kit
  - Altera Atlas Board
  - Altera Cyclone V SoC Board
  - Altera Arria V SoC Board
  - Arrow SoCKIT



- microSD Card



- Linux machine native or VM
  - 4GByte RAM minimum

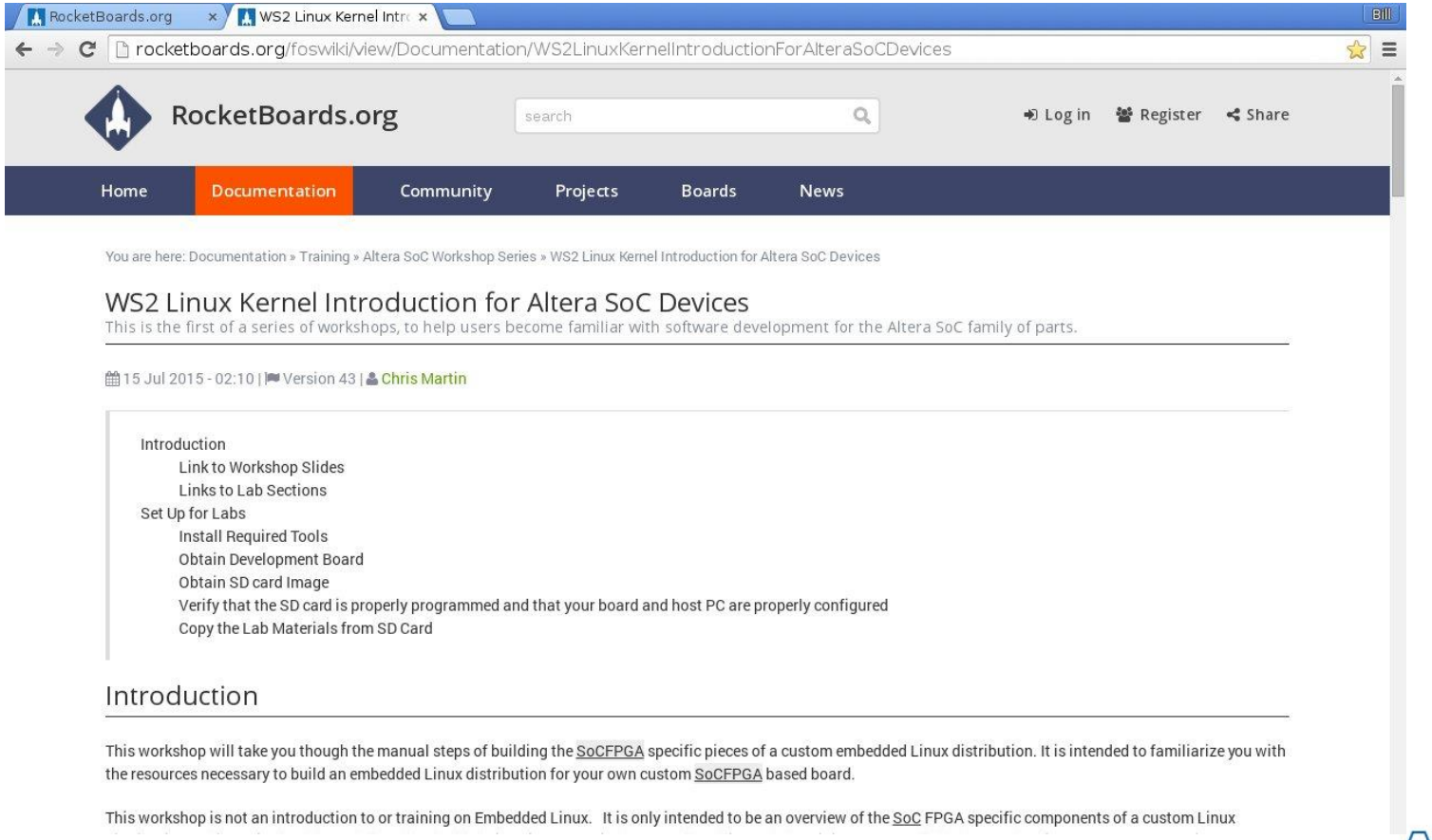


- Serial Terminal Application

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Obtaining Lab Files and Instructions

- Posted on RocketBoards
  - Link will be emailed out after class



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What You Will Accomplish

- Generate and build the preloader
- Generate and compile the device tree
- Obtain and build u-boot
- Obtain, configure, and build the kernel
- Program to an SD card
- Run a Simple Linux App
  - Verifies the lab was completed
- Submit results & feedback

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Thank You

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